

Single-Stage PFC LLC Converter Design

with the STMicroelectronics L6790A

Worked design

90 to 264 Vac, 47 to 63 Hz
 $25\text{ V} / 26.3\text{ A} = 657.5\text{ W}$

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Before you start

The L6790A behaviour described here comes from a **draft** datasheet dated 30 April 2026. That draft has items marked TBD and at least one place where it contradicts itself. Check every controller constant against the released datasheet before the design goes to production.

The worked design has not been built. Only the numbers marked as measured are measured; the rest are calculated.

All figures were drawn for this note. Where a figure follows the argument of a published application note, the caption says so.

Revision history

Version	Date	Change
V1.0	August 2026	First issue. Worked design: 25 V / 26.3 A, 657.5 W.
V1.1	September 2026	Background section on LLC and power factor correction added, with circuit figures. Transformer design, the closed form for $f_{sw}(\theta)$, the compensator design method, the gain-margin check and the input capacitor added. Worked design moved to the $N_p:N_s = 15:2$ transformer.
V1.2	September 2026	Renamed single-stage PFC LLC throughout. Verification margins defined. Worked values moved out of the design procedure and into the design example, where each one names the equation it comes from and shows the numbers put into it.
V1.3	September 2026	Design example on one transformer (ETD 49/25/16DG) with the winding and pin assignment drawn. Compensator chapter rewritten with the loop equations, the op-amp equivalent of the TL431 network and a worked loop design. Text shortened throughout. Korean edition to follow.

Conventions

- Frequencies are switching frequencies unless written f_l , the line frequency.
- $f_n = f_{sw}/f_r$, the switching frequency normalised to the series resonance; gain curves are drawn against it. Other documents may write f_s for f_{sw} .
- M is the gain of the resonant network: the output referred to the primary over the fundamental of the drive. **$M = 1$ at the series resonance** at any load.
- $Z_0 = \sqrt{L_r/C_r}$ and $Q = Z_0/R_{ac}$: $Q = 0$ is no load; larger Q is more power and a flatter curve.
- $V_{o,eff} = V_{out} + N_{rect} V_f$, with N_{rect} the rectifier drops in the conduction path: 1 for a centre tap, 2 for a full bridge.
- n is the model turns ratio in the gain equation; n_T the wound ratio with the leakage in L_r .
- $\lambda = L_r/L_m$; older literature uses $m = 1 + 1/\lambda$.
- θ is the line phase angle, zero at the mains zero crossing. “Equivalent input” is what the tank sees after topology morphing, not the mains voltage.

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1 Introduction

A conventional universal-input LLC supply has two stages: a boost power-factor corrector that holds a 400 V bus, and an LLC converter that steps the bus down. The bus capacitor stores the energy a unity-power-factor input cannot deliver evenly, and it gives the LLC an almost constant input, so the tank covers only a narrow gain range.

A **single-stage PFC LLC** removes the boost stage and the bus capacitor. The rectified mains feeds the resonant tank directly. The input is then a 100/120 Hz half sine, the gain is set by the input rather than by the controller, and the output capacitor must absorb what the bus capacitor used to. Most two-stage design habits do not apply.

This note explains how the converter works and how to design one with the STMicroelectronics **L6790A**. One design runs through it: **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out.**

Scope. Power stage, resonant tank, output bank, controller network and voltage loop. Not covered: EMI filter, magnetics construction, layout, safety approval.

2 The two converters this one is made of

This chapter covers an ordinary LLC and ordinary power factor correction. Readers who know both can skip to Section 3.

2.1 What an LLC converter is

An LLC converter is a square-wave generator driving a resonant network, which feeds a transformer and a rectifier. The switches only set the frequency. There is no duty cycle and no programmed inductor current.

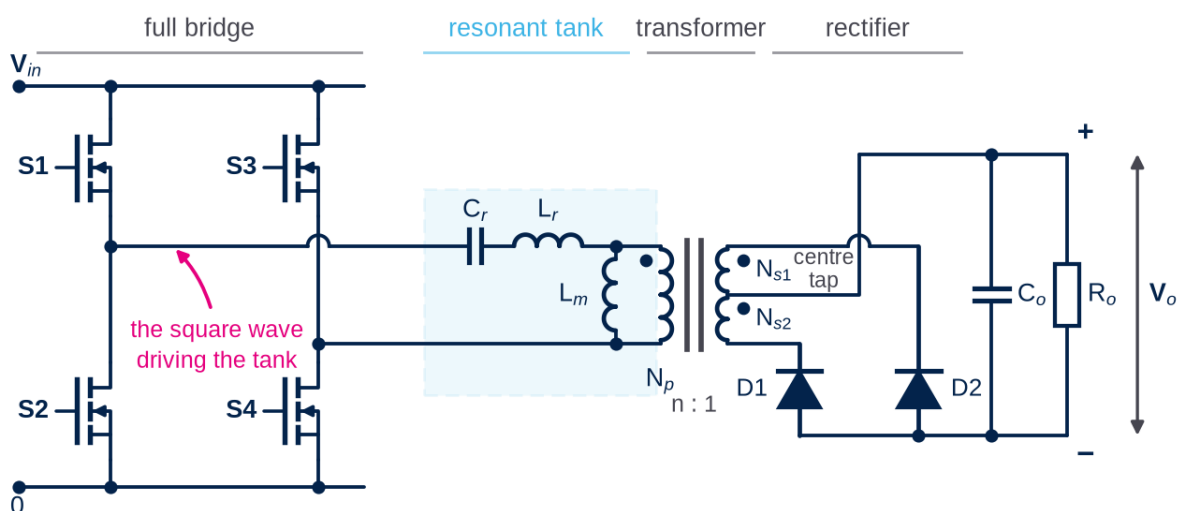


Figure 1: An LLC stage. The switches set the frequency; the tank sets the power; the transformer sets the voltage.

The three tank elements are the series inductance L_r , the magnetising inductance L_m of the transformer, and the series capacitance C_r . L_m is the transformer's own inductance and L_r is usually its leakage inductance, so only C_r is

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always an added part.

2.2 Why an LLC, and not something simpler

Compared with a forward or flyback converter, an LLC gives:

- **Soft switching at every load.** The primary switches turn on at zero voltage, and below resonance the rectifiers turn off at zero current. The magnetising current that does this is present at any load, so ZVS is not lost at light load.
- **Higher frequency**, because switching loss no longer limits it, so smaller magnetics and filters.
- **The leakage inductance is used as L_r** instead of needing a snubber, and there is no output inductor.

The costs: the frequency moves with line and load, the magnetising current circulates at any load so light-load efficiency is modest, and the design comes from a gain curve, not from a duty-cycle expression.

2.3 What resonance is for

In a hard-switched converter the energy in each switch's output capacitance is lost at every turn-on, and that loss limits the frequency.

An LLC running **in its inductive region** presents an inductive load to the bridge, so the tank current lags the drive voltage. In the dead time that lagging current discharges the mid-point by itself, and the next switch turns on at zero volts. **That is zero-voltage switching**; Section 2.10 gives the condition.

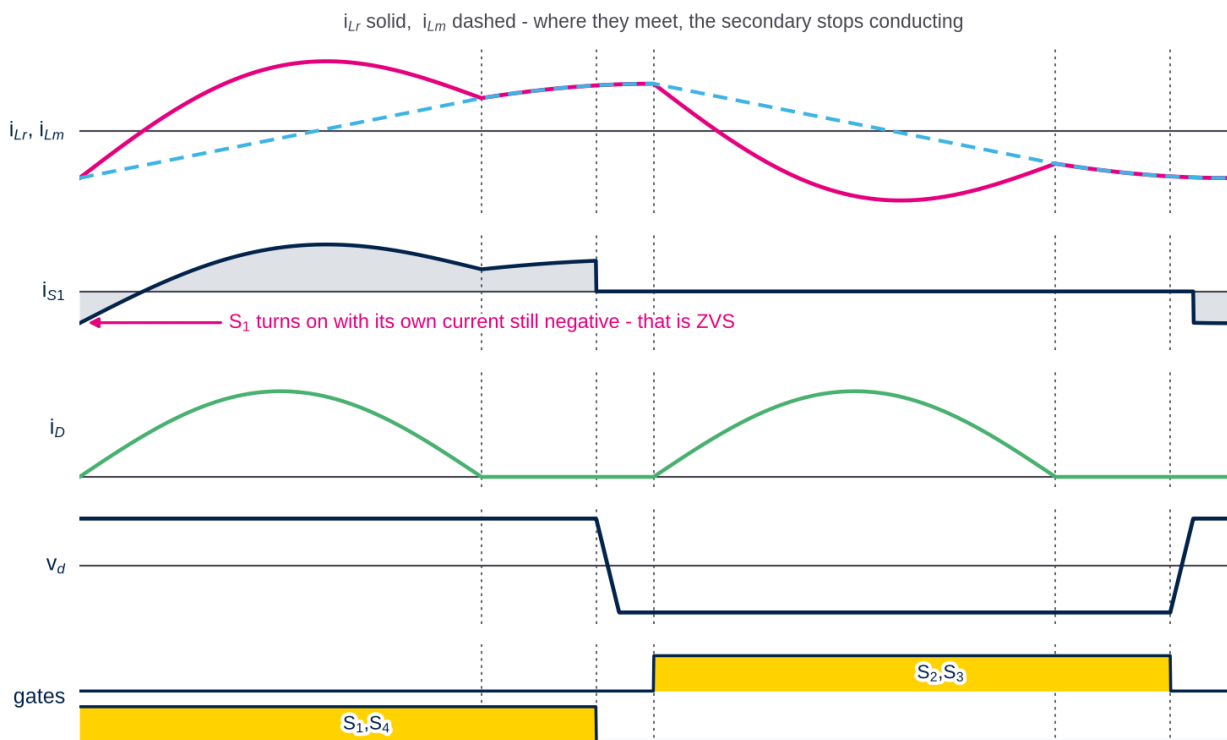


Figure 2: One switching period below resonance. The difference between i_{Lr} and i_{Lm} crosses to the secondary. The switch turns on while its current is still negative; that is what discharges the mid-point.

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The condition is *inductive operation*, not a fixed frequency. Below the capacitive boundary the current leads, the bridge hard-switches into a low impedance, and parts are destroyed rather than heated. The boundary moves with load (Section 2.9).

2.4 How the circuit becomes $M(f_n, Q)$

The gain curve comes from three simplifications, each of which discards something.

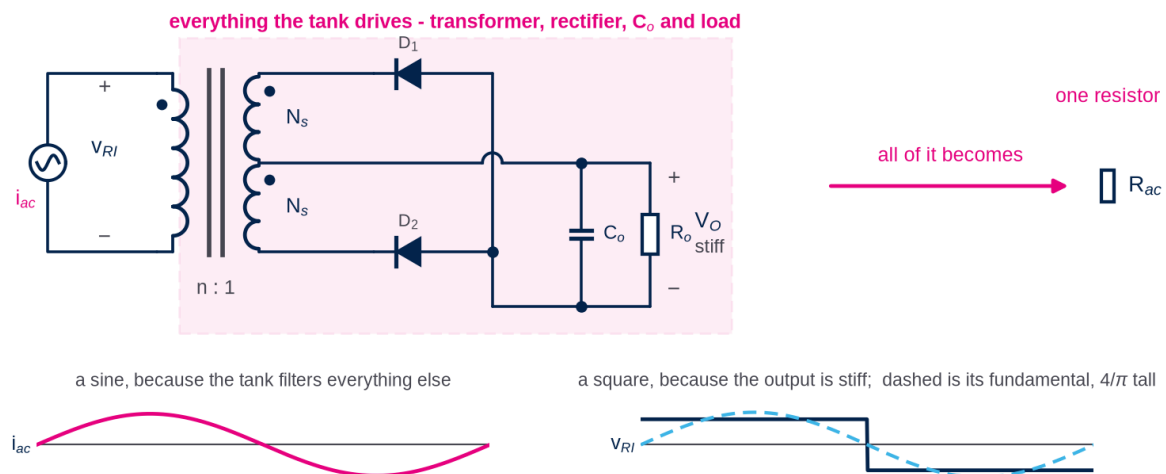


Figure 3: Why the rectifier and load can be one resistance. The output is stiff, so the tank sees a square voltage; the tank passes only the fundamental, so the current is a sine. At the fundamental the pair takes the same power as a resistor.

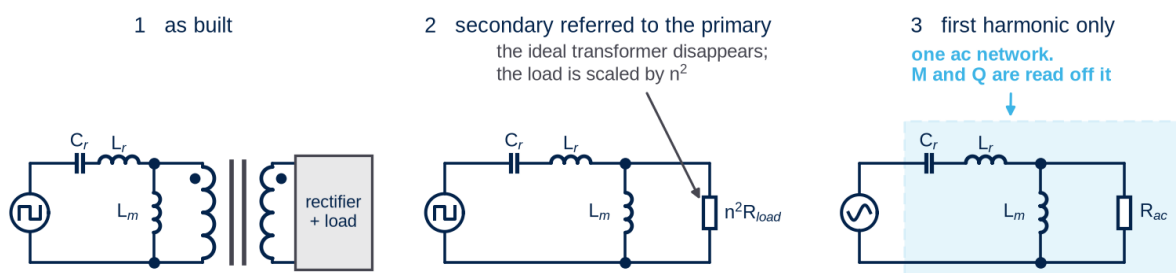


Figure 4: The reduction in three steps: as built; secondary referred to the primary; fundamental only. The result is one ac network with gain M and damping Q .

- **Refer the secondary to the primary.** The load resistance is multiplied by n^2 .
- **Replace the rectifier and output filter by one resistor.** At the fundamental they look like one.
- **Keep only the fundamental of the drive.** This is the *first harmonic approximation*. It is why an LLC design ends with a simulation or a measurement, not with the equations.

Two numbers describe the remaining network:

- **The gain M :** the voltage across R_{ac} over the fundamental that drives it. C_r and L_r cancel at the series resonance f_r , so **$M = 1$ there at any load**. Below f_r the tank can step up; above it, only down.
- **The quality factor $Q = Z_0/R_{ac}$,** with $Z_0 = \sqrt{L_r/C_r}$. **Q rises with output power.** At $Q = 0$ the curve is tall and peaked; load flattens it.

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With f_n normalised to the series resonance:

Equation 1

$$f_n = \frac{f_{sw}}{f_r}, \quad Z_0 = \sqrt{\frac{L_r}{C_r}}, \quad Q = \frac{Z_0}{R_{ac}}, \quad \lambda = \frac{L_r}{L_m}$$

$M(f_n, Q)$ always means the gain of this reduced circuit.

2.5 The two resonances

While the secondary conducts, the reflected output voltage clamps L_m , leaving the series resonance

Equation 2

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

When the secondary is off, both inductances resonate:

Equation 3

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}}$$

At f_r the gain is 1 for any load. At f_o the no-load gain is unlimited. The tank can boost between the two; above f_r it cannot.

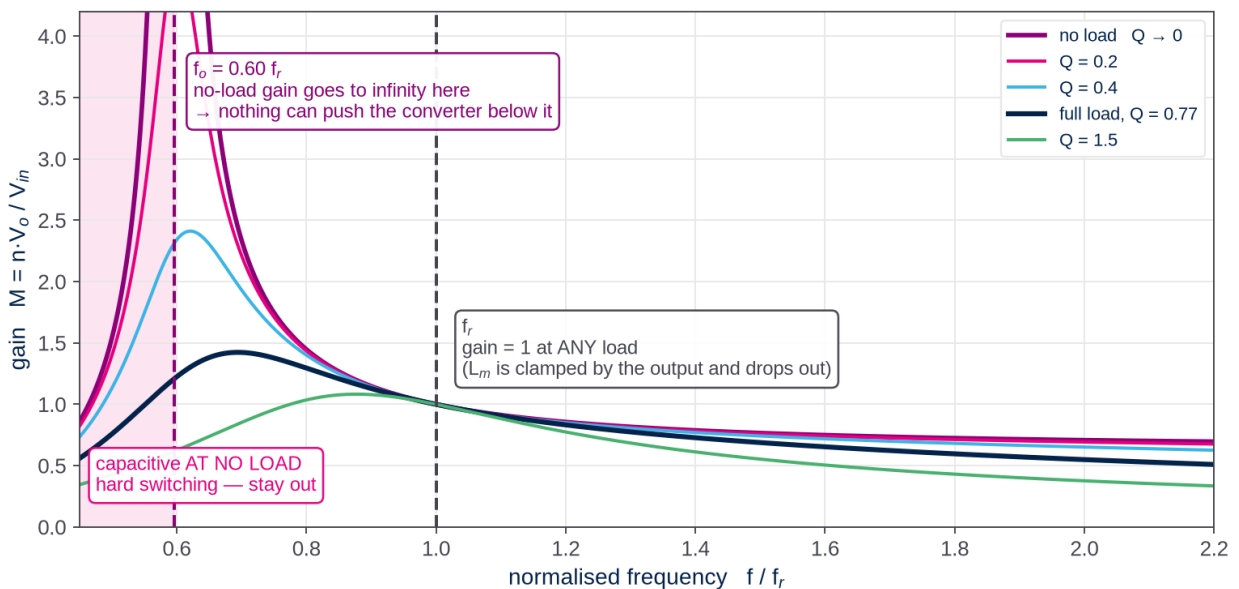


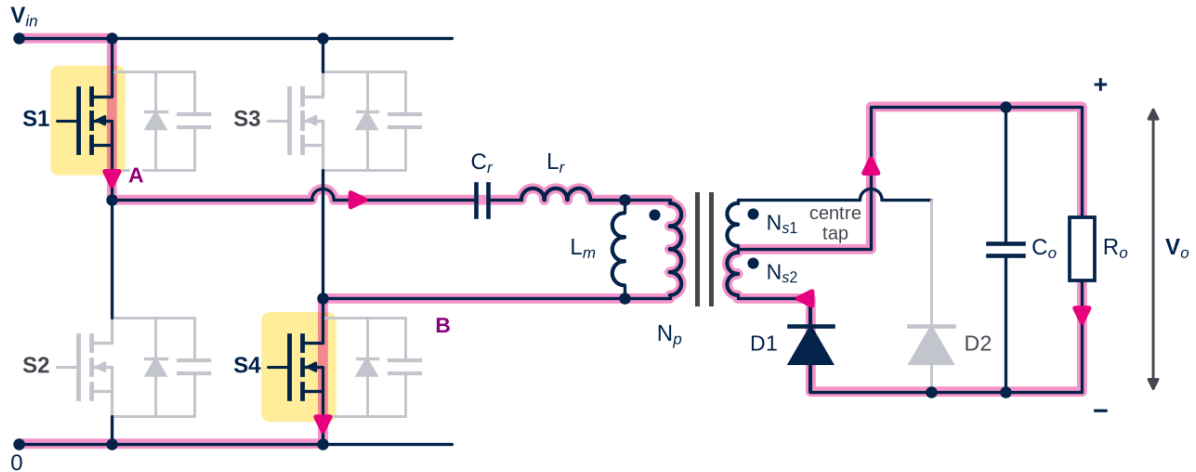
Figure 5: The two resonances and the band between them in which the tank can boost. The shaded edge is the no-load capacitive boundary; it moves up under load.

Below resonance each half cycle has two parts, with the same pair of switches on throughout: power delivery, then freewheeling. At resonance the half cycle is power delivery only; above resonance power delivery is cut short.

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1 POWER DELIVERY

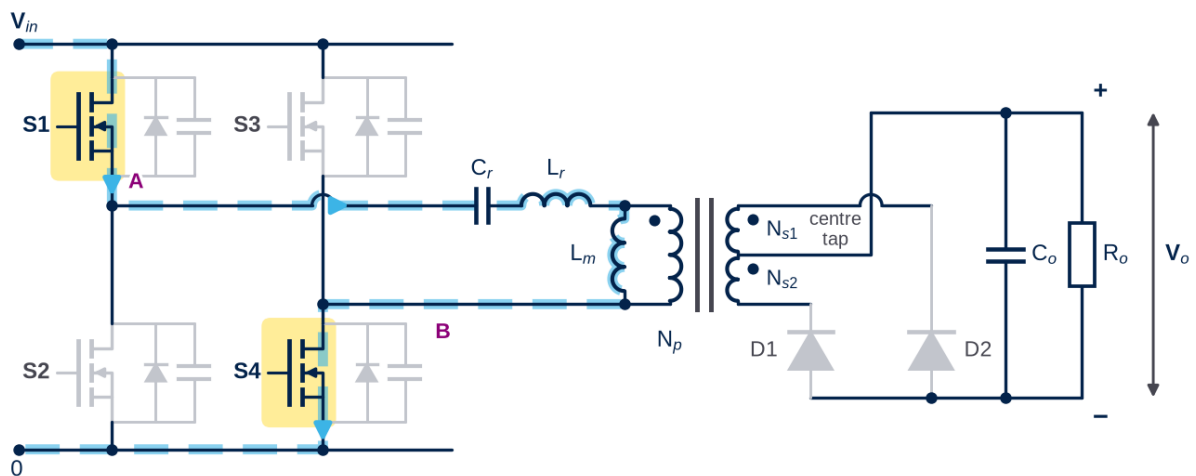
S1, S4 on · D1 conducts · lower half of N_s



L_m is clamped, so i_{Lm} is a straight ramp and $i_{Lr} - i_{Lm}$ is the half sine that crosses.
What is traced is the load component; i_{Lm} flows in L_m too, but it reverses inside this interval.

2 FREEWHEELING

S1, S4 STILL on · secondary off



i_{Lr} has fallen to i_{Lm} . Every rectifier is off, L_m is free again, and the tank rings at f_o .

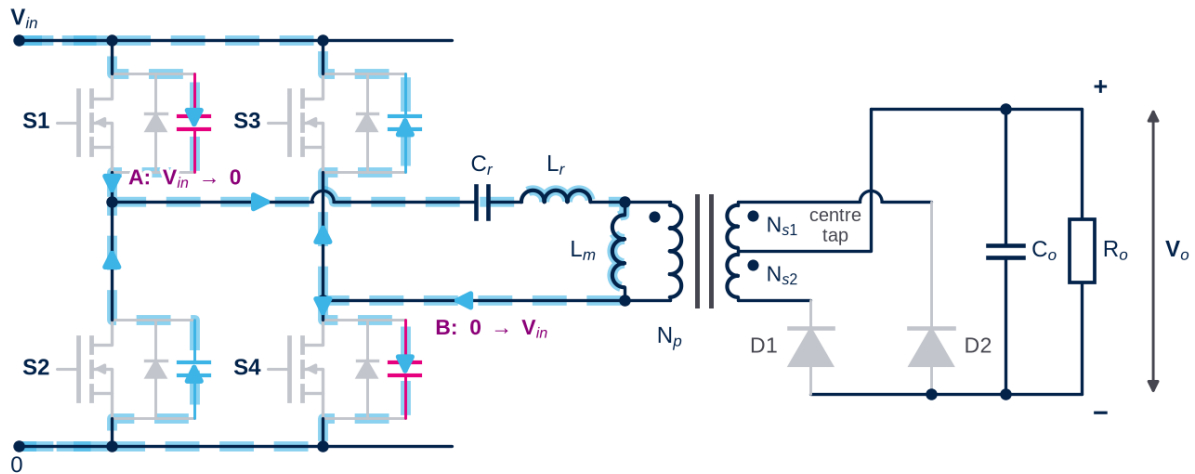
— load current
 — magnetising current only
 gated on
 body diode conducting = the ZVS window
— C_{oss} charging, V_{ds} rising
— C_{oss} discharging

Figure 6: First half, steps 1 and 2. 1 Power delivery: S_1, S_4 on, the resonant current exceeds the magnetising current and the difference reaches D_1 . L_m is clamped, so this interval resonates at f_r . **2 Freewheeling:** the resonant current has fallen to the magnetising current, the rectifiers are off, and L_m joins the resonance at f_o . The same switches are still on. Greyed body diodes and C_{oss} do nothing in that step.

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3 DEAD TIME (a)

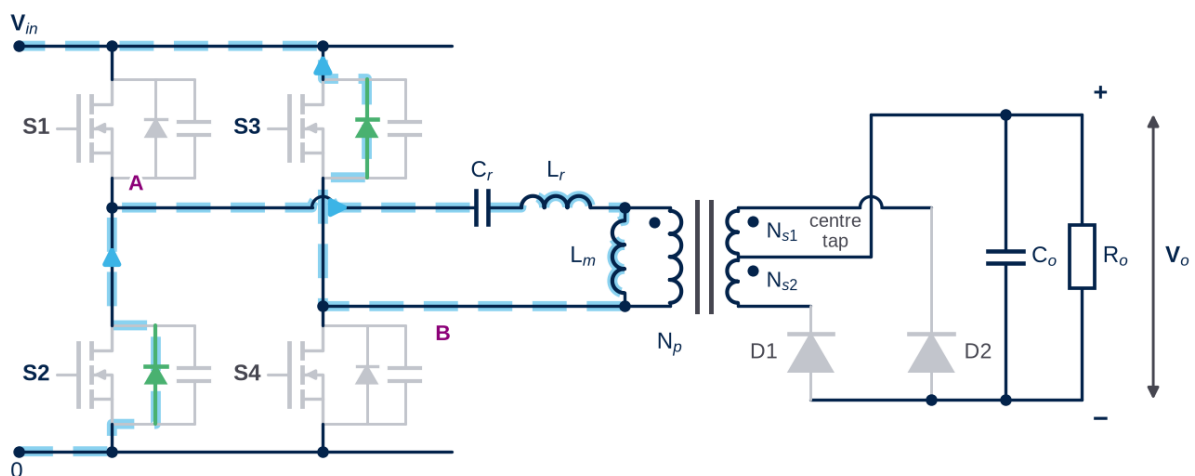
all four off · C_{oss} swaps the midpoints over



All four channels are off, so the ONE tank current can reach the rails only THROUGH the four C_{oss} . It arrives at A from both, so A is pulled down; it leaves B into both, so B is pushed up. That swap IS the transition.

4 DEAD TIME (b) — ZVS

body diodes of S2 and S3 clamp



$V_{ds} = 0$ on S2 and S3 now, because their own body diodes are clamping them. Gate them on HERE.

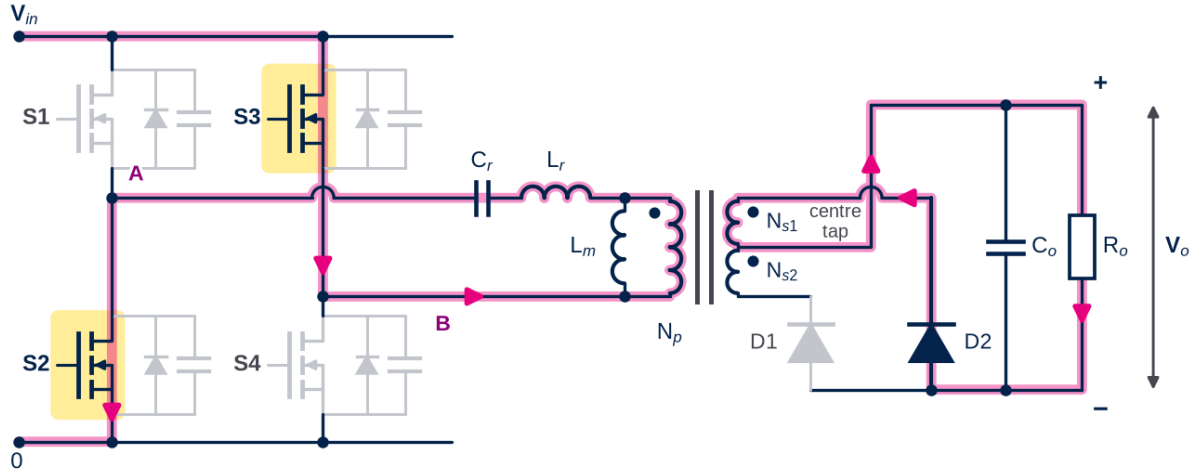
— load current — magnetising current only gated on
 body diode conducting = the ZVS window — C_{oss} charging, V_{ds} rising — C_{oss} discharging

Figure 7: Steps 3 and 4, the dead time. 3 All switches off; the magnetising current can only flow through the four C_{oss} , pulling A down and pushing B up. 4 The swing is complete and the body diodes of S_2, S_3 clamp V_{ds} at zero. The next gate must arrive inside this window.

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5 POWER DELIVERY

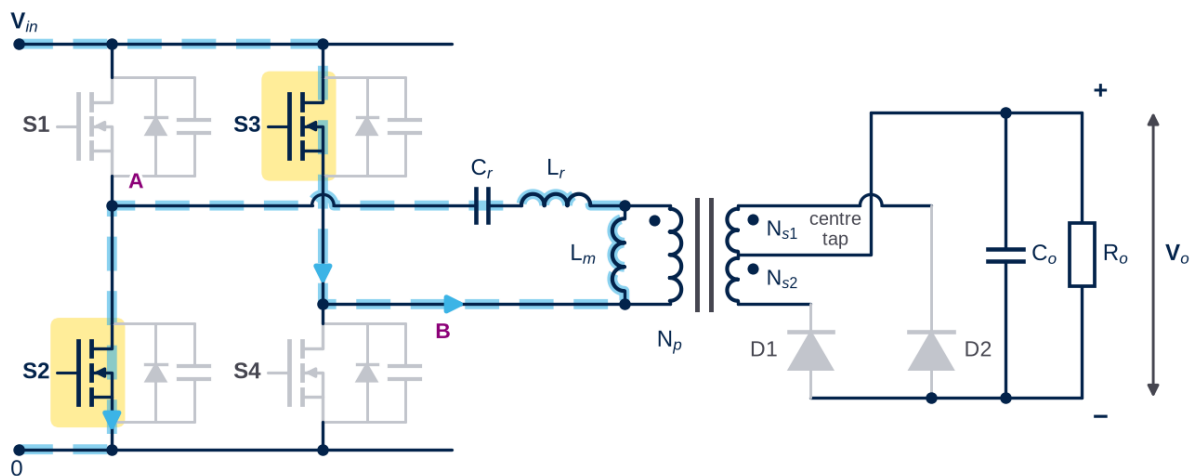
S2, S3 on · D2 conducts · upper half of N_s



The mirror of 1. Just after turn-on the old current still runs back through the two channels. Again only the load component is traced - i_{Lm} reverses inside this interval as well.

6 FREEWHEELING

S2, S3 STILL on · secondary off



The mirror of 2, and the easy one to miss: it ends at the right-hand edge of a waveform plot.

— load current — magnetising current only gated on
 body diode conducting = the ZVS window C_{oss} charging, V_{ds} rising C_{oss} discharging

Figure 8: Second half, steps 5 and 6: the mirror image of 1 and 2 with S_2, S_3 and D_2 .

13

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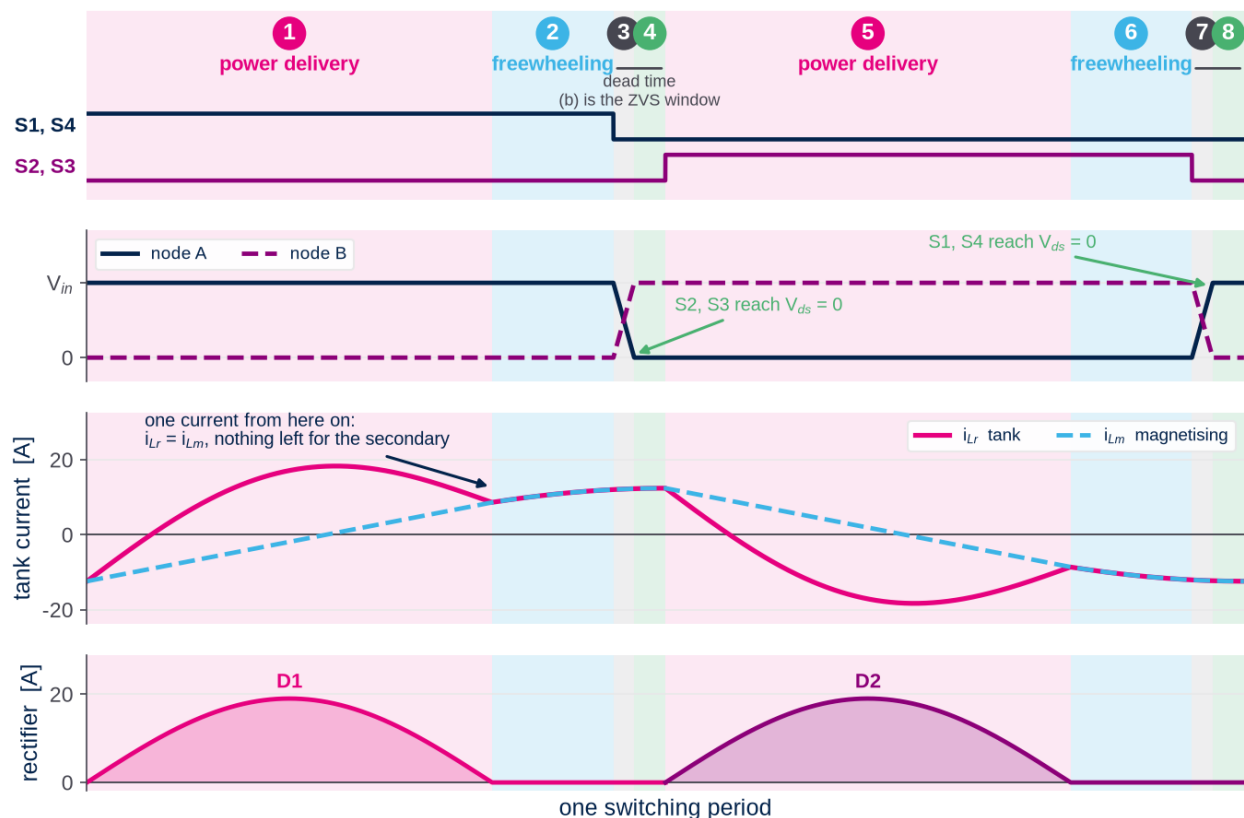


Figure 10: Where each step sits on the waveforms. The numbered bands are the eight panels above in time order. Drawn below resonance at $f_{sw}/f_r = 0.70$ with the dead time widened so that steps 3, 4, 7 and 8 are visible; the current amplitudes are this design's $I_{Lr,pk}$ and $I_{Lm,pk}$ at the lowest input voltage.

Steps 1 and 2 are the two resonances, and both are in one half cycle with the same switches on. ZVS happens in neither. It happens in the dead time, steps 3–4 and 7–8.

“Freewheeling” means two different intervals in the literature. Here it is steps 2 and 6: rectifiers off, L_m in the resonance, same switches on. Toshiba’s *Resonant Circuits and Soft Switching* (2019) uses it for the dead time, when the body diodes conduct. The dead time exists at every frequency; steps 2 and 6 exist only below resonance.

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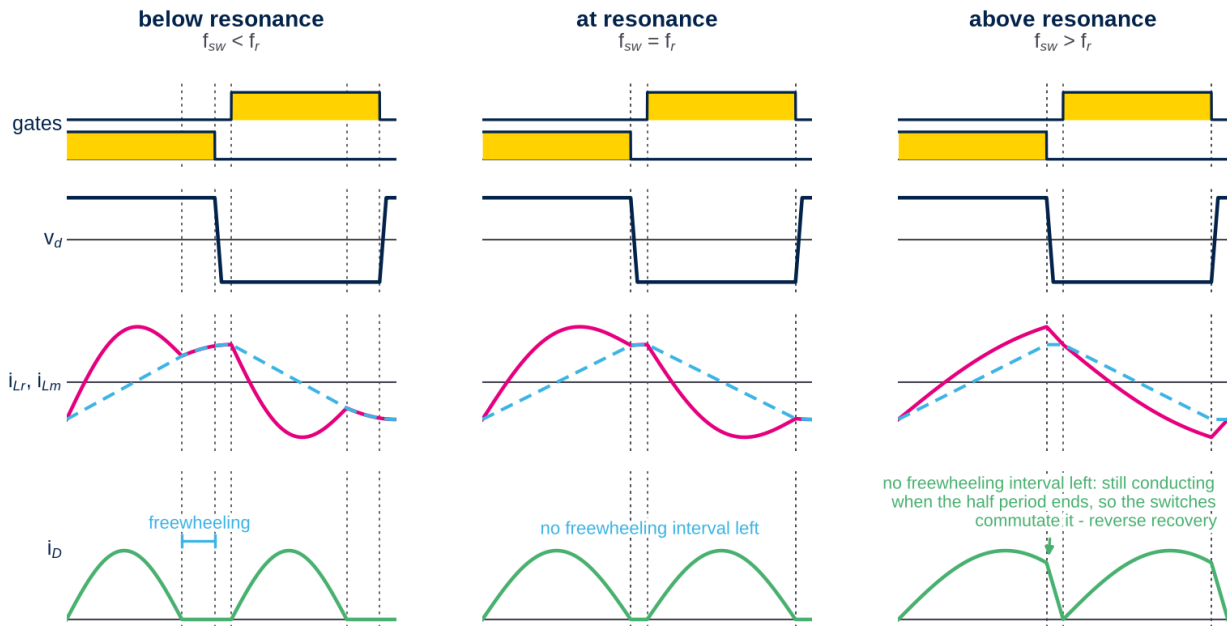


Figure 11: Below, at and above resonance: gates, bridge voltage, resonant current with the magnetising current dashed, and rectifier current. What changes is the resonant half period against the switching half period.

- **Below f_r .** The resonant half sine ends early and i_{Lr} **lands on i_{Lm}** . From then on no power crosses to the secondary: freewheeling.
- **At f_r .** i_{Lr} meets i_{Lm} as the bridge switches and the rectifier current reaches zero at the same instant. This is the most efficient point.
- **Above f_r .** The switching half period ends first, the resonant half sine is **cut off**, the switches turn off more current, and the rectifier is interrupted while conducting.

The freewheeling interval is f_o . Nothing clamps L_m , so $L_r + L_m$ rings with C_r ; it looks flat only because f_o is far below f_r . Lower f_{sw} towards f_o and the interval fills the whole half period, and no power is delivered. Everything useful happens between the two frequencies.

Symbols are not standard. At least one widely used guide calls the *series* resonance f_o , the opposite of the meaning here. Check before comparing.

A conventional LLC stays on one side, usually just above f_r at nominal input, where the circulating current is smallest. Which side this converter runs on is answered in Section 4.7.

2.6 The gain function and the three regions

With the first-harmonic approximation the tank gain is

Equation 4

$$M(f_n, Q, \lambda) = \frac{1}{\sqrt{\left(1 + \lambda - \frac{\lambda}{f_n^2}\right)^2 + Q^2 \left(f_n - \frac{1}{f_n}\right)^2}}$$

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Two checks by eye: at $f_n = 1$ the Q term vanishes and $M = 1$ at any Q , so every curve passes through that point. At no load ($Q = 0$) raising f_n only brings M down to $1/(1+\lambda)$; that floor matters in Section 4.10.

The curve has three regions. Below the capacitive/inductive boundary the bridge hard-switches: not allowed. Between the boundary and f_r the tank is inductive and boosts; above f_r it is inductive and bucks. The boundary is drawn for the full-load curve; at no load it sits at f_o and it moves up with load.

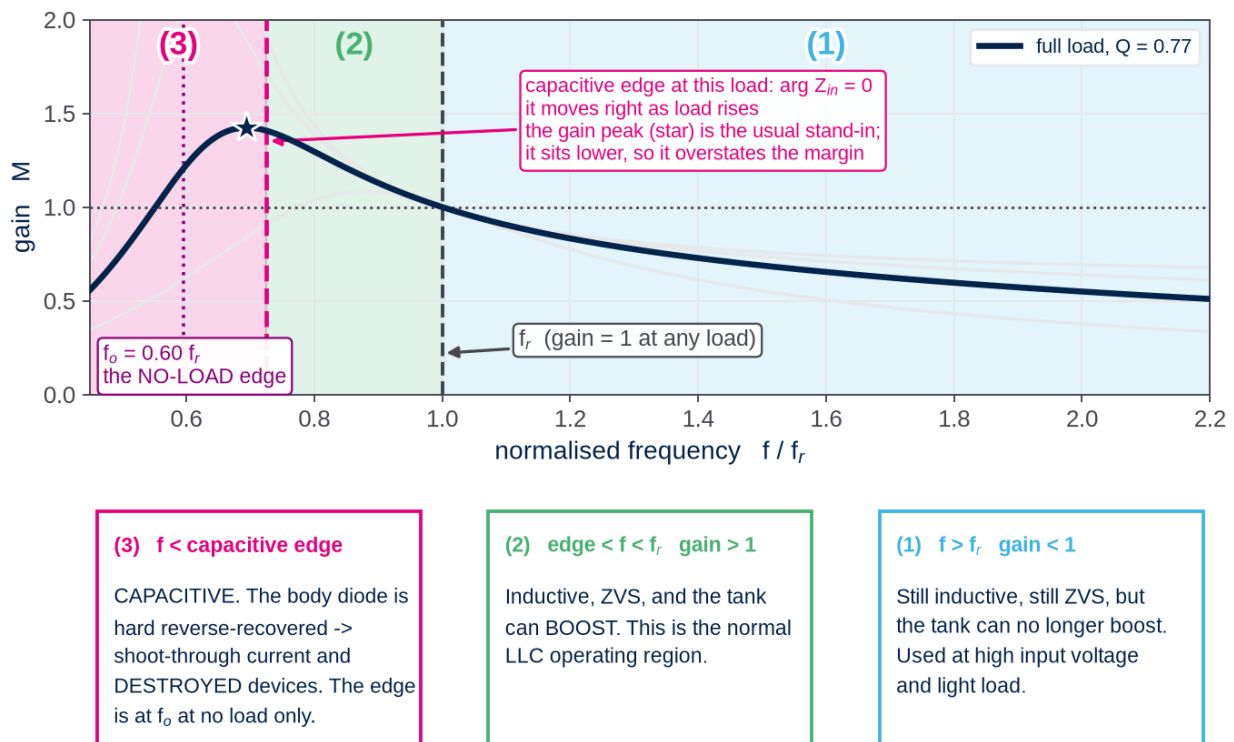


Figure 12: The three regions, shaded for the full-load curve. Only the inductive ones are usable. Framing after ROHM TechWeb.

Which side the converter is on can be read from the chart without a waveform. Draw the required gain as a horizontal line and find where it crosses the curve for the present load. Every curve passes through $M = 1$ at $f_n = 1$, so the rule is:

- **Required gain above 1:** the crossing lies left of $f_n = 1$. The converter is below resonance and boosts; the waveforms are the first row of Figure 11.
- **Required gain equal to 1:** the crossing is at f_r , at any load.
- **Required gain below 1:** the crossing lies right of $f_n = 1$. The converter is above resonance and bucks; the resonant half sine is cut off.

So the input voltage decides the side, not the load: the required gain is the reflected output voltage over the input the tank sees, and the load only picks which curve the line meets. A higher input lowers the line and moves the crossing right. The load matters for the other question, whether the crossing is still on the inductive side of the peak.

2.7 Capacitive and inductive, and why those words

The names describe **what the bridge sees**: the phase of the tank current against the square wave that drives it. At low frequency C_r dominates and the current *leads*; at high frequency the inductances dominate and it *lags*. That

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phase decides whether the switching is soft or hard.

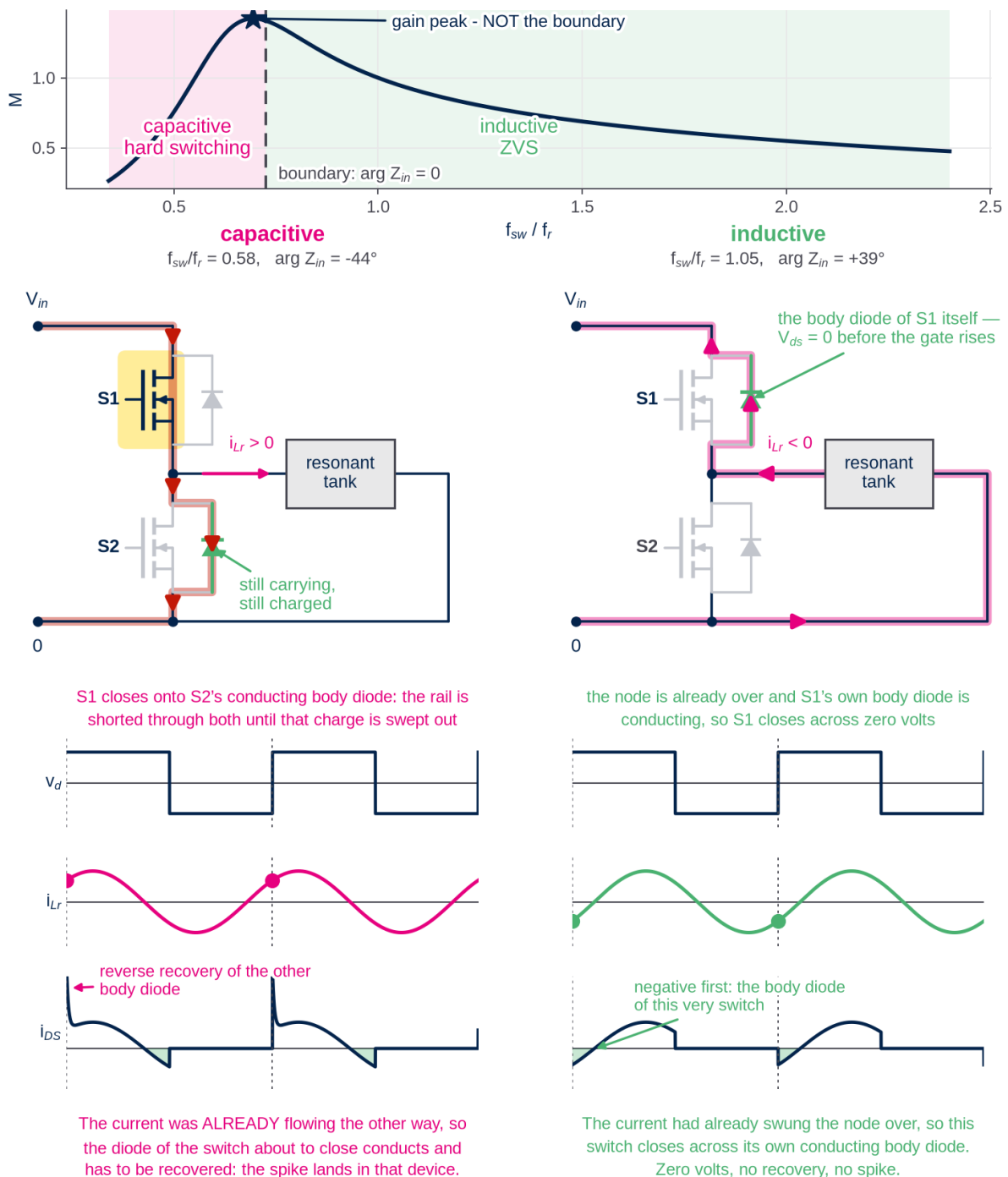


Figure 13: Either side of the boundary. Top: the boundary sits a little above the gain peak. Middle: the leg as S_1 is gated on. Left, the tank current is already positive, flowing in S_2 's body diode, so S_1 closes onto a conducting diode. Right, it is negative, has carried the mid-point up during the dead time, and S_1 finds its own body diode conducting at zero volts. Bottom: bridge voltage, tank current and the drain current of the closing switch, to one scale.

- **Inductive: the current lags.** At turn-off it still pushes the bridge node towards the other rail; the dead time lets it, and the next device turns on at zero volts. **ZVS.**

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- **Capacitive: the current leads.** It has already reversed, so it pushes the node back. The next device turns on into the full rail, and the other body diode is forced off with reverse recovery through a low impedance. **Hard switching, and the failure that destroys parts.** The capacitive region is a boundary the design stays on one side of; the controller has a protection for the case where it does not.

2.8 Why a diode has to recover on one side and not on the other

Take one leg: a high-side switch, a low-side switch and the tank on the mid-point. When either device turns off, **the tank current does not stop**. What carries it during the dead time decides everything. The middle row of Figure 13 shows the two cases, after ON Semiconductor AN-4151.

- **Inductive.** The current carries the mid-point to the *other* rail through the two C_{oss} , the incoming body diode picks it up, and that device is gated on at zero volts. Its own body diode then hands over to the channel with no reverse voltage: **nothing to recover**.
- **Capacitive.** The current has already reversed. It holds the mid-point on the rail it came from through the body diode of the device that just turned off. The incoming device is gated on across the full rail, onto a conducting diode.

That diode cannot block until its stored charge is swept out, so for that time both devices conduct and the rail is shorted through them. The current is a spike limited only by loop inductance, dissipated in the incoming device at full rail voltage; when the diode snaps off, the di/dt in the loop inductance appears as overshoot across it.

This is why a fast-recovery body diode is required on the primary (Section 6.24): in a full bridge that diode conducts before every turn-on even in normal operation. It is also why f_{Min} is kept above f_o , and why the anti-capacitive protection exists.

2.9 The two boundaries are not the same boundary

Both pairs of words describe positions on the same frequency axis, but they are different lines on it.

	capacitive / inductive	below / above resonance
the boundary	where $\arg Z_{in} = 0$	f_r , the series resonance
does it move?	yes — with load	no, it is fixed by L_r and C_r
what it decides	ZVS or hard switching, on the primary	boost or buck, and ZCS or not, on the secondary
which side is allowed	inductive only, always	both sides are legitimate

Table 1: Two classifications, two boundaries, two consequences.

The capacitive boundary sits between f_o and f_r and **rises with load**: at no load it is f_o , at full load it has moved up towards f_r . A frequency that is inductive at light load can be capacitive at overload.

The boundary is not the gain peak. It is where $\arg Z_{in} = 0$, a little *above* the peak. Using the peak is optimistic in the dangerous direction. Section 7.5 gives both frequencies for this design; the verdict comes from the ZVS check of Section 6.5.

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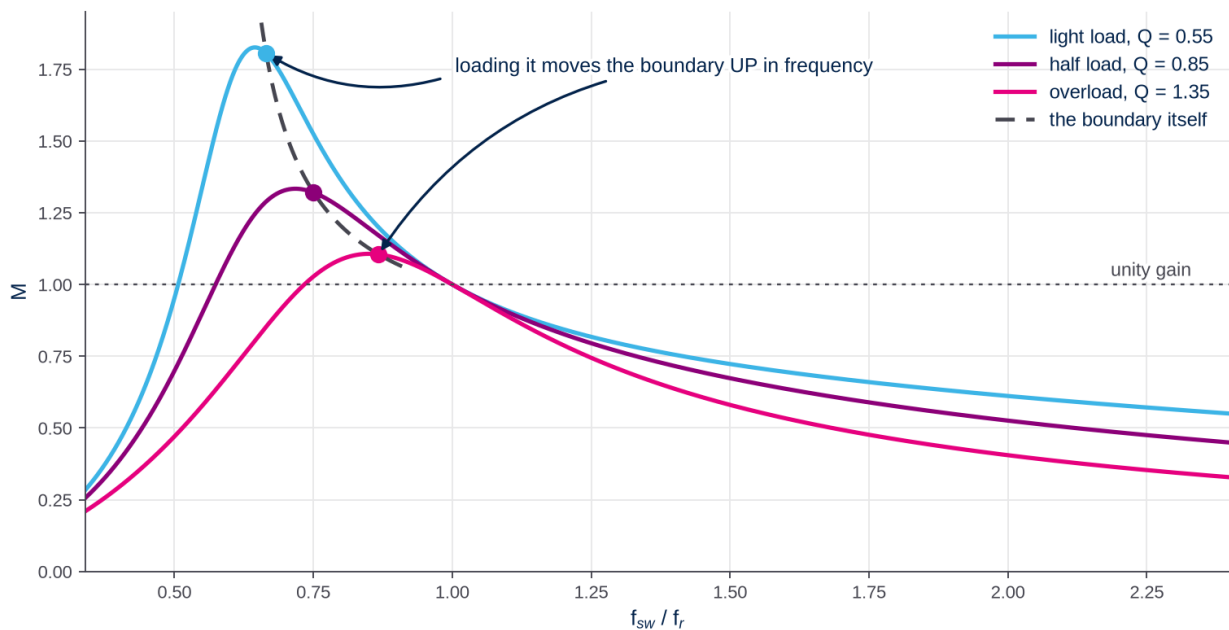


Figure 14: The boundary moves up with load. The dashed curve is the boundary itself, traced as Q varies.

- Below f_r and inductive is the normal boosting region.
- Below f_r and capacitive is the fault. Going below f_r is not the danger; going below the boundary is.
- Above f_r is inductive at every load.

Design rule. Keep f_{Min} above f_o . That is only the no-load boundary, so it is necessary but not sufficient; the real check is the ZVS sweep of Section 6.5, run under load at every line phase.

2.10 ZVS and ZCS are not the same thing

Both switch a device while its voltage or its current is zero. They apply to different devices, and only one is guaranteed.

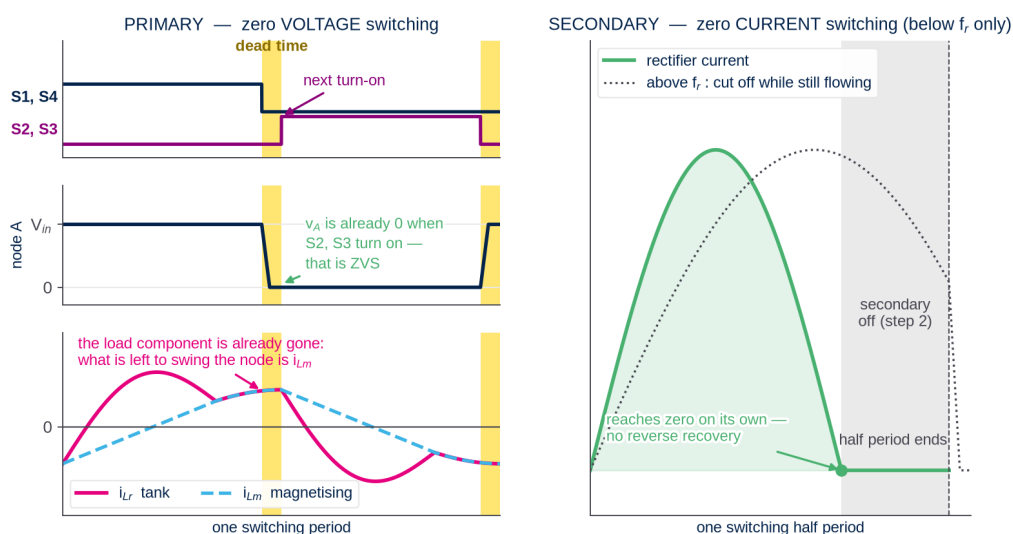


Figure 15: ZVS on the primary, ZCS on the secondary: different devices, different mechanisms, different conditions.

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- **ZVS on the primary switches** needs inductive operation, which holds on both sides of f_r . It is required everywhere; Section 6.5 gives the margin.
- **ZCS on the secondary rectifiers** happens when the resonant current reaches zero by itself, which is only **below** f_r . Above f_r the rectifier is cut off while conducting and reverse recovery returns.

A converter whose input crosses f_r must be checked on both sides: its rectifier body diode and synchronous-rectifier dead time matter whenever it runs above f_r .

2.11 Sizing the dead time so that ZVS actually happens

ZVS is a charge problem. During the dead time the magnetising current must move enough charge to swing the bridge node across the rail. The design quantity is T_{ZC} , the time the tank current takes to reach zero after the transition, and it must exceed the dead time:

Equation 5

$$T_{ZC} > t_D$$

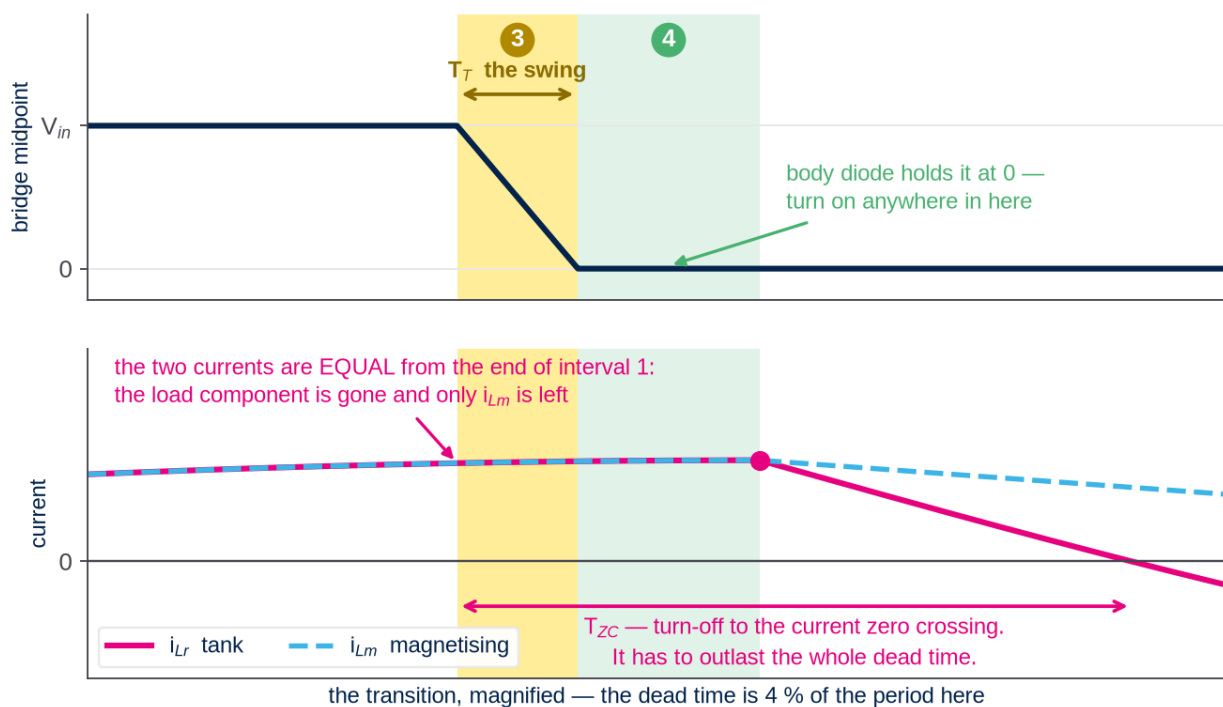


Figure 16: The magnetising current discharges the bridge node during the dead time. Two conditions: the current must still flow when the gates drop ($T_{ZC} > t_D$), and the swing must finish inside the dead time.

Use $C_{o(tr)}$ (equivalently Q_{oss}), not the headline C_{oss} : the datasheet values differ by three to five times, and ZVS is a charge question.

2.12 Not a flyback, and why that changes the core

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Flyback habits are wrong here. Two of them cost money: **why the core area matters in a converter that stores none of the energy it delivers**, and **why the saturation test current is nothing like the peak winding current**. Both come from when the two windings conduct.

In a flyback they never conduct together, so at every instant the whole winding current is magnetising current. Every joule delivered was first stored in the core, the flux follows the *current*, and A_e is sized from the peak current.

In an LLC they conduct together. Their ampere-turns oppose, and only the difference magnetises the core:

Equation 6

$$N_p i_p - N_s i_s = N_p i_\mu$$

The load current passes through by transformer action and never enters the core. What the core holds is the magnetising energy $\frac{1}{2} L_m i_\mu^2$, and that is not a by-product: i_μ is what swings the bridge node in the dead time. The gap exists to set L_m , not to store load energy.

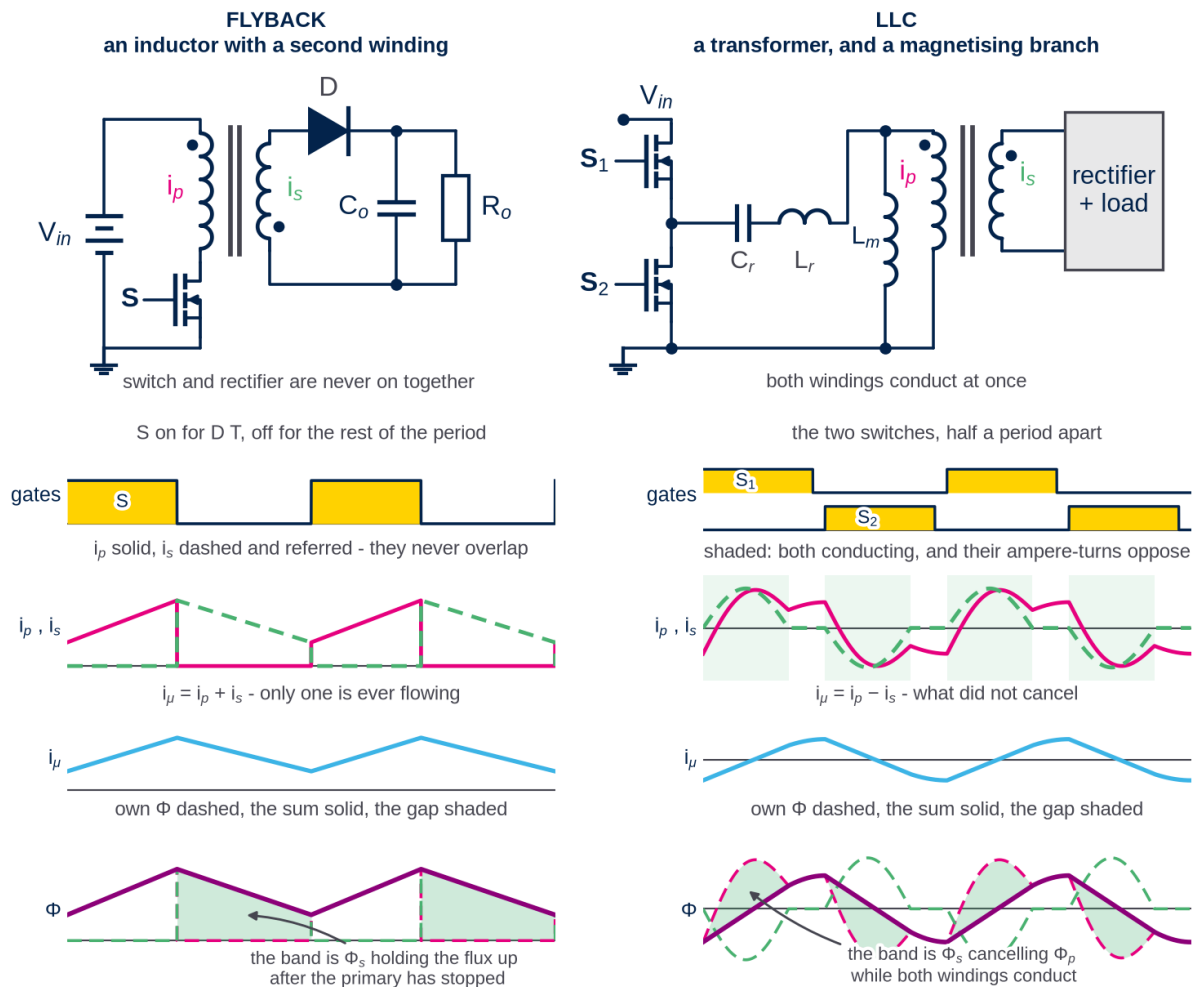


Figure 17: Left: the switch and the rectifier are never on together, so the conducting winding carries all the magnetising current and the flux ramps with it. Right: both conduct at once and only $i_\mu = i_p - i_s$ magnetises the core. i_s is referred to the primary. Bottom row: dashed, what each winding would drive alone; solid, their sum; shaded, what the secondary removed.

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Built in order. Figure 18 follows one period of each converter interval by interval: which winding conducts, what voltage that winding holds, and therefore which way the flux moves and what stops it.

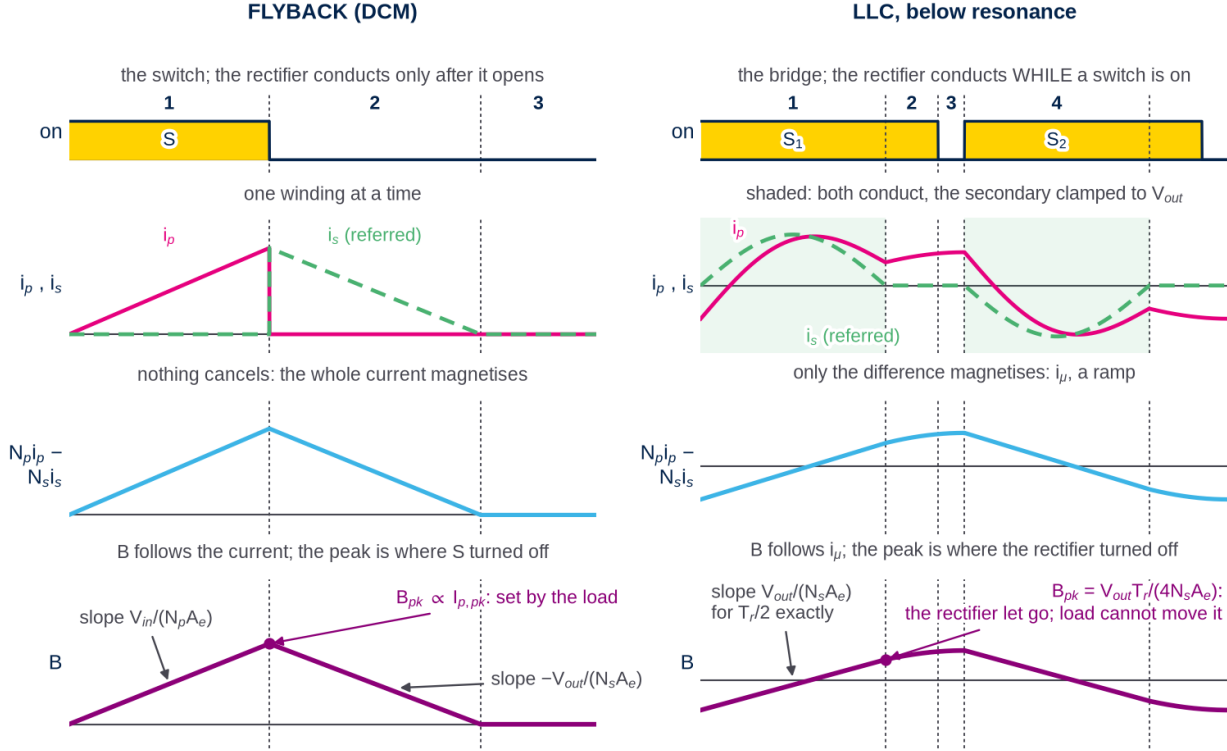


Figure 18: The flux built step by step. Left, a flyback in discontinuous conduction; right, an LLC below resonance, drawn from the eight-interval model with this design's currents. The last row is the point: in the flyback the flux tracks the winding current and peaks wherever the switch turns off; in the LLC it ramps at $V_{out}/(N_s A_e)$ for exactly $T_r/2$ and stops when the rectifier lets go.

- **Flyback, 1.** The switch closes. V_{in} sits on the primary, i_p ramps at V_{in}/L_p , and because no other winding conducts every ampere-turn magnetises: B rises at $V_{in}/(N_p A_e)$. The energy goes into the gap.
- **Flyback, 2.** The switch opens. The flux cannot jump, so the ampere-turns pass to the secondary at once, $N_s i_s = N_p i_{p,pk}$. The rectifier conducts, V_{out} sits on the secondary, and B falls at $V_{out}/(N_s A_e)$ while the gap energy leaves for the output.
- **Flyback, 3.** The secondary current reaches zero; both windings are off and the flux rests near zero until the next cycle. The peak flux was fixed at the turn-off instant by $i_{p,pk}$, which is whatever the load and the input asked for.
- **LLC, 1.** A bridge switch closes. As soon as the reflected voltage exceeds V_{out} the rectifier conducts, so the secondary is clamped to V_{out} and the magnetising branch sees nV_{out} . The load current flows in both windings at once, their ampere-turns cancel, and only i_μ ramps: B rises at $V_{out}/(N_s A_e)$.
- **LLC, 2.** The resonant half period ends: i_s reaches zero, the rectifier turns off and the clamp is gone. L_m joins the resonance with C_r , i_μ is nearly flat, and B holds at its peak. That peak was set by V_{out} and $T_r/2$ and by nothing else.
- **LLC, 3.** Dead time. i_μ swings the bridge node to the other rail; the flux hardly moves.
- **LLC, 4.** The other switch closes and the other rectifier clamps the secondary to $-V_{out}$; B ramps back down at the same slope to $-B_{pk}$.

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	Flyback	LLC
Which winding sets the flux	whichever one conducts; B follows its current	the clamped secondary; B follows its volt-seconds
What the peak flux depends on	$I_{p,pk}$: load, input voltage and conduction mode	V_{out} , T_r and N_s only
Flux against load	rises with load	does not move
Flux against switching frequency	falls as the frequency rises (shorter on-time)	constant below resonance, falls above it
What the gap does	stores the load energy; gap and A_e together set the power	sets L_m , i.e. the magnetising current and ZVS; holds only the magnetising energy
What threatens the core	overload, over-current	over-voltage on the output
Which protection guards the core	the current limit	the output OVP; the current limit does not
Saturation test current	the peak winding current, with margin	$i_{\mu,pk}$ scaled to the OVP ceiling; far below the winding peak
What sizes A_e	peak current and L_p : energy	output volt-seconds at f_r : Faraday

Table 2: The same two cores, compared.

“An LLC transformer stores no energy” is a slogan. It stores the magnetising energy twice per period; it does not store the load energy.

A_e still matters because Faraday's law does not ask whether energy is stored:

Equation 7

$$B(t) = \frac{1}{NA_e} \int v dt$$

with v the winding voltage, N its turns and A_e the core area. **Only the input to the calculation differs.** In a flyback the volt-seconds are the input voltage times the on-time. In an LLC the conducting secondary clamps the winding to the output for a resonant half period, so the flux is set by the **output voltage alone** (Section 6.11).

Three consequences:

- **The flux does not rise with load.** More load raises i_s and the reflected part of i_p together; their difference i_μ does not move. Overload threatens the copper and the semiconductors, not the core.
- **The flux rises with output voltage**, so the ceiling to design against is the over-voltage threshold, not the nominal output.
- **The flux does not fall with switching frequency** below resonance: the rectifier clamps the winding for $T_r/2$ whatever f_{sw} is, which is why f_r appears in the flux equation.

What to keep in mind when designing.

- **Size A_e from volt-seconds, not from current.** The input is V_{out} , N_s and f_r (Section 6.11); the winding current does not enter.
- **Do not add gap for margin.** In a flyback more gap means more stored energy. Here more gap means a smaller L_m : more magnetising current, more circulating loss, a different λ , and no change in the flux at all. The gap is set

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by L_m and checked by L_{open} ; the flux margin is set by the OVP ceiling and checked by the DC-overlap test.

- **The current limit does not protect the core.** OCP guards the copper and the silicon. What bounds the flux is the output OVP, so the saturation specification comes from the OVP threshold (Section 6.19).
- **The test current follows L_μ ; the flux does not.** A smaller gap raises L_μ and lowers $i_{\mu,pk}$ at the same flux. Recompute I_{eq} from the actual L_μ whenever the gap changes; never carry it over.
- **B_{pk} is fixed, so core loss follows frequency.** The flux is the same at every point of the line cycle below resonance, and the loss per cycle at that flux rises with f_{sw} : the worst core loss is at the top of the frequency sweep, the worst flux at the bottom.
- **The flyback habit that costs the most** is asking the supplier for saturation at the tank peak current. That asks for a core well over twice the size the converter needs, for a flux the converter can never produce.

A DC-overlap test is made with every other winding open, which removes the cancellation: all of the test current is magnetising current. Section 6.19 takes that to a number.

2.13 How an LLC is normally designed

The standard sequence. Section 6 follows it and departs from it in two places.

- **1 Turns ratio** so that the converter runs near $M = 1$ at nominal input, where circulating current is lowest.
- **2 Gain range** $M_{min} \dots M_{max}$ from the input range and output tolerance.
- **3 Load resistance referred to the primary, R_{ac} .**
- **4 m (or λ) and Q together** from the peak-gain chart. For each m there is a curve of attainable peak gain against Q , and the design must sit under it with margin. Larger m : less circulating current, less peak gain.
- **5 Component values.** Q and R_{ac} give Z_0 ; Z_0 and f_r give C_r and L_r ; m gives L_m .
- **6 Verify** gain margin at the worst corner, ZVS at the worst point, and currents and flux against real parts.

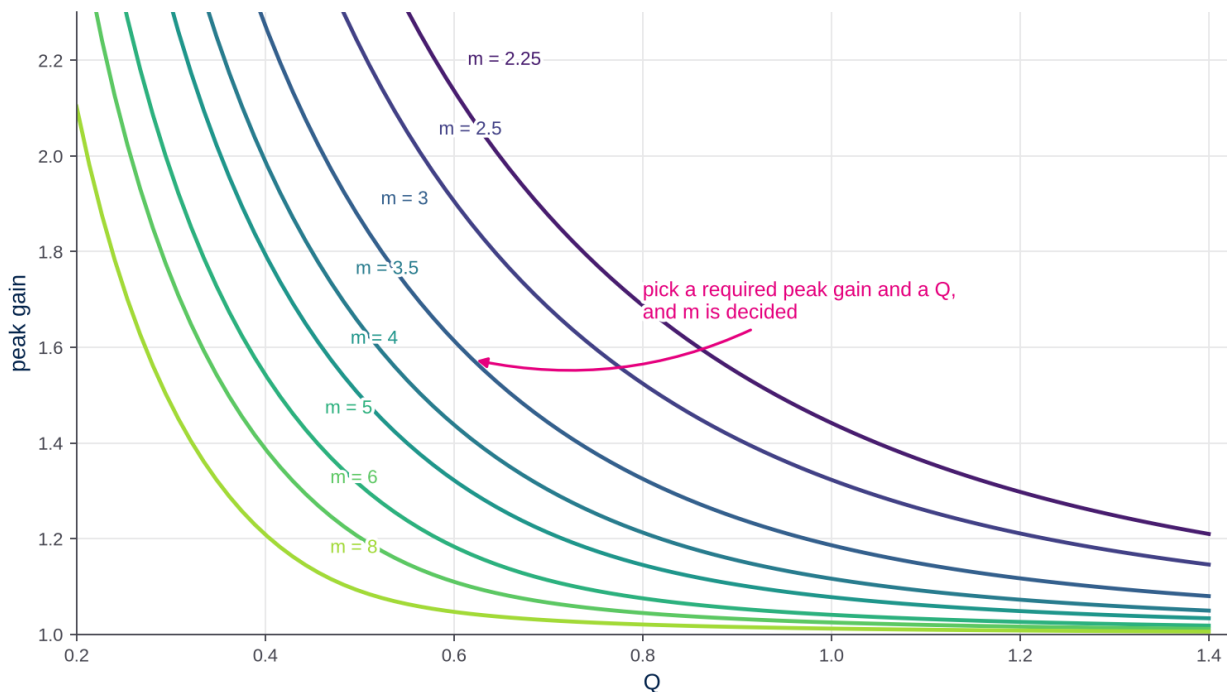


Figure 19: Step 4: the peak gain depends on both m and Q , so a required gain and a Q leave only a band of usable m .

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2.14 What power factor correction is

A bridge rectifier feeding a capacitor holds the bus near the mains peak, so the diodes conduct only in the short window when the mains is above the capacitor voltage. All the charge for the whole cycle arrives in that window.

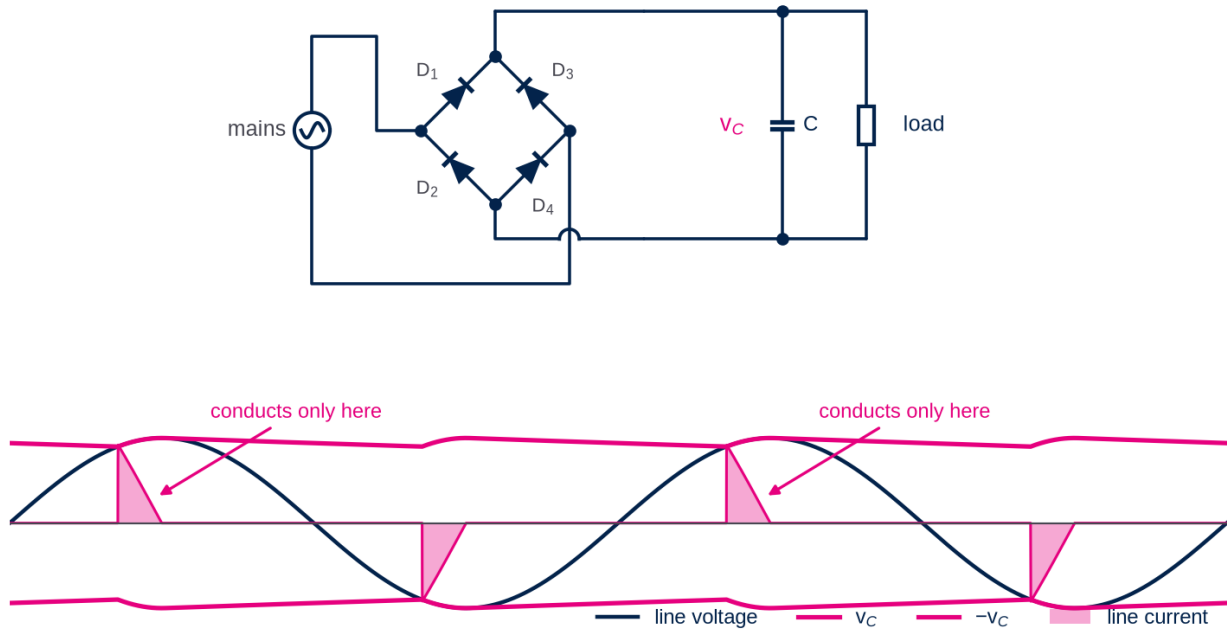


Figure 20: A capacitor-input rectifier: v_C stays near the crest, the diodes conduct in narrow windows, and the current inside them is tall.

The peak current is several times what a resistor of the same power would draw, and a pulse train that narrow is mostly harmonics.

2.15 Power factor, and why it is not the same as distortion

Power factor is the ratio of real power to apparent power:

Equation 8

$$PF = \frac{P}{V_{rms} I_{rms}} = \cos \phi_1 \times \frac{I_{1,rms}}{I_{rms}}$$

with P the average power, $I_{1,rms}$ the rms of the fundamental alone, and ϕ_1 its phase against the voltage. **Displacement** (the phase) is what a motor gets wrong; **distortion** (the harmonic share of the rms) is what a rectifier gets wrong. The current above is roughly in phase and still scores about 0.6.

Total harmonic distortion is the harmonic content as a fraction of the fundamental:

Equation 9

$$THD = \frac{\sqrt{I_{rms}^2 - I_{1,rms}^2}}{I_{1,rms}}, \quad \frac{I_{1,rms}}{I_{rms}} = \frac{1}{\sqrt{1 + THD^2}}$$

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A converter can hold PF above 0.99 and still fail IEC 61000-3-2, which limits *individual* harmonics in amperes. Class D applies to mains rectifiers up to 600 W; above that the absolute Class A limits apply. Settle the class first.

2.16 How a corrector fixes it

Force the input current to follow the input voltage and the converter looks like a resistor to the mains. The standard implementation is a **boost converter after the bridge**, switching fast enough that the mains is constant within one cycle.

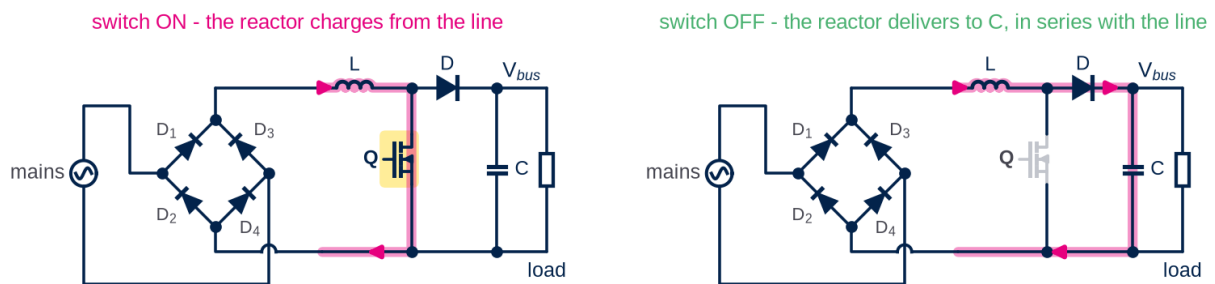


Figure 21: A boost corrector: the reactor charges from the line while the switch is on and delivers to the bus in series with the line while it is off.

- **Why a boost.** Near the zero crossing the stage must step up by an unlimited ratio; only a boost can, and its input current is continuous, which makes shaping possible.
- **The inner loop** regulates the inductor current, fast.
- **The outer loop** regulates the bus voltage and is made **slower than $2f_l$** . A fast voltage loop would fight the bus ripple and flatten the current shape the corrector exists to make.
- **The multiplier** joins them: shape from the mains, level from the output.

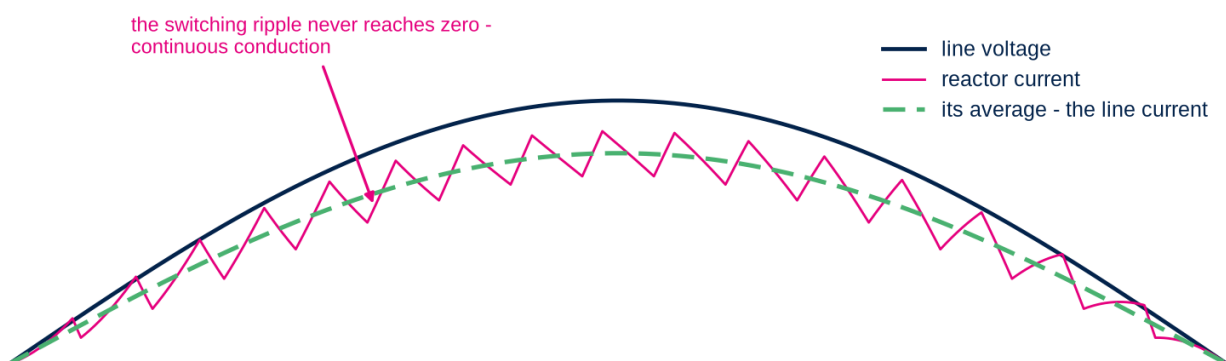


Figure 22: The result in continuous conduction: switching ripple rides on the reactor current, and its average follows the input voltage.

A current that follows the voltage means an input power that follows $\sin^2\theta$: zero twice per cycle, twice the average at the crests. The load wants constant power, so **something must store the difference**.

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2.17 Why they are normally two stages

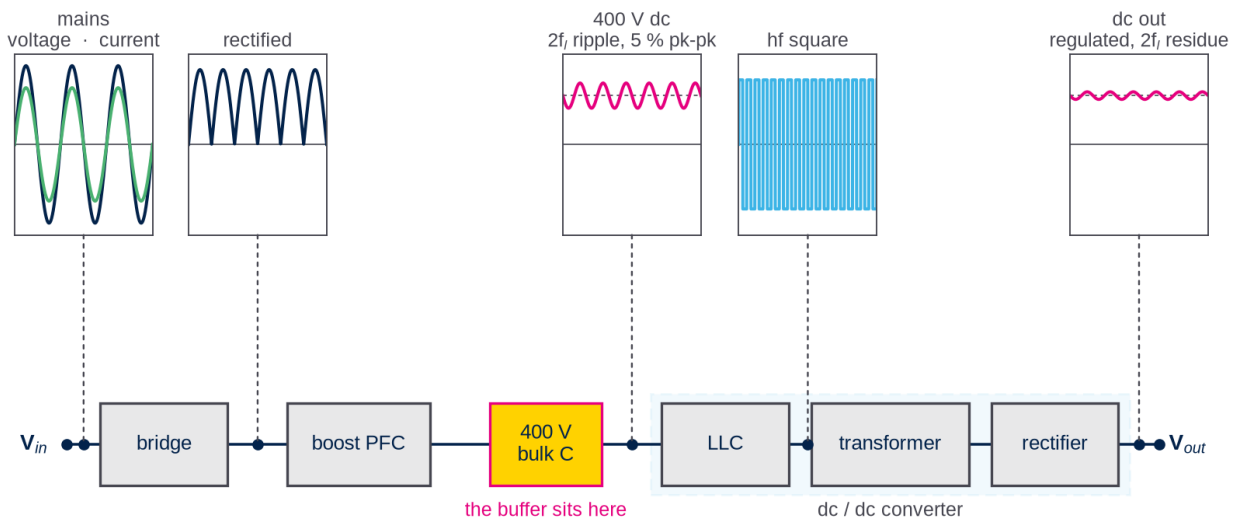


Figure 23: The two-stage arrangement with the waveform at every node. The buffer that absorbs the twice-line-frequency pulsation sits between the stages at 400 V. Ripple on the dc nodes is exaggerated.

The bus capacitor does two jobs:

- it **buffers** the difference between the pulsating input power and the constant output power;
- it **isolates** the LLC from the mains waveform, so the tank sees a nearly constant input and has to cover only a narrow gain range.

Removing the boost stage removes both. The buffering must be done elsewhere, and the tank must work from a rectified sine that goes to zero a hundred times a second.

3 Why single stage, and what it costs

Two steps of that sequence do not survive here. There is no nominal point, because the input sweeps a half sine; and no single tank covers the gain range, so the bridge itself changes (Sections 4.1 and 5).

3.1 The energy a unity power factor cannot deliver

Unity power factor makes the instantaneous input power a raised sine squared, swinging between zero and twice the average at twice the line frequency:

Equation 10

$$p_{in}(t) = V_{ac}I_{ac} [1 - \cos(2\omega_{lt})]$$

with V_{ac} and I_{ac} the rms mains voltage and current. The load wants constant power, so the difference must be stored and returned a few milliseconds later, whatever the topology.

Panel 2 of Figure 24 draws it: the two shaded areas between the $\sin^2$ curve and the flat load are equal, and a capacitor somewhere must absorb and return them.

3.2 Moving the buffer from 400 V to the output

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In the two-stage converter the buffer is the 400 V bus capacitor. Here there is no bus, so it moves to the output. **The joules do not change; the voltage they sit at does**, and a capacitor returns only the energy between its starting voltage and the lowest the load accepts:

Equation 11

$$E = \frac{1}{2} C (V^2 - V_{min}^2)$$

Hold-up shows the price. Both architectures must store the same energy for the same T_{hold} . On a 400 V bus that may fall to 320 V the window is $400^2 - 320^2$; on an output of a few tens of volts it is $V_{out}^2 - V_{o,min}^2$, smaller by a factor in the hundreds, so the capacitance grows by that factor. Settle this first: if the enclosure cannot house the bank, the output voltage must change. Section 7.11 sizes the bank for this design, where ripple, not hold-up, decides.

The trade. Out: a switch, an inductor, a diode, a 400 V electrolytic and their losses. In: a very large low-voltage capacitor bank, 2f_l ripple on the output, and a tank that must cover a much wider range. Whether it is a good trade depends on how much output ripple the load accepts.

3.3 What the load must tolerate

The output carries the 2f_l ripple directly. It is set by the bank and the load, not by the loop, and it cannot be regulated away: trying puts the distortion on the input current instead (Section 6.28). **The allowed ripple is a property of the load** and must come from the specification. This design allows a few per cent peak-to-peak.

4 Operating principle

4.1 Gain is a boundary condition, not a control variable

In a two-stage LLC the controller commands gain. Here it cannot, because **both ports are voltage sources**: the output is a millifarad capacitor and the input is the rectified mains. The instantaneous gain is forced:

Equation 12

$$M(\theta) = \frac{n V_{o, eff}}{V_{drive}(\theta)} = \frac{2 n V_{o, eff}}{\sqrt{2} V_{ac, eq} \sin \theta}$$

with θ the line phase and V_{drive} the amplitude the bridge applies to the tank: the rectified mains in full bridge, half of it in half bridge. Both voltages are pinned from outside, so their ratio is fixed moment by moment.

Moving the frequency therefore moves **Q**, that is, power. **Frequency is a power command, not a voltage command.**

4.2 Two divergences that cancel

Near the zero crossing the required gain rises as $1/\sin\theta$ without limit. The converter survives because the load disappears at the same time:

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Equation 13

$$Q(\theta) = Q_{pk} \sin^2 \theta$$

and an unloaded LLC has unlimited gain at f_o . The two infinities cancel, and the operating point walks down towards f_o as the mains approaches zero (Section 7.7). f_o is the **frequency floor of the whole design**; an oscillator clamp below it destroys hardware.

4.3 Frequency modulation is the power factor correction

Figure 24 follows the control law round one line half cycle. The first three steps are forced by the two voltage sources; only the fourth is something the controller does.

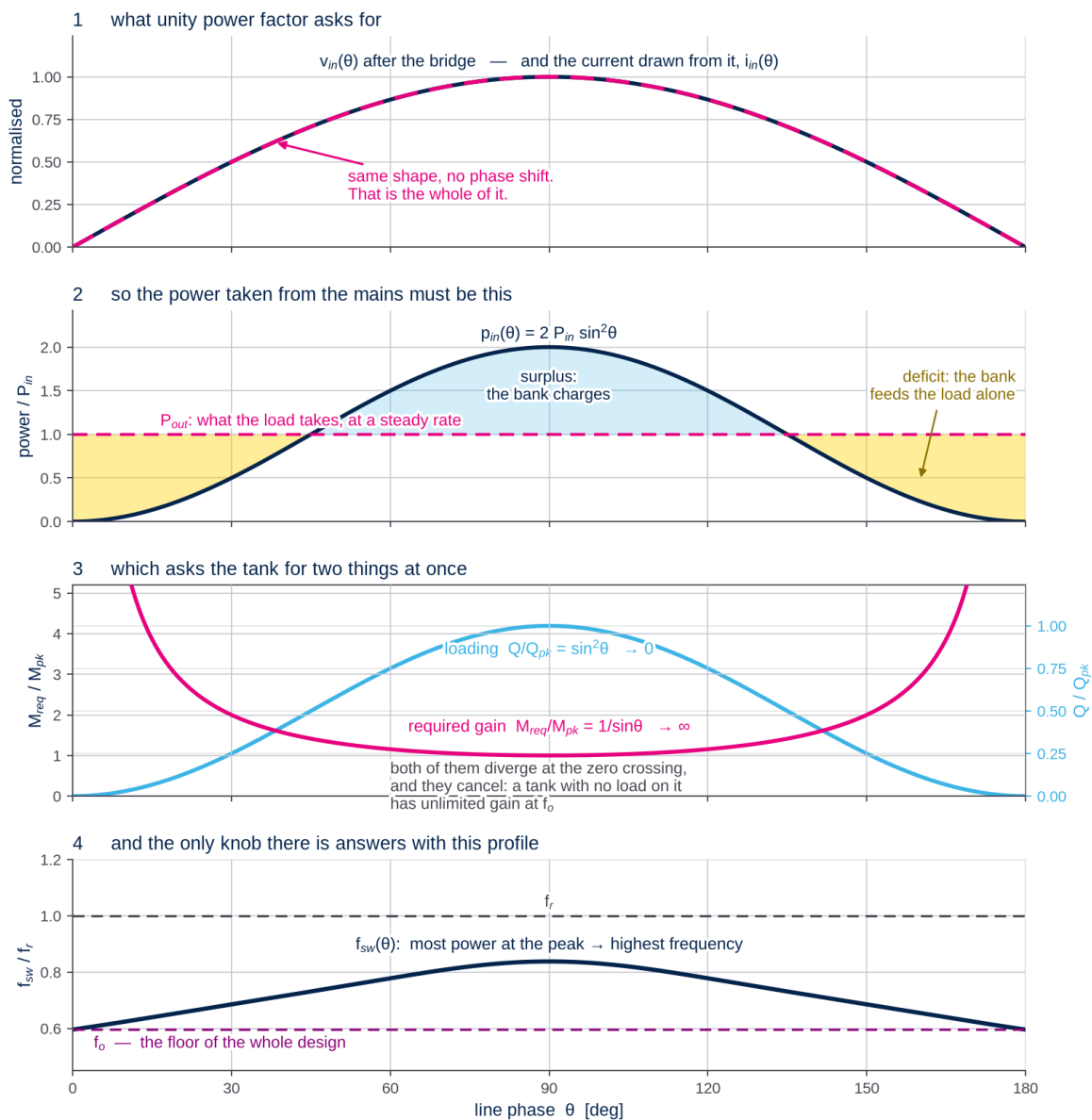


Figure 24: How the power factor is corrected, over one line half cycle. 1 Unity power factor: the current has the shape of the rectified mains. 2 The power drawn is $\sin^2$, while the load takes a steady P_{out} ; the shaded areas are what the bank buffers. 3 The tank is handed a gain and a load, and both run away at the zero crossing. 4 The switching frequency is the only free variable; this profile satisfies 1 to 3. Normalised; with numbers in Section 7.7.

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There is no bulk capacitor, so the tank sees the mains raw. Unity power factor fixes the instantaneous power at $2 P_{in} \sin^2\theta$, which fixes both the gain (Equation 12) and the loading (Equation 13) at every instant. The controller answers with the one variable it has. There is no current loop and no multiplier against a rectified reference: **the frequency profile $f_{sw}(\theta)$ is the power factor correction.**

At the line phase	$\sin\theta$	instantaneous power p/P_{in}	required gain M_{req}/M_{pk}	loading Q/Q_{pk}
$\theta = 90^\circ$, the peak	1.000	2.000	1.000	1.000
$\theta = 60^\circ$	0.866	1.500	1.155	0.750
$\theta = 45^\circ$	0.707	1.000	1.414	0.500
$\theta = 30^\circ$	0.500	0.500	2.000	0.250
$\theta = 10^\circ$	0.174	0.060	5.759	0.030
$\theta \rightarrow 0$	0	0	∞	0

Table 3: The half cycle at six instants, normalised as in Figure 24. Trigonometry only; no part value enters.

In the last two rows the gain demand runs away, but so does the ability to meet it: an unloaded tank has unlimited gain at f_o . The frequency walks down towards f_o and the current goes to zero with the voltage.

The controller has two signals on two time scales:

- **FB says how much.** It carries the error amplifier output through the optocoupler and commands an input *power* (Section 4.6). The loop behind it crosses well below the line frequency, so within one half cycle it is a constant: the *height* of the profile in panel 4.
- **HVSU says where in the cycle.** The start-up pin is also the input-voltage sense, so the controller knows the instantaneous rectified mains without a divider. That is the fast signal, the *shape* of the profile.
- **The internal PFC and THD block combines them** into I_{EA} , the one current that moves the oscillator (Equation 59).

The draft datasheet does not publish the law inside that block. This note therefore computes, from the tank, the profile that *must* come out if the current is to follow the voltage. Measuring $f_{sw}(\theta)$ on the first board is how the block gets characterised.

Zero-crossing dead zone. Very near $\theta = 0$ the profile asks for a frequency below f_{Min} , and the converter stops drawing current until the mains comes back. This is inherent and appears as third-harmonic distortion. The other THD mechanism is the voltage loop (Section 6.26).

4.4 Solving for the profile without a root search

The profile does not need a numerical root search. With $M_{req} = M_{pk}/\sin\theta$, $Q = Q_{pk}\sin^2\theta$ and $x = 1/f_n^2$ the gain equation is a cubic in x :

Equation 14

$$\lambda^2 x^3 + (q - 2\lambda(1 + \lambda))x^2 + ((1 + \lambda)^2 - 2q - \frac{u}{M_{pk}^2})x + q = 0, \quad u = \sin^2 \theta, \quad q = Q_{pk}^2 u^2$$

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One root is negative (the cubic is $+q$ at $x = 0$ and falls to $-\infty$). The other two are the capacitive and the inductive crossing of the same curve, and the inductive one is the **middle** root, because x falls as frequency rises. In the trigonometric form of Cardano it is always the $k = 1$ branch. No positive root means the no-solution case of Section 4.10, not a numerical failure. A grid search would also need an upper bound on f_n , and a bound taken from the ac maximum instead of the morphing corner misses the real operating point.

4.5 Reading the gain chart of a single-stage converter

The chart in Figure 5 is the standard picture: a family of curves, one required-gain line, one crossing. A single-stage converter is drawn the same way and **read differently**. Figure 25 puts the two side by side.

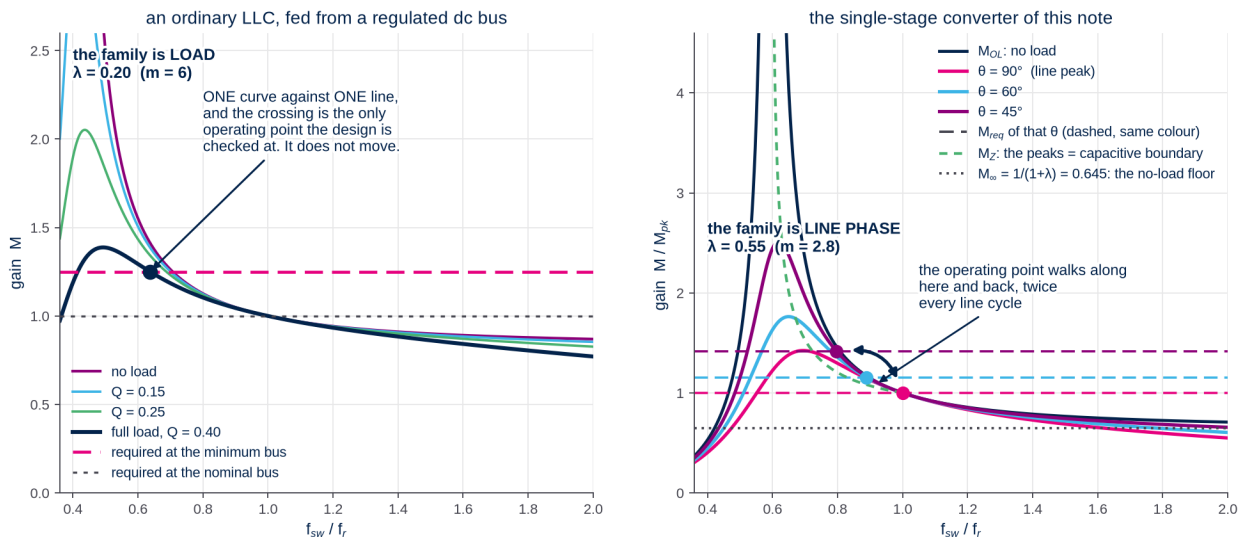


Figure 25: The ordinary chart and the single-stage one: same axes, same equation, read differently. Left, the family is load and the converter sits at one crossing. Right, the family is line phase, each curve has its own required-gain line in its own colour, and the converter travels the family twice per line cycle. The right panel is normalised to M_{pk} , so it holds for any input voltage.

Six differences:

- **The family parameter is line phase, not load.** The load is fixed at full and the curves are $\theta = 90^\circ, 60^\circ, 45^\circ, \dots$, because $Q = Q_{pk} \sin^2 \theta$ falls as the mains falls.
- **The operating point moves, 100 or 120 times a second.** There is no single point to check; the design is checked along a path.
- **There is a ladder of required gains, $M_{req}(\theta) = M_{pk}/\sin \theta$, one line per phase, climbing without limit towards the zero crossing. Each curve is compared with its own line only.**
- **The curves get taller as the lines get higher.** Halve $\sin \theta$ and the requirement doubles, but Q falls by four and the curve grows far more than twice as tall. The margin grows towards the zero crossing.
- **M_Z , the locus of the peaks, is drawn.** It is the capacitive boundary of Section 2.9, and every crossing must lie to its right. It has to be cleared at every phase, so it is a curve of its own.
- **$M_\infty = 1/(1+\lambda)$ is on the chart.** An ordinary LLC never needs a gain that small; a single-stage converter at high input voltage does. Below this line there is **no solution at any frequency** and burst mode takes over (Section 4.10).

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The mistake this chart invites: checking the tallest curve against the lowest line. There is no such operating point. θ sets the curve *and* the line together, so only the same-coloured pairs mean anything. That is also why this note draws no overview with several input voltages on one family: at the line peak every condition looks comfortable.

At the lowest equivalent input the chart answers “can the tank make the gain and stay inductive?” At the highest it answers “how high does the frequency go, and is there a solution at light load?” Section 7.6 draws it at all six input voltages for that reason.

4.6 The feedback pin is a power command

Combining the burst-mode expression with the multiplier in the block diagram gives the meaning of the feedback voltage:

Equation 15

$$V_{FB} = V_{os} + \frac{2K_{HV}}{K_M K_{FF}} R_{CS} P_{in} = 0.5 \text{ V} + 0.167 \frac{\text{V}}{\Omega \cdot \text{W}} R_{CS} P_{in}$$

V_{os} is the 0.5 V the pin sits at with no power commanded; K_{HV} , K_M and K_{FF} are the input-voltage sense, multiplier and feed-forward gains of the block diagram. The 2.8 V feedback span gives $2.8/0.167 = 16.8 \Omega \cdot \text{W}$, the maximum-power rule of the datasheet. **Maximum power, burst entry and overload detection all sit on one scale, R_{CS} .** The optocoupler loop commands an input power; the internal loop distributes it over the line cycle as $\sin^2\theta$.

This is why the current-sense pin tolerates no filter and no series resistor: the power law and the over-current threshold read the same pin.

4.7 Which side of resonance this converter runs on

A single-stage converter does not pick one side. The equivalent input follows a half sine, the required gain follows it, and the operating point crosses f_r and comes back within one line cycle. How much time it spends on each side depends on the mains voltage; Section 7.7 shows the profile at the six input voltages.

Both sides must be designed for. The boosting side sets the gain requirement and the ZVS margin. The bucking side sets the top frequency and takes the secondary out of ZCS, so the rectifier body diode and the SR dead time matter here.

4.8 Why λ must be about 0.5

A classic LLC uses $m = L_p/L_r$ of 5 to 10, that is λ of 0.11 to 0.25. A single-stage converter needs very high gain near the zero crossing, and peak gain falls as λ falls. ST's evaluation board runs $\lambda = 0.500$ and this design comes out close to it (Section 7.4): the value comes from the topology.

The price is circulating current: a small L_m means a large magnetising current with conduction loss at every load. **A single-stage PFC LLC is less efficient than a two-stage LLC, inherently.**

4.9 $f_{sw,max}$ does not check the tank — it computes it

Two of the four candidates for λ carry the specified maximum switching frequency in a denominator:

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Equation 16

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw, max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw, max}}\right)^2}$$

λ_1 is the plain minimum-gain condition; λ_2 and λ_{TD} fold in the frequency ceiling: the tank must reach the lowest required gain *before* it runs out of frequency. With the usual $f_{sw, max} = 1.5 f_r$ the squared ratio is 0.44 and **the value typed as $f_{sw, max}$ roughly doubles the λ asked for**. Set it near f_r and the required λ goes to infinity.

The common $1.5 \times f_r$ is a **choice**, not a limit, and it sits upstream of the tank: a round number typed into it changes L_m .

f_{Max} , the oscillator ceiling, is different. It *is* a limit, fixed by R_T , C_T and the idle time, and it is checked after the fact by k_{ceil} (Section 7.3). One decides the tank; the other reports whether the oscillator can serve it.

4.10 The other bound on λ , and where it has no solution

The ceiling on λ comes from the lowest required gain, at the highest equivalent input and no load. The no-load gain does not fall without limit; raising the frequency only brings it down to an asymptote:

Equation 17

$$M_\infty = \lim_{f \rightarrow \infty} M_{no\ load} = \frac{1}{1 + \lambda}$$

If the required minimum gain is below M_∞ , no frequency satisfies it. A spreadsheet solving for that frequency returns an error, and the error looks like a broken formula. It is not: the question has no answer. Whether a tank sits above or below the asymptote is a matter of a per cent, and no full-load check shows it; Section 7.4 reports it for this design.

Losing the solution is not a failure: no load at high input voltage belongs to **burst mode**. But it decides which candidate turns ratios can still regulate by frequency alone at no load, and no full-load number shows it.

5 Topology morphing

A resonant tank has a usable gain range of roughly 1.5:1 at full load. A universal mains asks for 2.93:1. No single tank covers that and keeps ZVS, so the L6790A changes the bridge instead.

Below a mains peak of 235 V the part runs a **full bridge**, which drives the tank with twice the rail. Above 245 V_{pk} it runs a **half bridge**. The tank therefore sees 173.2 to 332.3 Vac equivalent, a range of 1.92:1, and one tank can cover it.

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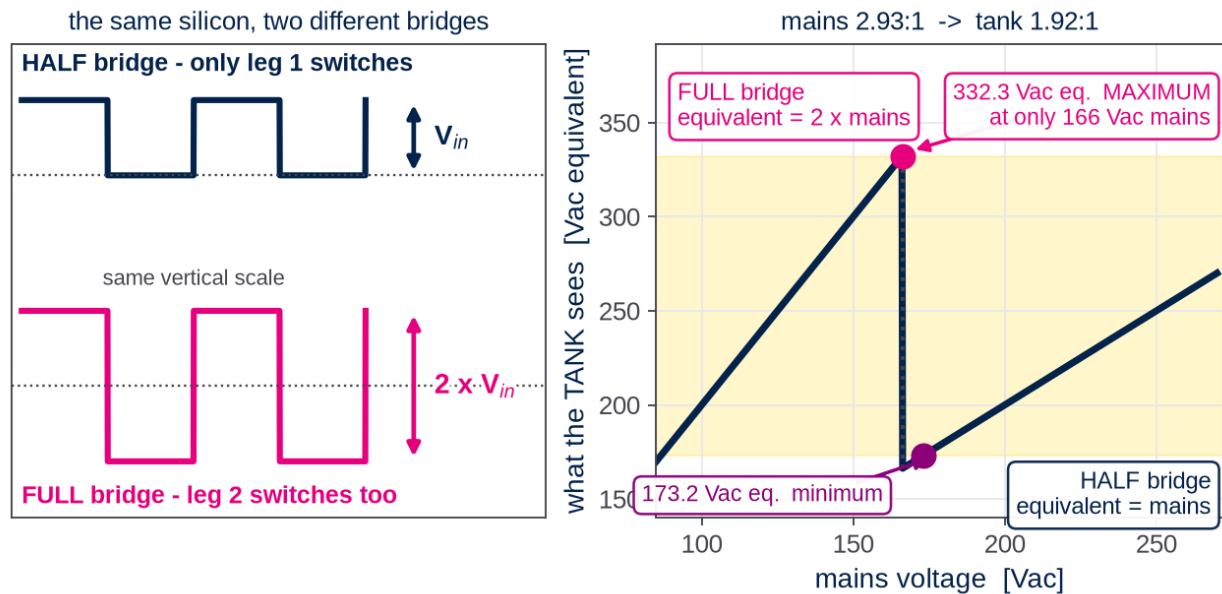


Figure 26: Morphing collapses a 2.93:1 mains range into a 1.92:1 range at the tank. The worst corners are the morphing edges, not the ends of the mains range.

5.1 Full bridge

Diagonal pairs conduct together, so the drive is a square wave of amplitude V_{in} with fundamental $(4/\pi)V_{in}$. All four devices switch.

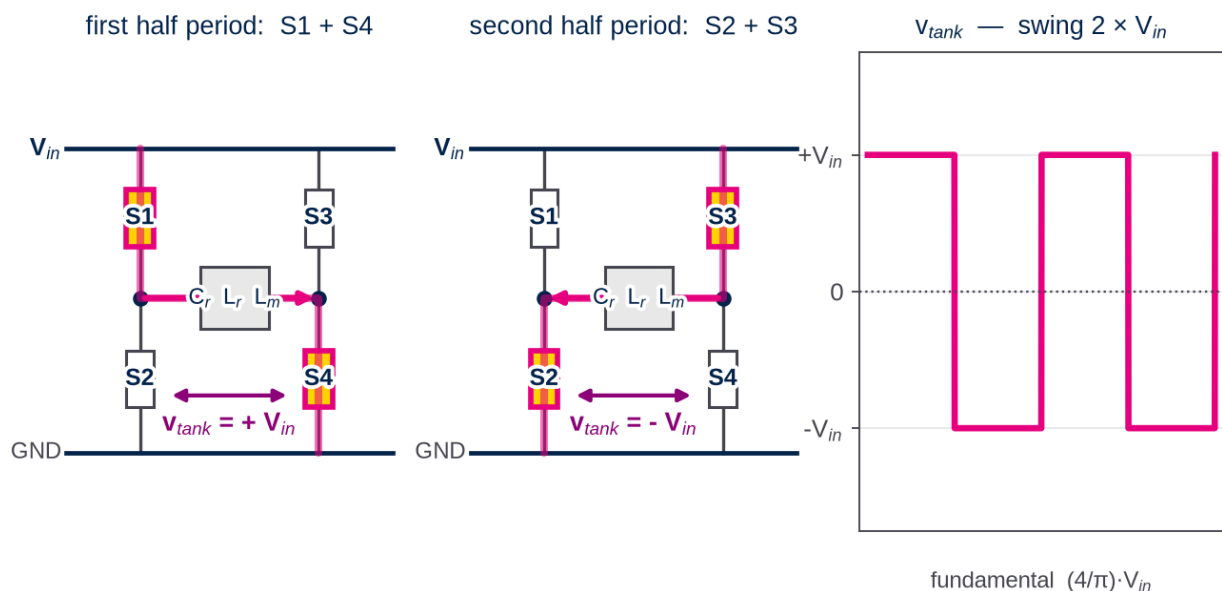


Figure 27: Full bridge: conduction path in each half period, and the resulting tank drive.

5.2 Half bridge

LOUT2 is held high: S3 never turns on and S4 never turns off, so the bridge output swings between 0 and V_{in} . C_r blocks the $V_{in}/2$ of dc, and the tank sees $\pm V_{in}/2$, a fundamental of $(2/\pi)V_{in}$. No extra hardware.

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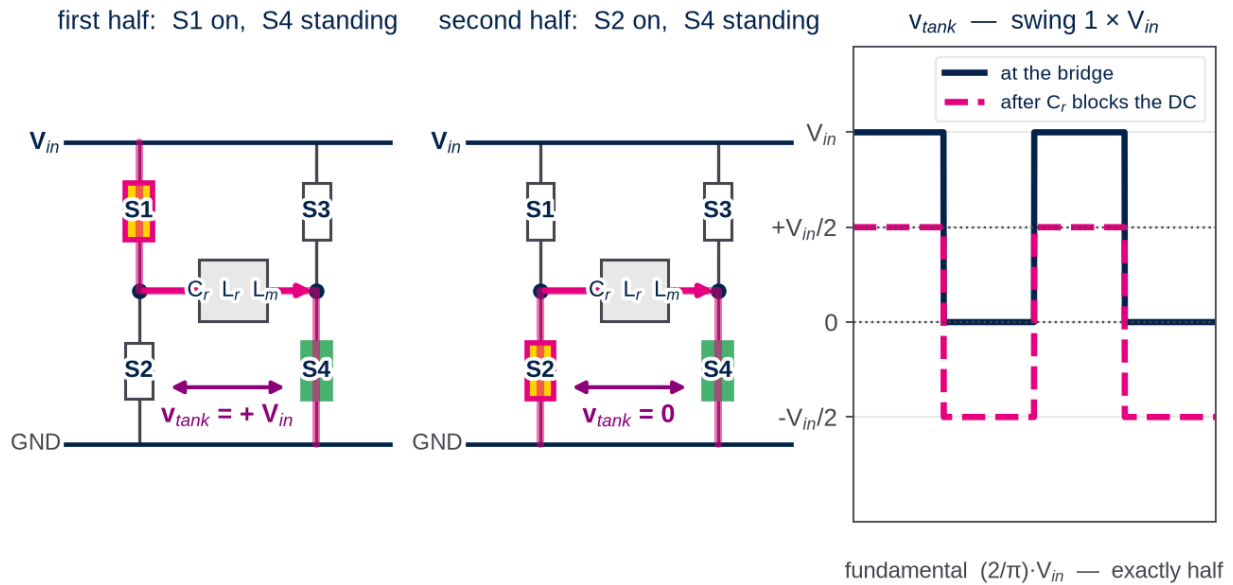


Figure 28: Half bridge: leg 2 stops switching, and C_r removes the dc that the asymmetric drive creates.

The standing device is the hottest one. S4 never switches but conducts the full tank current continuously, and in this design it is the largest single loss item (Section 7.12).

5.3 How the controller does it

Only one gate signal changes between the modes. The thresholds are **fixed inside the IC**; R_{CFG} only enables morphing and sets the brown-out threshold.

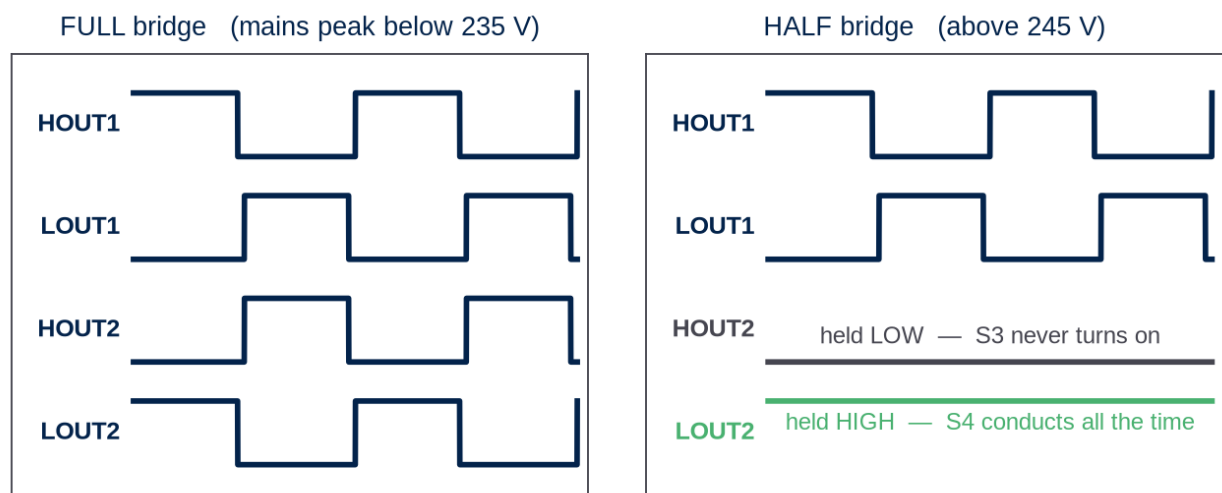


Figure 29: The four bridge drive signals in each mode. LOUT2 held high is the entire mechanism.

5.4 The hysteresis band, and the range it really implies

The part drops to half bridge at $245 V_{pk}$ going up and returns to full bridge at $235 V_{pk}$ coming down. The hysteresis stops toggling, but **between the two the mode depends on the direction of travel**, not on the voltage.

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Taking each threshold where its mode is guaranteed gives 173.2 to 332.3 Vac (1.92:1). Including the band gives **166.2 to 346.5 Vac (2.08:1)**. This design passes at the wider range too; do not use the narrower figure when tightening a specification.

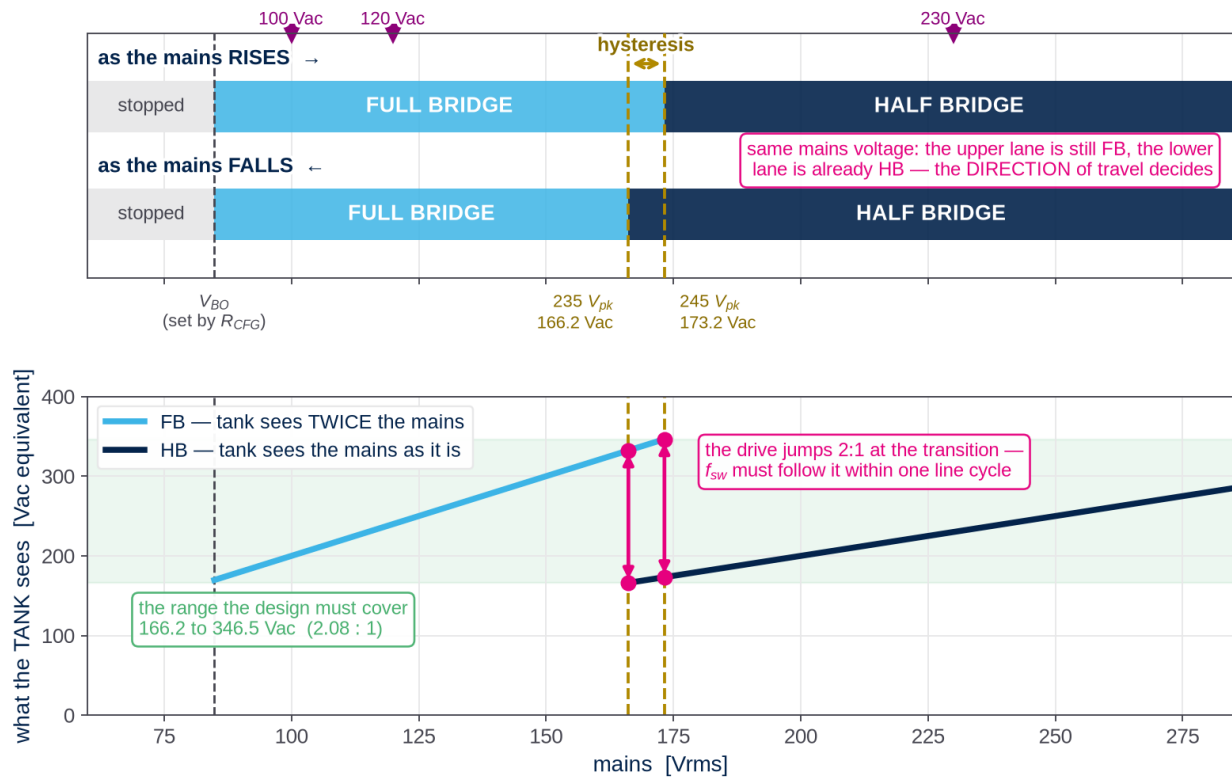


Figure 30: Where each mode applies. Inside the band the mode depends on which direction the mains arrived from.

Real mains never sits at 166 to 173 Vrms; the band is met on a programmable ac source and in dip and surge tests. Test it with a **step** across the threshold, not a ramp.

Transition transient. Crossing a threshold changes the drive 2:1 within one line cycle, against a loop that crosses in the tens of hertz, so the output droops until the loop recovers. This is less severe than hold-up, which is already met, so it is no reason to enlarge C_{out} ; it has not been checked in the time domain.

6 Design procedure

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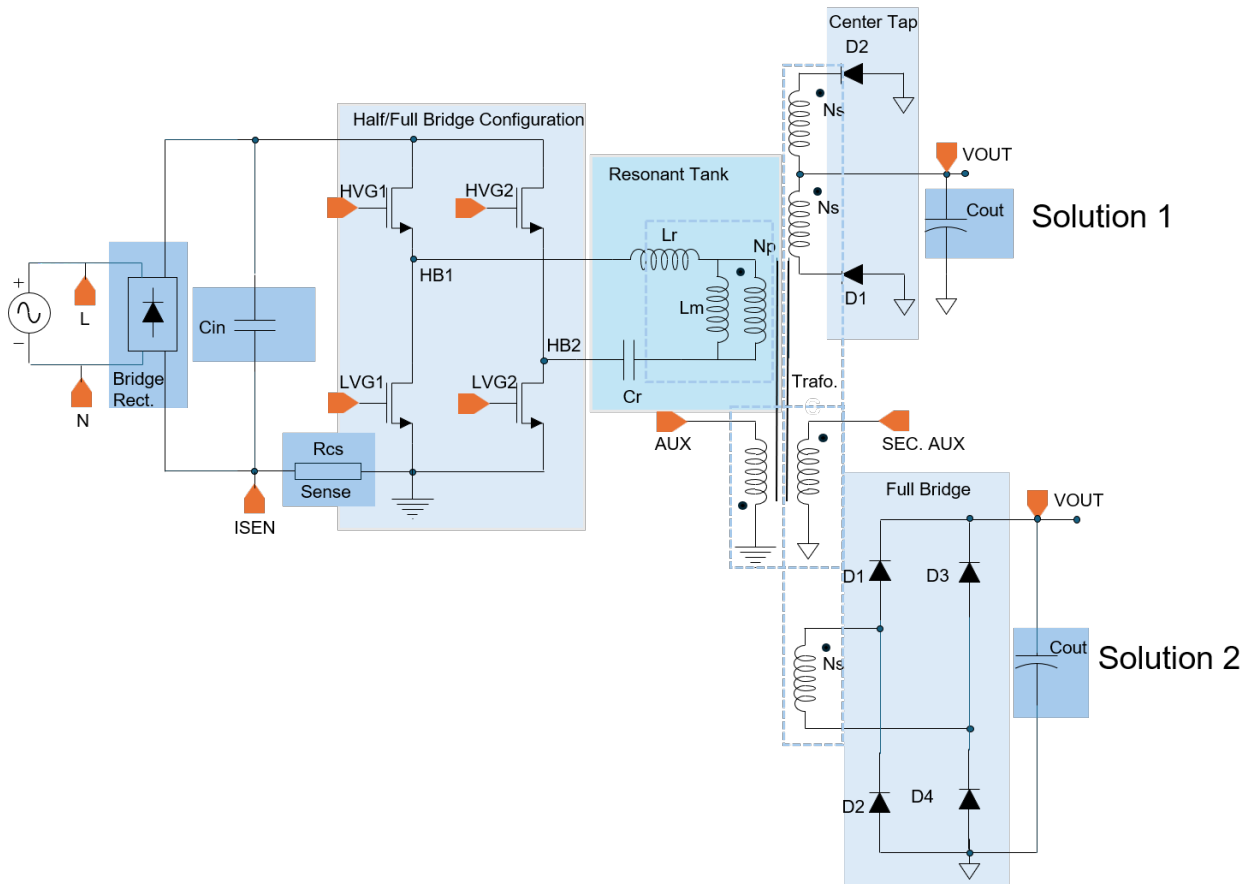


Figure 31: The power stage. The bridge rectifier feeds the leg pair directly: no bulk capacitor, no boost stage. The secondary is drawn as centre tap (Solution 1) and full bridge (Solution 2).

Pick the secondary first: it sets N_{rect} and N_{rect} sets the reflected voltage. A centre tap has one device in the conduction path, a full bridge two, so at a low output voltage the centre tap halves the secondary conduction loss at the price of twice the reverse voltage per device. This design uses the centre tap. The steps below are in dependency order.

6.1 The specification

The full specification is in Section 7.1. Two qualifiers apply to every step: **the worst line frequency is the lowest, and hold-up is specified at the ripple trough.**

6.2 The equivalent input range

With morphing the tank does not see 90 to 264 Vac. It sees an **equivalent** input given by the bridge mode:

Equation 18

$$V_{ac,eq} = 2 V_{ac} \text{ (full bridge)} \quad V_{ac,eq} = V_{ac} \text{ (half bridge)}$$

At the half-bridge edge ($245 V_{pk}$) the tank sees 173.2 Vac; at the full-bridge edge ($235 V_{pk}$) it sees 332.3 Vac. **These edges are the design corners:** the low one for gain and ZVS, the high one for switching frequency. The mains voltages a supply actually meets map onto the same axis: 90 and 110 Vac in full bridge become 180 and 220 Vac equivalent, 230 and 264 Vac in half bridge stay 230 and 264. All four lie inside the two edges, and this note reports at all six conditions.

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6.3 Turns ratio and reflected voltage

The turns ratio appears only multiplied by the effective output voltage, so what matters is the **reflected voltage**:

Equation 19

$$V_{refl} = n V_{o, eff} = n(V_{out} + N_{rect} V_f)$$

n is the model ratio in the gain equation; n_T is the wound ratio with the leakage folded into L_r . At $\lambda \approx 0.5$ they differ by more than 20 % (Section 6.9), so the drawing must carry open- and short-circuit inductance as well as turns.

6.4 The resonant tank

The equivalent ac load resistance for a centre-tapped secondary, evaluated at the line peak, is

Equation 20

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o, eff}^2}{P_{in, LLC}}$$

with $P_{in, LLC}$ the power into the resonant stage. Every LLC text writes $8/\pi^2$ with the one output power a two-stage converter has. Here the drawn power is $2P \sin^2 \theta$, so R_{ac} is pinned to the line peak, where $p = 2P$: the 8 becomes a 4, and Q means the quality factor at the peak, with $Q(\theta) = Q_{pk} \sin^2 \theta$ elsewhere. Using the two-stage value understates the loading by two.

The gain requirement at the low corner and the ZVS requirement cap the quality factor at Q_{ZVS} . $R_{ac} Q_{ZVS}$ is the impedance the tank is sized from: it fixes C_r through Equation 2, and C_r with f_r fixes L_r . Three rules matter more than the arithmetic:

- The three calculated figures are not one tank. L_r is whatever pairs with the **selected** C_r , and the L_m figure is L_r over the λ that *no load at the high corner* asks for, a condition a design may knowingly not meet (Section 4.10). It is not a value to round towards.
- **Never round L_m up.** That lowers λ and the ZVS margin. C_r above and L_m below their calculated values do the opposite.
- **Stay below the Q_{ZVS} cap.** That gap is most of the ZVS margin. Choose L_m with n so that $n_T = n/(1+\lambda_{act})$ is a ratio that can be wound.

6.5 ZVS verification

The closed-form ZVS estimate in most guides is a fitted approximation at the design Q_{ZVS} , not at the Q the converter runs at. Against a full sweep it errs in **either** direction by tens of per cent, and optimistic is the dangerous direction.

The shortcut is written with a bare λ and means the *design* λ . Read it with λ_{act} and the phase can come out **negative** for a tank well inside the inductive region. The sweep has no such ambiguity.

Sweep θ , input voltage and load with the **selected** tank and take the minimum. Section 7.5 does it for this design; the worst point is not where one would expect.

6.6 Which side of resonance the converter runs on

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Below f_r the secondary current is a truncated sine with a dead interval and the conduction ratio $d = f_{sw}/f_r$ is less than one; above f_r d clamps at one. The rms figures behind ratings and losses contain d .

The turns ratio decides the side. $M_{req} = 2nV_{o,eff}/(\sqrt{2} V_{eq})$, so a larger n asks for more gain and sits further below resonance. Two candidate transformers on the same tank can end up on opposite sides of f_r at the same mains voltage.

Settle this before the transformer is ordered: below resonance the rectifiers turn off at zero current, above it they do not.

6.7 Currents over the line cycle

Ratings come from the worst switching cycle; losses from the line-cycle rms. Two further rules:

- The primary devices and the sense resistor see the **composite** tank current: reflected load current plus magnetising current. Its peak is neither the sum nor the larger of the two peaks, because they occur at different instants.
- Below resonance the secondary conducts for d of each half period. Omitting d over-estimates the rms, and the loss by its square. Section 7.8 draws the composite current of this design.

6.8 The transformer

The tank fixed L_r , L_m and a ratio; the transformer has turns, an open-circuit inductance and a leakage. **Two turns ratios and two inductances carry almost the same names**, and that is where most errors are made. This section turns Section 2.12 into numbers a supplier can measure.

Every number on the drawing is read off the primary current, the current in each secondary winding, the secondary winding voltage, or one bench test. Figure 43 marks where on this design's waveforms; Table 4 says what each mark sets.

Mark	Item	Read from — and over what	What it sets
1	Primary peak current	Top of i_p : the half sine of load current riding on the magnetising ramp. Worst switching cycle: line peak, minimum equivalent input, full load.	Primary switch and OCP1 headroom. Not the saturation test.
2	Primary rms current	i_p^2 averaged over the whole switching period, then over the line half cycle. Every unit carries the same current: the primaries are in series.	Primary copper: $A_{cu} = I_{rms}/J$ per unit
3	Magnetising peak	The dashed i_{Lm} at the switching instant. Below resonance it depends on neither f_{sw} nor load, only on V_{out} , so it is the same on every cycle.	The peak flux; and, scaled to V_{OVP2} , the DC-overlap test current (mark 7)
4	Secondary peak current	Top of the pulse in either secondary winding, same worst cycle. Per unit it is $1/N_x$ of the rectifier-leg value: the secondaries are in parallel.	Rectifier peak rating; the foil termination
5	Secondary rms current, each winding	One pulse per period in each winding, so the rms is taken over the whole period with the idle half counted, then over the line half cycle.	Secondary copper: $A_{cu} = I_{rms}/J$, per winding and per unit
6	Winding volt-seconds	The secondary winding voltage is V_{out} for the resonant half period $T_r/2$ while the rectifier conducts. That area is the flux swing $2N_s A_e B_{pk}$; f_r is the worst case because the interval never gets longer than $T_r/2$.	Core area and N_s : $A_e \geq V_{out}/(4 f_r N_s B_{max})$

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Mark	Item	Read from — and over what	What it sets
7	DC-overlap (saturation) test	Not a waveform. LCR meter across NP1, every other winding open (NS2, NS3, NAUX); a dc current overlapped on the primary and the primary inductance read against it. The test current is mark 3 scaled to the highest output the controller allows, V_{OVP2}/V_{out} .	Core saturation margin; the number the supplier tests to
—	L_{open}, L_{short}	LCR meter across NP1 at 100 kHz, 1 V. L_{open} : every other winding open. L_{short} : NS2 and NS3 both shorted, NAUX open. Terminal quantities, so a supplier can test them.	$L_m + L_r$ and L_r : the tank itself

Table 4: What each mark in the figure sets. Currents are per unit of the assembly: the primaries are in series and the secondaries in parallel.

6.9 Two ratios, two inductances

With all leakage referred to the primary the tank sees an ideal transformer of ratio n , a series L_r and a shunt L_m . That n is **not** the wound ratio. With k the coupling coefficient,

Equation 21

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}}$$

so $n_T = n\sqrt{1+\lambda}$: more than 20 % apart at $\lambda \approx 0.5$, far more than any gain margin.

The inductances split the same way. The tank model uses L_r and L_m ; the physical transformer has a magnetising inductance L_μ with leakage on both sides:

Equation 22

$$L_\mu = \sqrt{L_m (L_m + L_r)}, \quad L_{L1} = L_m + L_r - L_\mu, \quad L_{L2} = \frac{L_{L1}}{n_T^2}$$

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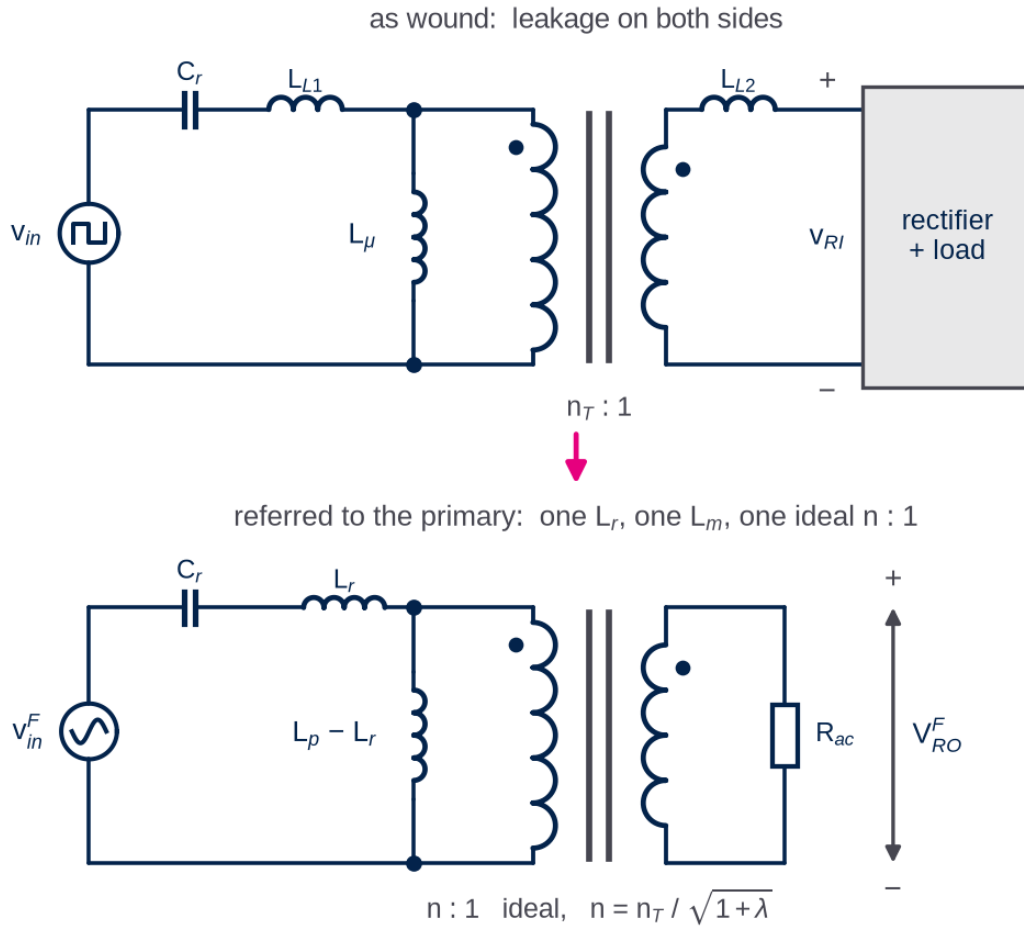


Figure 32: The same transformer drawn both ways. Above, as wound: leakage on both sides, the physical L_μ in shunt, ratio $n_T : 1$. Below, referred to the primary: one L_r , one L_m , ratio $n : 1$. Neither the shunt elements nor the ratios are the same numbers.

The three-element form assumes the leakage splits evenly. The real split does not matter: the tank sees only the two quantities a meter can read.

6.10 What the transformer specification must say

Only two inductances are measurable at the terminals, and they are the two the tank needs:

Equation 23

$$L_{open} = L_\mu + L_{L1} = L_m + L_r, \quad L_{short} = L_{L1} + \frac{L_\mu n_T^2 L_{L2}}{L_\mu + n_T^2 L_{L2}} = L_r$$

Both are read across the primary. L_{open} : every other winding open. L_{short} : every secondary shorted (on a centre-tapped part both halves), the auxiliary open. **Specify those two and the turns, never L_μ** : no terminal pair exposes it. State the turns as numbers, because with one or two secondary turns the lead-out loop dominates that winding's inductance and $\sqrt{L_1/L_2}$ does not give the ratio.

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6.11 Peak flux is set by the secondary, not the primary

Mark **6** of Figure 43: the secondary winding voltage is a rectangle of height V_{out} and width $T_r/2$, and the flux swings $\pm B_{pk}$ across it. Above resonance the secondary conducts for a shorter switching half period, so f_r is the worst case everywhere on the line cycle:

Equation 24

$$B_{pk} = \frac{V_{out, eff}}{4 f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4 f_r N_s B_{max}}$$

N_p does not appear. Only secondary turns reduce the flux. For a low-voltage, high-current output, where the secondary wants to be one turn, halving N_s doubles the core area.

From an inductance the same flux is $B_{pk} = L_{\mu} i_{\mu, pk} / (N_p A_e)$ with the **physical** L_{μ} . Using the tank L_m understates it by $\sqrt{(1+\lambda)}$.

6.12 Choosing the core: two areas, and the one that usually decides

A core must satisfy two independent conditions: enough cross-section A_e for the flux at the chosen turns, and enough window A_N for the copper. Their product is the **area product**, and either can bind.

Check the flux against A_{min} , not A_e . On a PQ core they differ by ten to twenty per cent and A_{min} saturates first: the real peak is $B_{pk} A_e / A_{min}$.

In a single-stage PFC LLC **a third condition normally decides**: the leakage must come out at L_r . As a fraction of what a meter reads,

Equation 25

$$\frac{L_{short}}{L_{open}} = \frac{L_r}{L_m + L_r} = \frac{\lambda}{1 + \lambda}$$

At $\lambda \approx 0.5$ that is **about a third**. A conventional transformer leaks a few per cent; this one must leak ten times more, on purpose. Leakage that large is a winding **geometry**, and geometry costs window. A core that passes on A_e and copper can still fail here.

6.13 The winding arrangement is the leakage

Leakage is set by how far apart the windings are and how much window separates them:

- **Interleaved** (primary, secondary, primary): one or two per cent. What a forward or flyback transformer wants; the opposite of what is needed here.
- **Concentric, not interleaved**: a few per cent. Still short.
- **Side by side**, primary on one half of the width and secondary on the other: tens of per cent, adjustable by the gap between them. **This is what a single-stage tank asks for**, and why the window must be generous.
- **A magnetic shunt** in the window: leakage without winding width, at the cost of a part and a harder tolerance.

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Every transformer text says to interleave. Here L_r is a design value, and a supplier who “improves” the coupling breaks the converter. Say so on the drawing, next to L_{short} .

6.14 The gap, and why the drawing must not name it

L_m is far below what an ungapped core gives, so the core is gapped:

Equation 26

$$A_L = \frac{L_{\text{open}}/N_x}{N_p^2} \quad g \approx \frac{\mu_0 A_e}{A_L}$$

with N_x the number of units in the assembly and g the total centre-leg gap. The gap expression ignores fringing, which always makes the real gap larger. **Specify A_L , or better L_{open} , and leave the gap to the supplier:** that is what they grind to and what a meter can check.

6.15 The window: wire, current density and what actually fits

Marks **2** and **5** of Figure 43. Copper is sized on the **line-cycle** rms: the rms of each switching period, averaged in I^2 over the line half cycle. The secondary value is per winding and per unit, with the idle half inside the average.

Each winding needs a copper area set by its rms current and the current density the design can cool:

Equation 27

$$A_{cu} = \frac{I_{\text{rms}}}{J} \quad \sum_w N_w A_{cu, w} \leq k_u A_N$$

with N_w the turns of winding w and $A_{cu, w}$ its copper area. J is an assumption: 4 to 5 A/mm² in free air, less enclosed. k_u , the window utilisation, is **0.3 or less** with Litz wire once insulation, bobbin wall, margins and tape are counted. Sizing on bare copper overstates what fits by three times.

With primaries in series every unit carries the whole primary current; only the secondaries share. That is why splitting helps a low-voltage, high-current output.

6.16 Skin depth, and why the wire is not a wire

At the switching frequency current crowds into a surface layer of depth

Equation 28

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$

with ρ the resistivity of copper at the winding temperature and f the frequency it carries, here f_r : about **0.2 mm** at 100 °C. A conductor thicker than 2δ carries no more current. The field from the other turns also drives circulating current in each conductor (the proximity effect), and in a side-by-side winding that term can dominate.

So a winding is not one wire. This note uses both ways out:

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- **Litz wire** on the primary: a bundle of many fine wires, twisted so that each takes a turn at every position, and each insulated by its own enamel. One fine wire is a **strand**; d_s is its diameter and a_s its copper cross-section. The rule is $d_s \leq 2\delta$ **per strand**.
- **Copper foil** on a low-voltage, high-current secondary: thickness t_f near δ , width does the carrying. It also terminates well into a centre tap.
- The ac resistance heats the winding; the dc resistance is a floor, not an answer.

Litz. Pick a standard strand at or below 2δ ; the strand count is the copper needed over what one strand gives:

Equation 29

$$a_s = \frac{\pi d_s^2}{4}, \quad n_s = \left\lceil \frac{A_{cu}}{a_s} \right\rceil$$

with A_{cu} from Equation 27. The bundle is wider than its copper: gaps between round strands, enamel and serving are collected into a fill factor k_{litz} of about 0.5 to 0.6, and the outside diameter is

Equation 30

$$d_{litz} = \sqrt{\frac{4A_{cu}}{\pi k_{litz}}}$$

Foil. Choose the thickness from the skin depth and what the supplier stocks; the width follows. If it exceeds the bobbin width, use several narrower strips in parallel, stacked radially:

Equation 31

$$w_f = \frac{A_{cu}}{t_f}, \quad n_f = \left\lceil \frac{w_f}{w_{f,max}} \right\rceil$$

These size the copper for current density and skin depth. The ac resistance, skin plus proximity, is a separate question, and on a side-by-side winding proximity is usually the larger term.

6.17 Isolation, margins and what they cost in window

The transformer is the isolation barrier. **Margin tape** leaves a creepage margin at each end of the bobbin, typically 3 mm a side for reinforced isolation from universal mains, so 6 mm of winding width is lost. **Triple-insulated wire** needs no margin but is thicker for the same copper and costs more.

On a side-by-side winding the width is spent twice: on the isolation margin and on the gap that sets the leakage. Budget both.

6.18 Loss, and the temperature the drawing has to survive

- **Core loss** comes off the material curve at the operating flux and frequency, not off the headline figure.
- **Copper loss** is $I^2 R_{ac}$ for both windings, and in a side-by-side arrangement the proximity term is not small.

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- **Temperature rise** is what limits the design, and it depends on surface area and airflow. The computed loss is an input to a thermal measurement, not an answer.

B_{pk} is fixed by the secondary volt-seconds at f_r , but core loss rises with frequency, and the frequency sweeps over the line cycle: worst flux at the bottom of the sweep, worst loss at the top.

6.19 The saturation test is not the peak winding current

What the line on the specification says. Every transformer specification carries a line of the form “DC overlap: inductance more than 90 % of initial, test current I_{sat} , normal temperature”. The name is the measurement. An LCR meter reads the primary inductance with a small ac signal (100 kHz, 1 V is a common setting, the same one the leakage test uses) while a bias source overlaps a dc current on the same winding; every other winding is open. The dc current is stepped up and the inductance is read at each step. The same test is called dc bias or dc superposition.

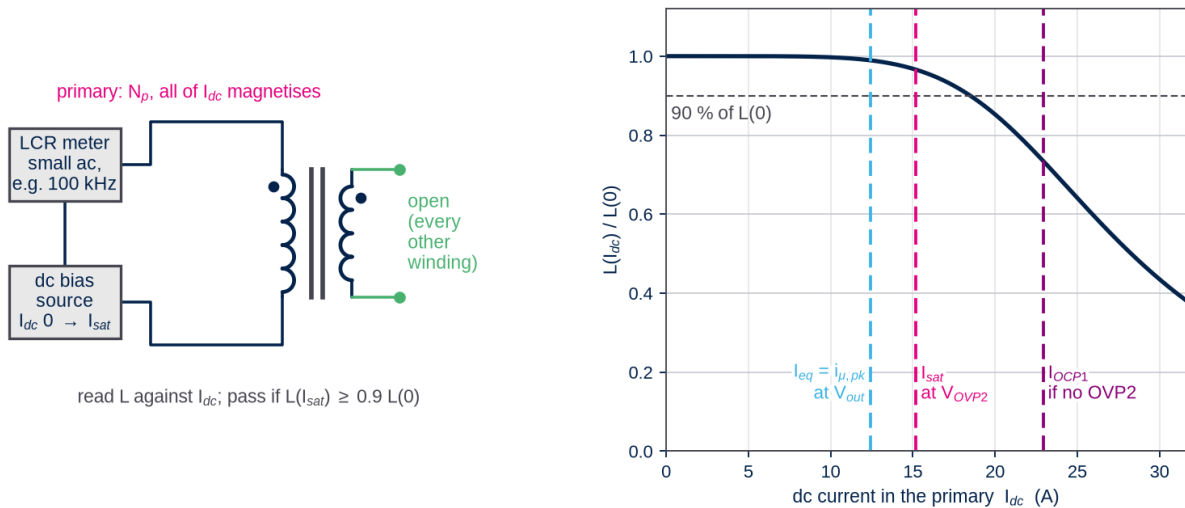


Figure 33: The DC-overlap test. Left: the bench. Right: inductance against the dc current, an illustrative shape, with this design's three currents marked: the operating magnetising peak, the test current the specification asks for, and the current that would have to be used if the controller had no voltage ceiling.

What it reveals. With the other windings open the dc current alone sets the flux, so the inductance stays flat while the core is linear and falls as the core approaches saturation; the 90 % point marks the knee. It is the only test at the terminals that reads the flux margin. Turns and A_L do not: the same A_L comes from a different gap, a different A_{min} or a different material, and L_{open} confirms the inductance, not the margin. A part with the right L_{open} and the wrong core is caught here and nowhere else.

Which current to write on the line. Mark 3 and the bench panel of Figure 43. The current to ask for is the peak of the magnetising trace, scaled to the over-voltage ceiling, not the peak winding current at mark 1.

In the test nothing cancels the primary ampere-turns, so the same current makes far more flux than in operation. Ask for the current that reproduces the operating flux in that test:

Equation 32

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_{\mu}}$$

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The denominator is L_{μ} , not L_{open} : the primary leakage links no core. Dividing by L_{open} understates the current by $1/\sqrt{1+\lambda}$. A free check: the answer **must** equal $i_{\mu,pk}$.

The peak tank current would ask for a flux no ferrite reaches. No arbitrary margin is wanted on I_{eq} either: the flux ceiling is already defined by the highest output the controller allows before it shuts down.

Why the ceiling is OVP2. The flux follows the output voltage alone (Section 6.11), so the largest flux the core ever carries comes at the largest output the controller lets stand. The L6790A has two levels on the ZCD pin. At OVP1 (2.3 V on the pin) it keeps switching at reduced power, so the output can sit between OVP1 and OVP2 with the transformer still driven. At OVP2 (2.5 V) it stops switching for 100 ms and restarts with a soft start. Above OVP2 the core is never driven, so OVP2 is the ceiling, and the ratio of the two voltages is the whole margin. No arbitrary factor goes on top of it:

Equation 33

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out,eff}}$$

What does not raise the flux: overload (the copper and the silicon take it), start-up (the output is lower), burst mode, and the switching frequency (the clamp lasts $T_r/2$ whatever f_{sw} is). What does: an output over-voltage, and temperature. B_s of a MnZn power ferrite falls by about a fifth between 25 and 100 °C, so a test made at normal temperature is read against the material's hot B_s , and the design flux at OVP2 has to sit under it with margin. The 90 % criterion is that margin.

If the controller has no OVP2-like ceiling. Then something else has to bound the output voltage, and the specification follows whichever bound exists.

- **A protection that stops the drive**, on either side of the barrier: a secondary crowbar or latch, an SR-controller OVP, a primary OVP that latches. Use its threshold in place of V_{OVP2} .
- **A protection that only reduces power**, like OVP1, is not a ceiling: the output can rest at it while the core is driven. Do not use it.
- **No voltage ceiling at all.** The output is then bounded only by what the tank can deliver at the oscillator floor into the lightest load, and with the floor near f_o that bound is not useful. What remains is the current limit: with nothing clamping the secondary the primary current is the magnetising current, so the core must not saturate at the largest current the controller ever lets through the primary, $I_{sat} = I_{OCP1}$. Safe, and expensive; Section 7.9 gives the ratio for this design.
- **The rule of thumb**, 1.2 to 1.5 times $i_{\mu,pk}$ or the winding peak at normal temperature, is what a specification carries when nobody asked the question. It names no event, so it is neither safe nor economical. Replace it with one of the three above.

6.20 The specification, and what it is not allowed to leave out

The open-circuit inductance tolerance cannot be the usual round number. f_o goes as $1/\sqrt{L_{open}}$, so a low part raises the frequency floor towards the oscillator clamp, and a part that meets $\pm 10\%$ can hard switch. Section 6.25 gives the figure and the two ways out.

Core, bobbin, wire and winding order are the supplier's choice; insulation and creepage follow the safety standard. Everything above is measurable at the terminals.

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6.21 If one transformer is not practical

At low voltage and high current the secondary is one heavy turn and the core comes out large. The transformer can then be built as N_x identical units with **primaries in series and secondaries in parallel**; series primaries carry the same current, so the secondaries share without balancing resistors.

- Open-circuit and leakage inductance divide by N_x , and so do the secondary currents. **The primary current does not.**
- The DC-overlap test current does not change.
- The wound ratio becomes $N_x N_p / N_s$, so **achievable ratios are quantised in steps of N_x / N_s** . Respect that when the tank is chosen.

6.22 The input capacitor

C_{in} is a film capacitor that absorbs switching ripple only. A large value holds charge across the zero crossing and distorts the line current, so it is a THD term:

Equation 34

$$C_{in} = 3 \frac{\text{nF}}{\text{W}} \times P_{in}$$

That is a **lower** bound; round up to the next standard value and rate it for the full input voltage.

6.23 The output capacitor bank

Two conditions apply, and the larger wins. The ripple condition is

Equation 35

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2}$$

and the hold-up condition is

Equation 36

$$C_{out} \geq \frac{2P_{out}T_{hold}}{\left(V_{out} - \frac{1}{2}\Delta v_{pp}\right)^2 - V_{o,min}^2}$$

Hold-up starts at the worst line phase. If the mains fails at the ripple trough the bank is already half a ripple down, hence the $-\frac{1}{2}\Delta v$ term; it costs about a tenth of the ride-out. Δv_{pp} is the ripple the bank produces, not the allowance, so this is checked once the bank is chosen.

Both conditions scale as $1/V_{out}^2$, so which wins depends only on the specification. With $k = V_{o,min}/V_{out}$, ripple dominates when

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Equation 37

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold}$$

using the *allowed* Δv , since it is asked before the bank exists. Section 7.11 evaluates it.

- Include the **2f_l component** in the ripple current; it is comparable to the switching part and adds in quadrature.
- With a centre-tapped secondary, judge capacitor ripple current at the **output node**, not per winding.
- Use the ESR at the **switching frequency**; the 120 Hz tanδ figure is five to ten times larger.

The controller start-up window is finite and this bank is a far heavier start-up load than a conventional design. Cold start into the full bank should be measured early.

6.24 Semiconductor requirements

State requirements rather than pick parts. Four rules:

- Rate the primary switches on the **composite tank peak**; the load component alone under-rates them by more than a tenth.
- Divide by the number of devices in parallel. Tabulated currents are **per switch position** on the primary and **per leg** on the secondary; a centre-tapped leg carries the whole secondary current.
- Compute loss from $R_{DS(on)}$ at $T_{j,max}$. The datasheet maximum is 25 °C process spread; temperature is a separate multiplier, about two for a superjunction device.
- A curve against T_a may be read as T_j **only if it is a pulse test**.

The standing device in half-bridge morphing (Section 5.2) never switches and carries the whole tank current; Section 7.12 shows what that cost here.

6.25 Controller network

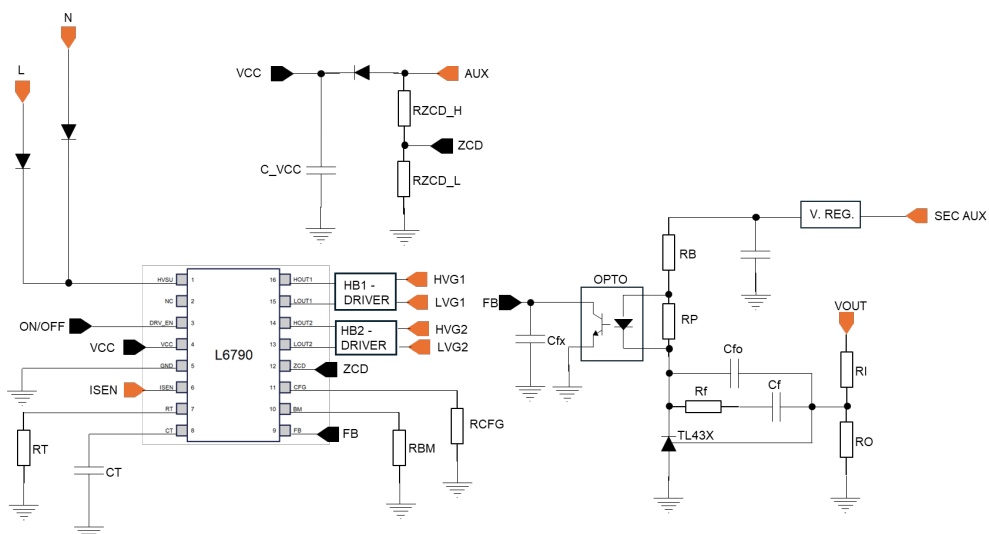


Figure 34: The controller and its passive network. HVSU is fed from the AC side of the bridge; the auxiliary winding drives ZCD through a divider; R_T and C_T set the oscillator limits; R_{CFG} and R_{BM} are read at power-up.

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Decide first what the auxiliary winding is for. If it supplies V_{CC} , its voltage at OVP2 must stay under the V_{CC} rating, which caps its ratio. Here V_{CC} comes from elsewhere, so the winding only senses and the only constraint is whole turns (Section 7.20).

Section 7.15 sizes the parts in order. R_T deserves the most attention: it sets the oscillator floor, and that floor must stay above f_o .

That floor is usually the thinnest margin. It also depends on the oscillator idle time, which the draft datasheet states inconsistently, so it must be confirmed by measuring $f_{sw}(\theta)$ on hardware.

The transformer tolerance must fit inside the same margin. A **low** L_{open} raises f_o ; the fall that can be tolerated is

Equation 38

$$\frac{\Delta L}{L} = 1 - \frac{1}{k_{floor}^2}$$

A floor margin of a few per cent allows a fall of a few per cent. The usual $\pm 10\%$ does **not** fit: at the bottom of the tolerance the converter hard switches. Either tighten the low-side tolerance or lower R_T to lift f_{Min} . **Check before the transformer is ordered.**

6.26 Voltage loop and compensation

The loop is the chain of Figure 35: a divider measures the output, a TL431 compares it with its reference and drives an optocoupler LED, the optocoupler transistor pulls the FB pin, the FB voltage commands power (Section 4.6), and that power goes into the output capacitor.

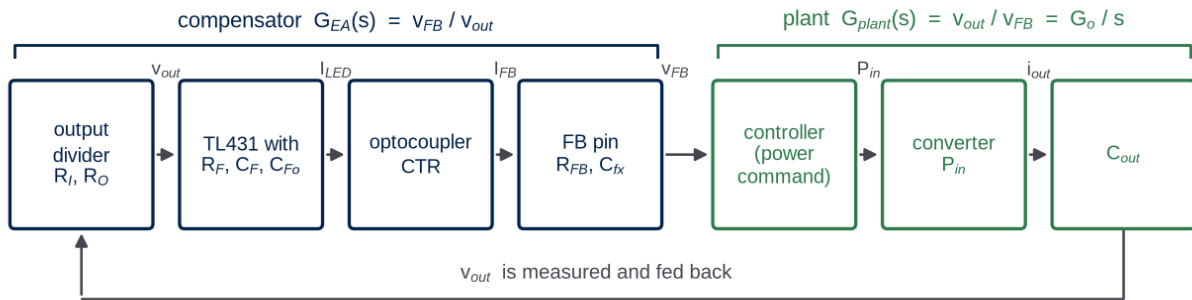


Figure 35: The voltage loop as blocks. Everything left of the FB pin is the compensator $G_{EA}(s)$; everything right of it is the plant $G_{plant}(s)$. The loop gain is $T(s) = G_{plant}(s) G_{EA}(s)$.

The plant. The FB voltage sets the power, the power over the output voltage is the current into C_{out} , and a current into a capacitor integrates:

Equation 39

$$G_{plant}(s) = \frac{v_{out}(s)}{v_{FB}(s)} = \frac{G_o}{s}, \quad G_o = \frac{P_{out}}{V_{out} V_{FB} C_{out}} \quad [\text{rad/s}]$$

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V_{FB} is the feedback voltage above its 0.5 V offset at rated power (Equation 15). An integrator falls at 20 dB per decade with -90° everywhere and would cross 0 dB at $f_{cto} = G_o/2\pi$. There is no second pole and no right-half-plane zero, so the loop is easy to stabilise. What makes it hard to place is the output ripple (Section 6.28).

6.27 What the loop must achieve, and what goes wrong when it does not

Three numbers are read from the Bode plot of $T(s)$: the **crossover** f_c where $|T| = 1$, the **phase margin** $180^\circ + \arg T$ at f_c , and the **gain margin**, how far $|T|$ is below 0 dB at f_{180} where $\arg T = -180^\circ$. Table 5 gives the aims.

Quantity	Aim	If it is too low	If it is too high
Crossover f_c	15 to 20 Hz in this converter (Section 6.28). The general rules are below $f_{sw}/5$ and below the ripple frequency the loop must not follow.	The loop is slow. The output dips further and recovers later after a load step or a morphing transition.	The loop follows the $2f_l$ output ripple. The power command, and so the input current, is modulated at $2f_l$; third harmonic on the input current, and chatter across the burst threshold.
Phase margin Φ_M	45° is the floor; 50 to 60° is the target. At 76° the step response has no overshoot; at 45° it rings ($Q \approx 1.2$) [onsemi TND381].	Ringing after every disturbance. At 0° the loop oscillates at f_c .	The response is over-damped and slow. Not a fault, but no longer free: the crossover has to drop to get it.
Gain margin GM	6 dB is the floor; 10 dB is comfortable.	Part spread moves $ T $ up. The optocoupler CTR alone spreads by 2:1 over bins, current and temperature; a loop with a small gain margin oscillates at f_{180} on a warm board with a high-CTR part.	—
G_{EA} at $2f_l$	At or below the value that keeps the third harmonic inside its budget (Equation 42).	Nothing wrong with the loop; the crossover is lower than it needs to be.	Third harmonic above budget, burst chatter.

Table 5: What the loop must achieve.

6.28 Why the crossover must be low: the $2f_l$ ripple

The output ripple at $2f_l$ (Section 3.1) has the peak-to-peak value

Equation 40

$$\Delta V_{loop} = \frac{P_{out}}{V_{out}} \frac{1}{2\pi f_l C_{out}}$$

The compensator passes a fraction $G_{EA}(2f_l)$ of it to the FB pin, which is a power command, so the input current is modulated at $2f_l$. A sine modulated at twice its own frequency gains a third harmonic; a modulation of the command by m puts $m/2$ into it, and the ripple amplitude is half the peak-to-peak, so

Equation 41

$$D_3 = \frac{G_{EA}(2f_l) \Delta V_{loop}}{4 V_{FB}}$$

Turned around, the third-harmonic budget fixes the largest compensator gain at $2f_l$:

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Equation 42

$$G_{EA}(2f_i) \leq \frac{4 V_{FB} D_3}{\Delta V_{loop}}$$

A small gain at $2f_i$ means a crossover well below it; with 5 % third harmonic allowed, 15 to 20 Hz. A faster loop would give less output ripple and a quicker load-step response, and more input distortion. The distortion limit wins. onsemi TND381 calls the failure “tail chasing”.

6.29 The compensator: TL431, optocoupler and the FB pin

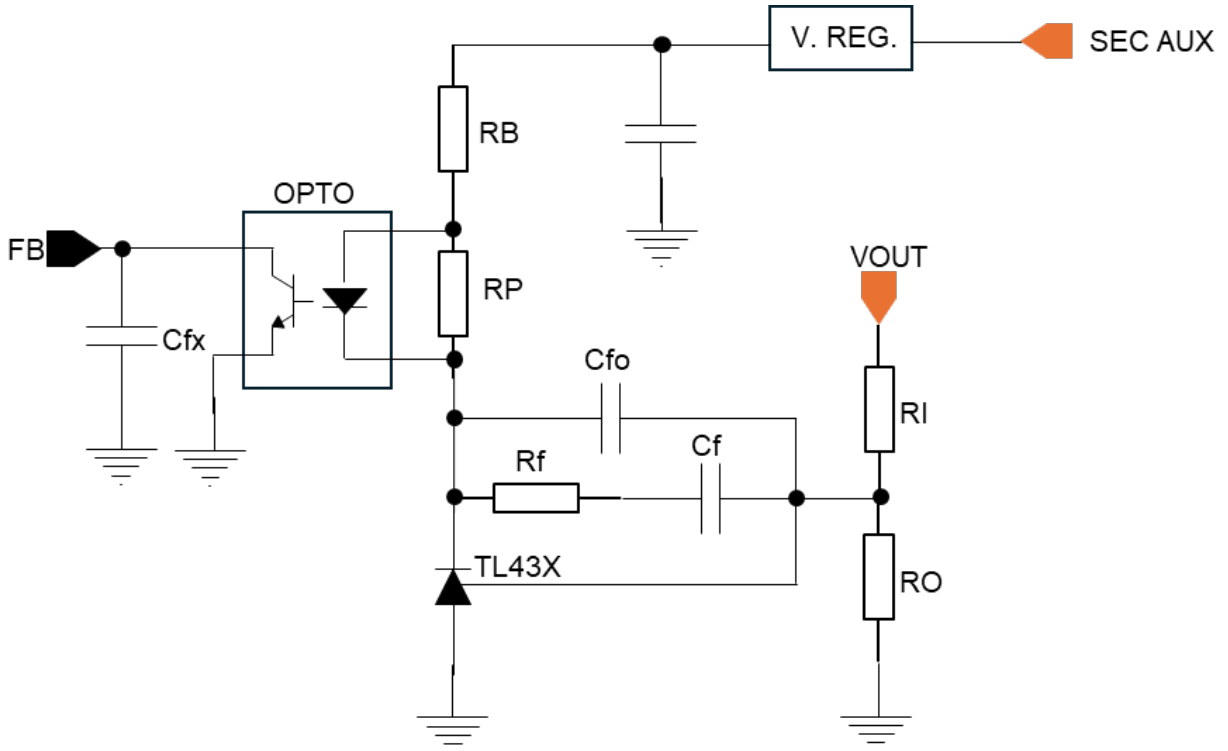


Figure 36: The compensator on the secondary side. R_I and R_O set the regulated output; the TL431 is the error amplifier; R_f , C_f and C_{fo} shape its gain; R_B feeds the optocoupler LED from the regulated V_Z rail; R_P keeps the TL431 above its minimum current; C_{fx} sits on the FB pin.

The divider R_I , R_O brings the output down to the TL431 reference $V_R = 2.495$ V:

Equation 43

$$V_{out} = V_R \left(1 + \frac{R_I}{R_O} \right)$$

When the output rises the TL431 sinks more cathode current through the LED, which R_B feeds from a regulated rail V_Z ; the optocoupler transistor pulls FB down against the internal pull-up R_{FB} and the power command falls. R_P across the LED carries the minimum cathode current the TL431 needs, so it has an upper limit:

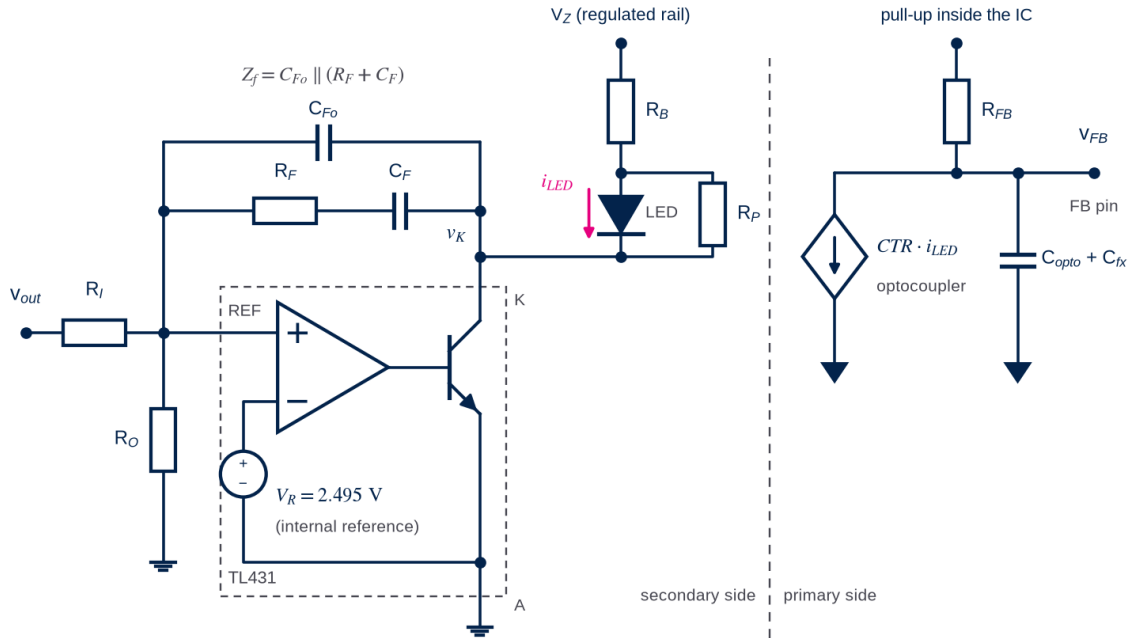
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Equation 44

$$R_P \leq \frac{V_{Fo}}{I_{min}}$$

Because the LED is fed from V_Z , the output ripple reaches it only through the TL431: there is no second path (TND381's "fast lane").

The same network as an op-amp. Figure 37 redraws it. Inside the TL431 an amplifier compares REF with the internal V_R and drives an NPN whose collector is the cathode; the transistor inverts, so from REF to cathode the part is a high-gain inverting amplifier that holds REF at V_R . R_I is its input resistor and C_{Fo} in parallel with $R_F + C_F$ its feedback impedance Z_f , so the cathode moves by $-Z_f/R_I$ times the output change; R_O sets only the dc point. The cathode current is the LED current, and the optocoupler transistor is a current-controlled current source sinking $CTR i_{LED}$ from the FB node into R_{FB} with the pole of $C_{opto} + C_{fx}$. The three blocks multiply, with the negative sign the loop needs.



$$\frac{v_K}{v_{out}} = -\frac{Z_f}{R_I}, \quad i_{LED} = -\frac{v_K}{R_B}, \quad v_{FB} = -CTR i_{LED} \left(R_{FB} \parallel \frac{1}{s(C_{opto} + C_{fx})} \right)$$

R_O carries no signal: the REF input is held at V_R

$$G_{EA}(s) = \frac{v_{FB}}{v_{out}} = \frac{Z_f}{R_I} \cdot \frac{CTR R_{FB}}{R_B} \cdot \frac{1}{1 + s R_{FB}(C_{opto} + C_{fx})}$$

Figure 37: The TL431 compensator as an op-amp circuit: R_I and Z_f make an inverting amplifier; R_B , R_P and the LED turn the cathode voltage into i_{LED} ; the optocoupler is a current source $CTR i_{LED}$ into R_{FB} and $C_{opto} + C_{fx}$. The two sides return to different grounds across the isolation barrier.

Its transfer function from output to FB pin is a Type II network with one more pole:

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Equation 45

$$G_{EA}(s) = \frac{EA_o}{s} \cdot \frac{1 + s/\omega_z}{(1 + s/\omega_p)(1 + s/\omega_{px})}$$

Equation 46

$$EA_o = \frac{CTR R_{FB}}{(C_F + C_{Fo}) R_I R_B} \quad [\text{rad/s}], \quad f_z = \frac{1}{2\pi R_F C_F}$$

$$f_p = \frac{1}{2\pi R_F C_{ser}}, \quad C_{ser} = \frac{C_F C_{Fo}}{C_F + C_{Fo}}, \quad f_{px} = \frac{1}{2\pi R_{FB} (C_{opto} + C_{fx})}$$

$\omega = 2\pi f$ throughout, and C_{ser} is C_F and C_{Fo} in series. $C_F + C_{Fo}$ make the pole at the origin (zero static error); $R_F C_F$ the zero f_z that returns the phase; R_F with C_{Fo} the pole f_p that brings the gain down again by $2f_i$; $C_{opto} + C_{fx}$ with R_{FB} a third pole at about a kilohertz that removes switching noise. CTR multiplies the whole gain, which is why its spread matters.

The bias window. The LED must drive the FB pin at its steady-state current with the lowest CTR (upper bound), and must not exceed the pin's maximum current with the highest CTR when the TL431 is fully on (lower bound):

Equation 47

$$R_{B,max} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,steady}/CTR_s}$$

$$R_{B,min} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,max}/CTR_m}$$

EA_o uses the CTR at the steady-state LED current, CTR_s ; the lower bound of R_B uses the CTR at maximum LED current, CTR_m . Take both from the bin that will be fitted, and re-check the loop at the top of the bin: crossover rises with $\sqrt{CTR}$ and phase margin falls.

6.30 Placing the zero and the pole: the K-factor method

The Venable K-factor method sets EA_o , the zero and the pole from two targets: the phase margin Φ_M and the gain allowed at $2f_i$. A spread factor K_v comes from the phase margin, with α_v a weighting constant (1.2 in the ST tool):

Equation 48

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right]$$

The zero goes K_v below the crossover and the pole K_v above it. Above the pole the gain is $EA_o K_v^2/\omega$, so the limit at $2f_i$ fixes EA_o :

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Equation 49

$$EA_o = \frac{2\pi(2f_l) G_{EA}(2f_l)}{K_v^2}$$

The centre point follows from the plant gain, with $\Gamma_v = 0.744 V_{eq,max}/V_{eq,min}$ an input-voltage margin factor of the ST tool (Appendix 10.3):

Equation 50

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v EA_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v}$$

The parts follow from Equation 46. The high-frequency pole is placed at a chosen f_{pHF} , and C_{fx} is what is left after the optocoupler capacitance:

Equation 51

$$C_{Fo} = \frac{f_z}{f_p} \cdot \frac{CTR_s R_{FB}}{R_l R_B EA_o}, \quad C_F = C_{Fo} \left(\frac{f_p}{f_z} - 1 \right)$$

$$R_F = \frac{1}{2\pi f_z C_F}, \quad C_{fx} = \frac{1}{2\pi f_{pHF} R_{FB}} - C_{opto}$$

Round each value to a standard part, then compute the zero, pole and gain the standard parts give and check the loop on those.

6.31 Checking the loop: crossover, phase margin and gain margin

The loop gain (Equations 39 and 45) is two integrators, one zero and two poles:

Equation 52

$$|T(\omega)| = \frac{G_o EA_o}{\omega^2} \cdot \frac{\sqrt{1 + (\omega/\omega_z)^2}}{\sqrt{1 + (\omega/\omega_p)^2} \sqrt{1 + (\omega/\omega_{px})^2}}$$

$$\arg T(\omega) = -180^\circ + \arctan \frac{\omega}{\omega_z} - \arctan \frac{\omega}{\omega_p} - \arctan \frac{\omega}{\omega_{px}}$$

The -180° is the two integrators, so the **phase margin is the compensator phase at the crossover**. With $A(\omega)$ the fraction in Equation 52, the crossover condition $\omega_c^2 = G_o EA_o A(\omega_c)$ is solved by repeating

Equation 53

$$\omega_c \leftarrow \sqrt{G_o EA_o A(\omega_c)}, \quad \text{starting from } \omega_c = \sqrt{G_o EA_o}$$

a few times; A changes slowly, so no solver is needed. Then

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Equation 54

$$\Phi_M = \arctan \frac{\omega_c}{\omega_z} - \arctan \frac{\omega_c}{\omega_p} - \arctan \frac{\omega_c}{\omega_{px}}$$

Figure 38 shows where the three numbers are read, in units of the loop's own crossover, so it is the shape of every loop of this kind. Section 7.14 puts this design's numbers on the same plot.

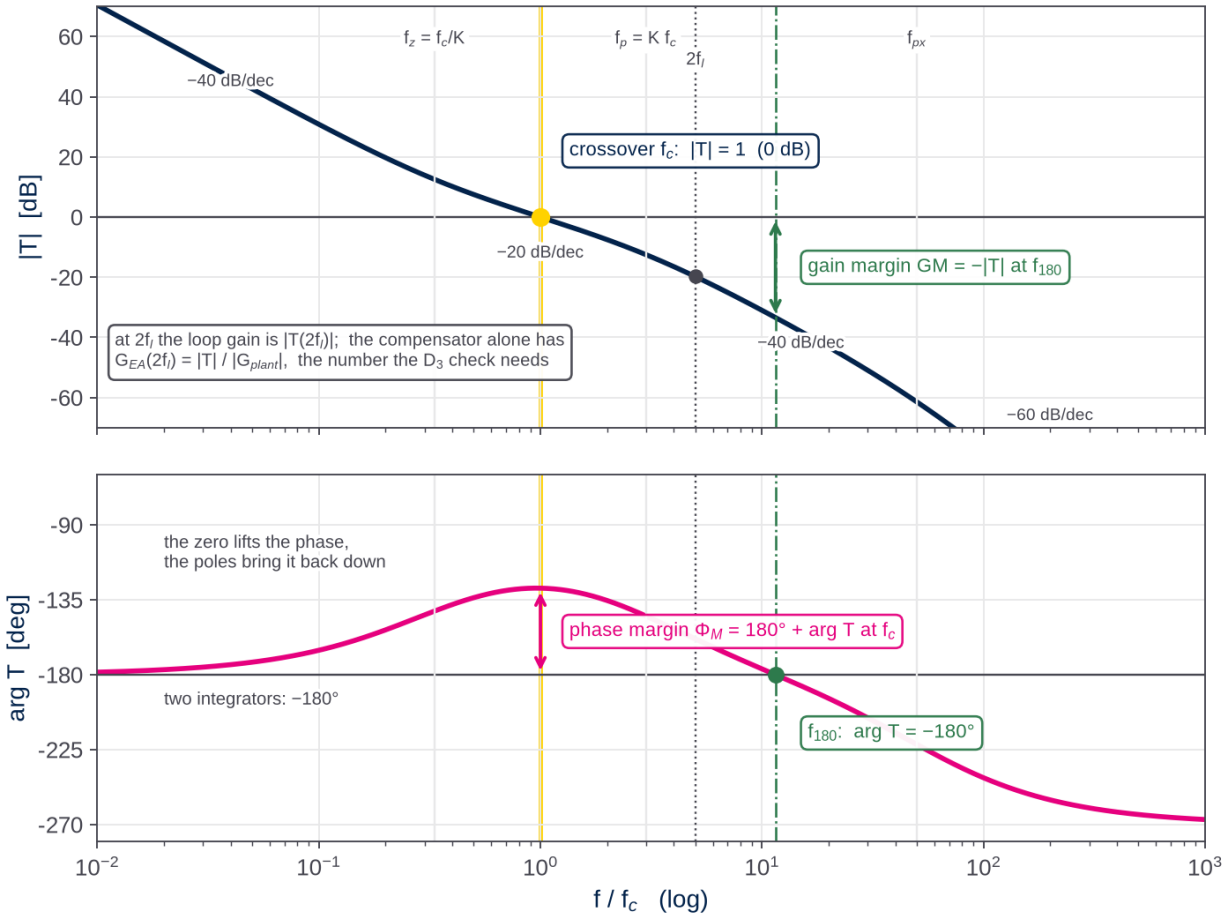


Figure 38: A two-integrator loop with a Type II compensator, in units of its crossover. Phase margin is the distance from -180° at f_c ; gain margin is $|T|$ below 0 dB at f_{180} . The dot at $2f_l$ is the gain the third-harmonic check reads.

6.32 Gain margin, and why it needs checking

Above f_{px} one zero faces two poles, so the phase keeps falling and -180° is crossed at a finite frequency, which also needs no search:

Equation 55

$$f_{180} = \sqrt{f_p f_{px} - f_z(f_p + f_{px})} , \quad GM = -20 \log_{10} |T(f_{180})|$$

The root is real whenever the zero sits well below both poles, which the K-factor placement gives; if it were negative the phase margin would be negative at any crossover. Judge against 6 dB as a floor and 10 dB as comfortable. With a crossover in the tens of hertz the margin is usually large; that is a reason to compute it, not to assume it.

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The same expression at $2f_l$ gives the third harmonic through Equation 41, which closes the check.

6.33 Feedback ripple against the burst threshold

A crossover this low leaves $2f_l$ ripple on the FB pin, which moves the commanded power every half cycle. If it crosses the burst-entry threshold the converter chatters. At the burst point the ripple is

Equation 56

$$\Delta V_{FB} = \frac{P_{in,BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l)$$

Lowering R_{BM} moves the burst entry below that ripple, at 0.01 V per k Ω , and the ripple must clear it on both sides:

Equation 57

$$\Delta R_{BM} = \frac{\Delta V_{FB}}{2 \times 0.01 \text{ V/k}\Omega}$$

Raising the crossover would shrink the ripple but worsen the distortion, so R_{BM} is the practical answer.

A loop this slow cannot recover from a large load step by itself; the anti-saturation circuit does, and it needs the FB pin to reach its full current range. That is why R_B must sit inside its window (Equation 47).

7 Design example

One design from specification to component values; every computed value shows its arithmetic. **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, centre-tapped synchronous rectification, no bulk capacitor, no boost stage.

7.1 The specification, sorted by what kind of number it is

Each number is **given** by the load and the mains, **chosen** by the designer, or **assumed** in place of a measurement not yet made.

Item	Symbol	Value	Kind	Note
Mains input	V_{ac}, f_l	90 to 264 Vac, 47 to 63 Hz	given	the worst line frequency is the lowest, so every f_l term uses 47 Hz
Output	V_{out}, I_{out}	25 V, 26.3 A (657.5 W)	given	$V_{o,eff} = V_{out} + N_{rect} V_f = 25.0 \text{ V}$ with $V_f = 0$
Output ripple, $2f_l$ pk-pk	Δv	$\leq 5\%$ of V_{out} (1.25 V)	given	a property of the load, not of the converter
Hold-up	$T_{hold}, V_{o,min}$	12 ms down to 19 V, at 100 Vac	given	starts at the ripple trough, the worst line phase
Secondary rectifier	N_{rect}	1 (centre tap, SR)	chosen	halves the secondary conduction loss; doubles the device reverse voltage
Target series resonance	f_r	150 kHz	chosen	sets the magnetics size
Specified maximum f_{sw}	$f_{sw,max}$	225 kHz (1.5 f_r)	chosen	an input to λ , Section 4.9

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Item	Symbol	Value	Kind	Note
Bridge dead time	t_D	220 ns	chosen	what the ZVS sweep has to beat
V_{CC} source	—	external 12 V rail	chosen	no turns-ratio ceiling on the auxiliary winding
LLC stage efficiency	η_{HB}	98 %	assumed	a 95 % outcome moves R_{ac} and R_{CS} by about 3 %; nothing downstream is sensitive
Oscillator idle time	T_{idle}	250 ns	assumed	the draft datasheet also says 700 ns, which would close the floor margin — the first thing to measure
$R_{DS(on)}$ temperature factor	k_T	1.8 / 1.9	assumed	primary / secondary, read off the datasheet curves at $T_{j,max}$
Burst entry point	r_{BM}	75 W (11 % of rated)	assumed	sets R_{BM} , then checked against the feedback ripple

Table 6: The specification. Everything else in this chapter is computed from these.

Harmonic class. IEC 61000-3-2 Class D reaches only 600 W. This design draws 685 W at the input, so the absolute Class A limits apply.

7.2 What the design came out as

Block	Values
Tank	C_r 100 nF, L_r 11 μ H, L_m 20 μ H, λ 0.55, f_r 151.7 kHz, f_o 90.4 kHz
Transformer	$N_p:N_s$ 15:2, n 6.02, n_T 7.50, L_{open} 31.0 μ H, L_{short} 11.0 μ H, $A_e \geq 103$ mm ² at B_{pk} 0.20 T
Frequency	f_{sw} at the line peak, full load: 127.1 kHz at the low corner, 249.0 kHz at the high corner
Output	470 μ F $\times$ 160 = 75.2 mF, ripple 1.18 V (4.74 %), hold-up 13.42 ms
Controller	R_T 11 k Ω , C_T 470 pF, R_{CS} 24.0 m Ω , R_{CFG} 30 k Ω , R_{BM} 30 k Ω
Loop	f_{cross} 17.57 Hz, Φ_M 49.64°, third harmonic 4.23 %

Table 7: Principal values. The rest of the chapter shows where each one comes from.

7.3 What a verification margin is

Every check is what the design has divided by what it needs:

Equation 58

$$k = \frac{X_{act}}{X_{req}} \implies \text{pass when } k > 1$$

$k = 1.05$ is five per cent of room; the smallest k is the thinnest part of the design.

Check	$k = \text{has} / \text{needs}$	Substituted	k
ZVS at the worst point of the sweep	$T_{ZC,min} / t_D$	401 ns / 220 ns	1.824

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Check	k = has / needs	Substituted	k
Oscillator floor clears the lower resonance	f_{Min} / f_o	92.25 kHz / 90.39 kHz	1.021
Oscillator ceiling clears the highest operating frequency	$f_{\text{Max}} / f_{\text{sw}} \text{ (FB edge, line peak)}$	319.61 kHz / 249.03 kHz	1.283
Over-current threshold clears the tank peak	$I_{\text{OCP1}} / I_{\text{Lr,pk}}$	22.92 A / 18.32 A	1.251
Hold-up achieved against hold-up required	$t_{\text{hold}} / T_{\text{hold}}$	13.42 ms / 12 ms	1.119
Secondary loss per rectifier leg against its budget	$P_{\text{budget}} / P_{\text{SR,leg}}$	3.00 W / 2.83 W	1.063
Loss of the standing primary device against its budget	$P_{\text{budget}} / P_{\text{mos,dc}}$	3.00 W / 9.65 W	0.311

Table 8: The verification margins.

$k_{\text{Ploss}} = 0.311$ is a result, not a failed calculation: the standing primary device spends 9.65 W against a 3 W budget; no single 600 V device meets that budget in the standing position, so the design goes ahead and the heatsink question is settled by measurement (Section 6.24). $k_{\text{floor}} = 1.021$ looks safe, but the transformer tolerance and the idle time both move it and neither is known to 2 % yet.

7.4 Following the numbers through

The design in the order it was computed; each step shows its equation and the numbers in it. Table 9 collects the results.

Step 1 — the power the tank passes. The bridge, the EMI filter and the LLC stage each get a loss budget, the last as η_{HB} :

$$P_{\text{in}} = P_{\text{out}} + P_{\text{LLC}} + P_{\text{EMI}} + P_{\text{BR}} = 657.5 + 13.42 + 8.34 + 5.52 = \mathbf{684.8 \text{ W}}$$

$$P_{\text{in,LLC}} = P_{\text{in}} - P_{\text{BR}} - P_{\text{EMI}} = 684.8 - 5.52 - 8.34 = \mathbf{670.9 \text{ W}}$$

Step 2 — the two corners of the equivalent input. They are the morphing thresholds, not the mains limits:

Equation 18, repeated

$$V_{\text{ac,eq}} = 2 V_{\text{ac}} \text{ (full bridge)} \quad V_{\text{ac,eq}} = V_{\text{ac}} \text{ (half bridge)}$$

$$V_{\text{ac,eq,low}} = \frac{245 V_{\text{pk}}}{\sqrt{2}} = \mathbf{173.2 V_{\text{ac}}} \quad V_{\text{ac,eq,high}} = \frac{2 \times 235 V_{\text{pk}}}{\sqrt{2}} = \mathbf{332.3 V_{\text{ac}}}$$

Step 3 — the turns ratio. The wound ratio 15 : 2 is the input, because it is what can be built (the plain calculation gives $n = 6.046$); the model ratio follows from it and the realised λ (step 8):

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Equation 21, repeated

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}}$$
$$n = \frac{n_T}{\sqrt{1 + \lambda_{act}}} = \frac{7.500}{1.2450} = \mathbf{6.024}$$

Equation 19, repeated

$$V_{refl} = n V_{o, eff} = n(V_{out} + N_{rect} V_f)$$
$$V_{refl} = n V_{o, eff} = 6.024 \times 25.0 = \mathbf{150.6 \text{ V}}$$

Step 4 — the gain the tank is asked for, at the line peak ($\theta = 90^\circ$) of each corner:

Equation 12, repeated

$$M(\theta) = \frac{n V_{o, eff}}{V_{drive}(\theta)} = \frac{2 n V_{o, eff}}{\sqrt{2} V_{ac, eq} \sin \theta}$$
$$M_{low} = \frac{2 \times 6.024 \times 25.0}{\sqrt{2} \times 173.24} = \mathbf{1.2294} \quad M_{high} = \frac{2 \times 6.024 \times 25.0}{\sqrt{2} \times 332.34} = \mathbf{0.6409}$$

Step 5 — the load the tank sees. The 4, not the textbook 8, because $P_{in, LLC}$ is the line-peak power (Section 6.4):

Equation 20, repeated

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o, eff}^2}{P_{in, LLC}}$$
$$R_{ac} = \frac{4}{\pi^2} \frac{6.024^2 \times 25.0^2}{670.9} = \mathbf{13.70 \text{ } \Omega}$$

Step 6 — the Q cap and the design impedance. At the low corner, the Q at which the tank still reaches M_{low} and the Q at which ZVS still completes inside t_D ; the smaller is kept (Equation 1):

$$Q_{ZVS} = \mathbf{1.7019} \quad Z_{0, design} = R_{ac} Q_{ZVS} = 13.70 \times 1.7019 = \mathbf{23.32 \text{ } \Omega}$$

Step 7 — C_r and L_r . C_r is taken up on purpose for Q margin; L_r pairs with the selected C_r :

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Equation 2, repeated

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

$$C_r = \frac{1}{2\pi f_r Z_{0, design}} = \frac{1}{2\pi \times 150 \text{ kHz} \times 23.32 \text{ } \Omega} = 45.50 \text{ nF} \rightarrow \mathbf{100 \text{ nF}}$$

$$L_r = \frac{1}{(2\pi f_r)^2 C_r} = \frac{1}{(2\pi \times 150 \text{ kHz})^2 \times 100 \text{ nF}} = 11.26 \text{ } \mu\text{H} \rightarrow \mathbf{11 \text{ } \mu\text{H}}$$

Step 8 — λ and L_m . Four candidates for λ , the largest binding:

Equation 16, repeated

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw, max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw, max}}\right)^2}$$

$$\lambda_1 = 0.560, \quad \lambda_2 = 1.009, \quad \lambda_{TD} = \mathbf{1.241} \quad \Rightarrow \quad L_m = \frac{L_r}{\lambda_{TD}} = \frac{11}{1.241} = 8.87 \text{ } \mu\text{H}$$

That 8.9 μH is the no-load condition at the high corner, which this design knowingly does not meet (Section 4.10), so it is not rounded to. L_m is chosen with n so that n_T is windable:

$$L_m = \mathbf{20 \text{ } \mu\text{H}} \quad \lambda_{act} = \frac{L_r}{L_m} = \frac{11}{20} = \mathbf{0.550}$$

Step 9 — what the selected parts make. Every later check is measured against these:

Equation 3, repeated

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}}$$

$$f_r = \frac{1}{2\pi\sqrt{11 \text{ } \mu\text{H} \times 100 \text{ nF}}} = \mathbf{151.7 \text{ kHz}} \quad f_o = \frac{1}{2\pi\sqrt{31 \text{ } \mu\text{H} \times 100 \text{ nF}}} = \mathbf{90.4 \text{ kHz}}$$

$$Z_0 = \sqrt{\frac{11 \text{ } \mu\text{H}}{100 \text{ nF}}} = 10.49 \text{ } \Omega \quad Q_{pk} = \frac{Z_0}{R_{ac}} = \frac{10.49}{13.70} = \mathbf{0.766}$$

$$n_T = n\sqrt{1 + \lambda_{act}} = 6.024 \times 1.2450 = \mathbf{7.500} = 15 : 2$$

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Step	Quantity	Eq.	Result
1	$P_{in}, P_{in,LLC}$	—	684.8 W, 670.9 W
2	$V_{ac,eq}$, low and high corner	18	173.2, 332.3 Vac
3	n, V_{refl}	21, 19	6.024, 150.6 V
4	M_{low}, M_{high}	12	1.2294, 0.6409
5	R_{ac}	20	13.70 Ω
6	$Q_{ZVS}, Z_{0,design}$	1	1.7019, 23.32 Ω
7	C_r, L_r	2	45.5 $\rightarrow$ 100 nF, 11.26 $\rightarrow$ 11 μ H
8	$\lambda_{req}, L_m, \lambda_{act}$	16	1.241, 20 μ H, 0.550
9	f_r, f_o, Q_{pk}, n_T	2, 3, 1	151.7 kHz, 90.4 kHz, 0.766, 7.500

Table 9: The design, step by step: the results of the nine steps. In a $\rightarrow$ b, a is the calculated value and b the part selected.

Steps 7 and 8 are the judgement calls: C_r 100 nF against a calculated 45.5, L_m 20 μ H against 8.9. Both lower Q_{pk} and hold λ , which is where the ZVS margin comes from.

No-load solution: none. The no-load gain of this tank bottoms out at $M_{\infty} = 0.6452$, and the high corner asks for 0.6409, below it. No frequency reaches that gain without load, and the margin is under one per cent, so no full-load check would show it. Burst mode owns that region (Section 4.10).

7.5 ZVS over the whole operating space

The closed form of Section 6.5 reads 292 ns on this tank against a swept 460 ns (36 % pessimistic); on another tank in the same family it is 23 % optimistic. Read with $\lambda_{act} = 0.550$ instead of the design $\lambda = 1.241$ it returns -471 ns, a capacitive answer for an inductive tank. This design is judged by the sweep.

The capacitive edge of the full-load curve ($\arg Z_{in} = 0$) is at 110.1 kHz; the gain peak is lower, at 105.4 kHz, with 8.4° of capacitive phase still there (Section 2.9). Neither is used for the verdict.

The worst point is not the corner the tank was designed at. Full load at the low equivalent input is where the *gain* requirement is set, and it gives $T_{ZC} = 460$ ns, a margin of 2.09. The smallest number in the table is **401 ns at 25 % load and the 332.3 Vac equivalent corner**, a margin of 1.82 against $t_D = 220$ ns. That is the number this design is held to.

Why light load at high input voltage can be the ZVS corner. The charge that swings the bridge node is carried by the magnetising current, and its peak goes as $V_{in}/(4f_{sw}L_m)$. Shed load and the controller raises f_{sw} to cut the gain, so that current falls while the node still has the same capacitance to move. Sweeping only the design corner therefore reports a margin that is not there.

7.6 The gain chart of this design

Figure 39 is the chart of Section 4.5 on this tank ($\lambda_{act} = 0.550$, $Q_{pk} = 0.766$), one panel per input voltage: the two morphing edges and the four mains voltages a supply meets. The curves are the same in every panel; only the required-gain lines move. The marked crossings are the operating points the sweep is built from.

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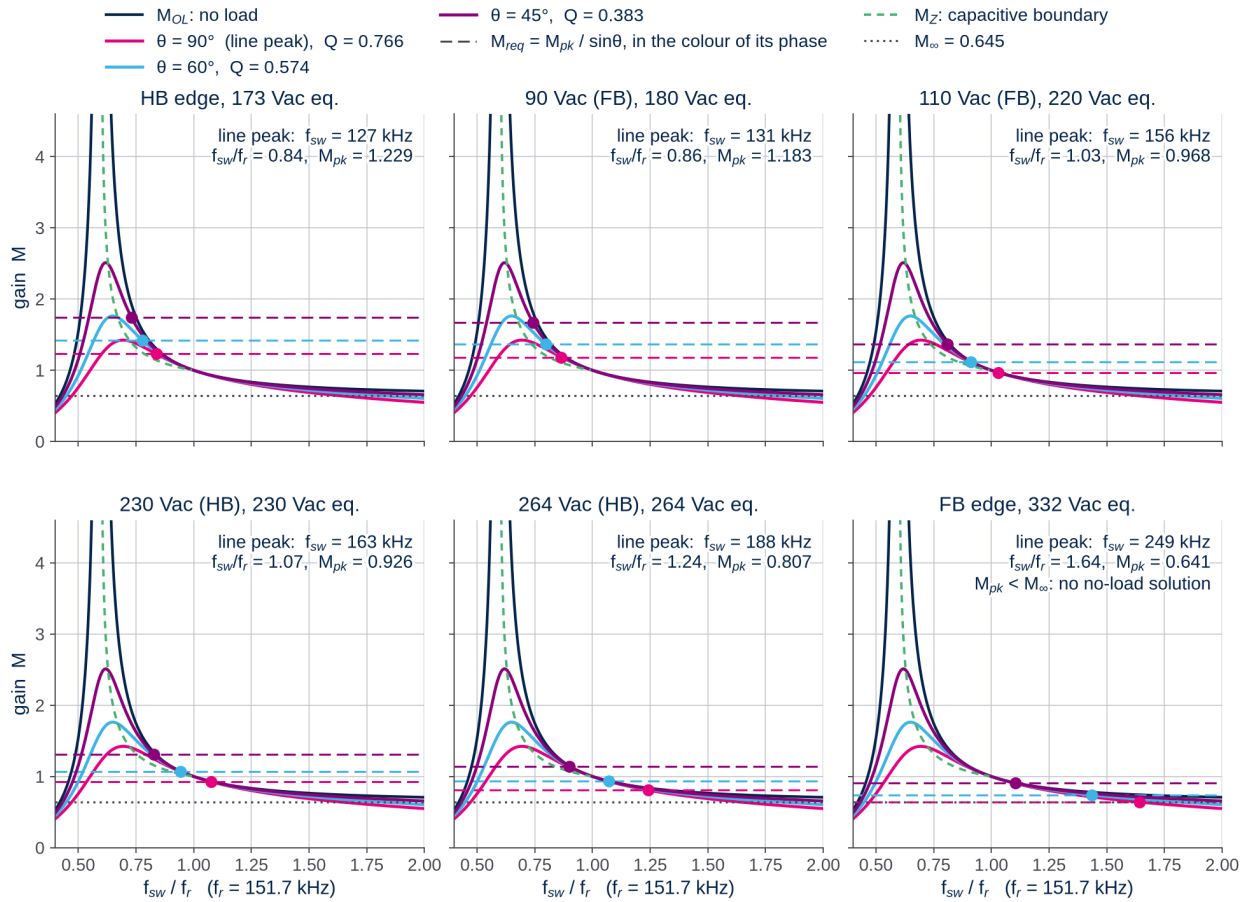


Figure 39: The gain chart of this design at the six input voltages, low to high. Top left, the HB edge (173 Vac eq.): highest required gain, crossings below f_r . Bottom right, the FB edge (332 Vac eq.): lowest required gain, crossings above f_r . 90 Vac in full bridge lands close to the HB edge, and 110 Vac in full bridge close to 230 Vac in half bridge, so the two mains systems look alike to the tank. Each phase is compared with the dashed line of its own colour only.

Input voltage	θ	$Q(\theta)$	$M_{req}(\theta)$	f_{sw}/f_r	f_{sw}
HB edge, 173 Vac eq.	90°	0.766	1.229	0.838	127.1 kHz
	60°	0.574	1.420	0.778	118.1 kHz
	45°	0.383	1.739	0.731	110.9 kHz
90 Vac (FB), 180 Vac eq.	90°	0.766	1.183	0.865	131.3 kHz
	60°	0.574	1.366	0.796	120.8 kHz
	45°	0.383	1.673	0.742	112.5 kHz
110 Vac (FB), 220 Vac eq.	90°	0.766	0.968	1.030	156.3 kHz
	60°	0.574	1.118	0.912	138.3 kHz
	45°	0.383	1.369	0.810	122.9 kHz
230 Vac (HB), 230 Vac eq.	90°	0.766	0.926	1.075	163.1 kHz
	60°	0.574	1.069	0.944	143.3 kHz
	45°	0.383	1.310	0.829	125.8 kHz

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Input voltage	θ	$Q(\theta)$	$M_{\text{req}}(\theta)$	f_{sw}/f_r	f_{sw}
264 Vac (HB), 264 Vac eq.	90°	0.766	0.807	1.242	188.5 kHz
	60°	0.574	0.932	1.071	162.5 kHz
	45°	0.383	1.141	0.901	136.8 kHz
FB edge, 332 Vac eq.	90°	0.766	0.641	1.641	249.0 kHz
	60°	0.574	0.740	1.435	217.8 kHz
	45°	0.383	0.906	1.106	167.8 kHz

Table 10: The marked crossings of Figure 39, at every input voltage. $Q = Q_{\text{pk}} \sin^2\theta$ and $M_{\text{req}} = M_{\text{pk}}/\sin\theta$; M_{pk} scales as $1/V_{\text{eq}}$, from 1.229 at the HB edge to 0.641 at the FB edge.

Four readings matter:

- **The HB edge sets the tank.** 173 Vac equivalent is the lowest voltage the tank sees, so its line-peak requirement $M_{\text{pk}} = 1.229$ is the largest gain ever asked for. It is met at $f_{\text{sw}}/f_r = 0.838$, well to the right of M_z : inductive.
- **The FB edge sets the frequency.** At 332 Vac equivalent the line-peak crossing is 249 kHz, and that, not the mains maximum, is what the oscillator ceiling must clear (Section 7.16).
- **The four mains voltages sit between the edges.** 90 Vac in full bridge (180 Vac eq.) is only 4 % above the HB edge, so a 90 Vac system runs close to the gain worst case. 110 Vac in full bridge (220) and 230 Vac in half bridge (230) are within 5 % of each other: morphing makes the two mains systems nearly alike to the tank. 264 Vac (264) is still 21 % below the FB edge.
- **The FB edge also judges λ .** The line-peak requirement there is 0.641 and the no-load floor $M_{\infty} = 0.645$. The requirement is **below** the floor, so at no load and high input voltage there is no frequency that satisfies it, and burst mode carries the converter (Section 4.10).

The six panels are six places on one axis, not six designs. The curves do not depend on input voltage: $M(f_n, Q)$ is fixed by λ_{act} and Q_{pk} . **The input voltage enters only through M_{req} .**

7.7 Which side of resonance this design runs on

With $n_T = 7.50$, at all but the lowest input voltages the tank spends part of every line cycle above f_r , where the secondary loses zero-current turn-off, so the rectifier body diode and the SR dead time need checking.

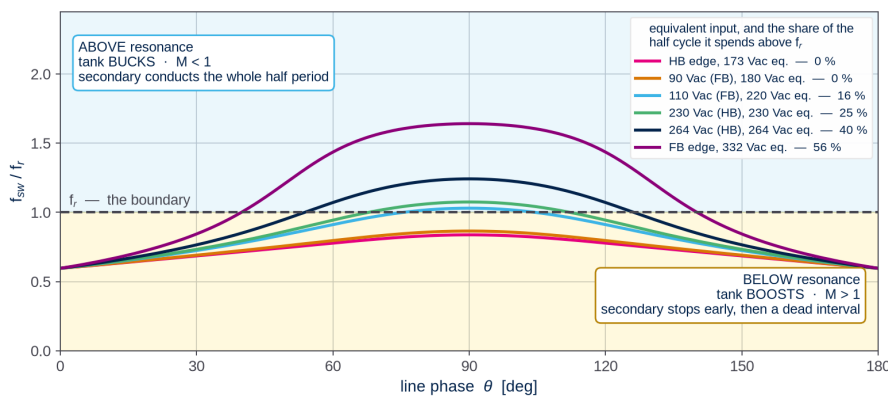


Figure 40: Which side of f_r the converter is on over a line half cycle, at the six input voltages. Every curve converges on f_0 at the zero crossing. The percentage in each legend entry is the fraction of the half cycle spent above f_r , without zero-current turn-off.

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Input voltage (bridge mode)	V _{eq} (Vac rms)	peak f _{sw}	side of f _r
HB edge, 173 Vac eq.	173 Vac	127.1 kHz	below
90 Vac (FB), 180 Vac eq.	180 Vac	131.3 kHz	below
110 Vac (FB), 220 Vac eq.	220 Vac	156.4 kHz	above
230 Vac (HB), 230 Vac eq.	230 Vac	163.2 kHz	above
264 Vac (HB), 264 Vac eq.	264 Vac	188.5 kHz	above
FB edge, 332 Vac eq.	332 Vac	249.0 kHz	above

Table 11: Peak f_{sw} over the half cycle against f_r = 151.7 kHz. 4 of the 6 input voltages cross into above-resonance operation for part of the cycle.

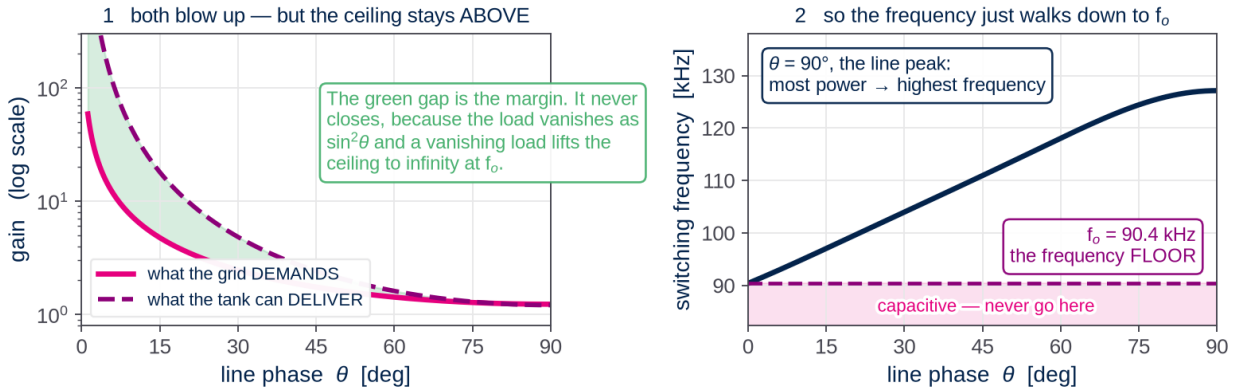


Figure 41: Near the zero crossing the required gain diverges and the load vanishes together, so the operating point converges on f_o = 90.4 kHz (Section 4.2).

7.8 The currents this design has to carry

Ratings come from the worst switching cycle and losses from the line-cycle rms, so both are listed. The peaks are read off one cycle at the low equivalent corner, line peak ($\theta = 90^\circ$ in Table 10). The composite peak is the peak of the sum, because the two components peak at different instants:

$$I_{Lr,pk} = \max_t |i_{Lm}(t) + i_{trafo}(t)| = 18.32 \text{ A} < 12.41 + 16.37 = 28.78 \text{ A}$$

The rms values are the rms of one switching period, averaged in I^2 over the line half cycle by the five-point Simpson rule:

$$I_{lc} = \sqrt{\frac{2}{\pi} \int_0^{\pi/2} I_{rms}^2(\theta) d\theta} \simeq \sqrt{\frac{1}{12} \sum_k w_k I_{rms}^2(\theta_k)}, \quad w = \{1, 4, 2, 4, 1\}$$

That is why the line-cycle primary rms, 10.93 A, is lower than the worst cycle's 13.76 A.

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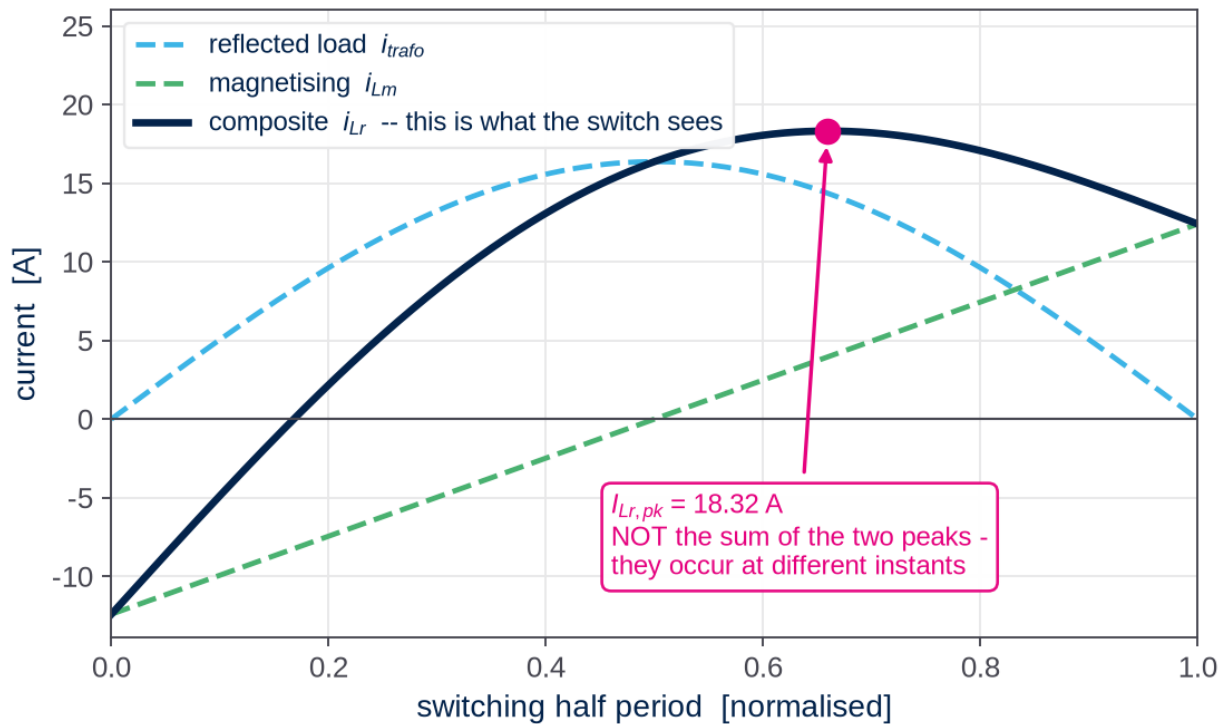


Figure 42: The composite tank current at the low equivalent corner, full load, and why its peak is not the sum of the two component peaks: they occur at different instants.

Quantity	Value	Where it is used
f_{sw}	127.1 kHz	the switching cycle the peaks below are read in
$I_{sec,pk}$	98.6 A	secondary rectifier peak
$I_{trafo,pk}$	16.37 A	reflected load component
$I_{Lm,pk}$	12.41 A	magnetising component
composite tank peak	18.32 A	primary device rating, OCP margin
$I_{pri,rms}$ (worst cycle)	13.76 A	primary rms rating
I_{pri} (line-cycle rms)	10.93 A	primary conduction loss
I_{rect} per leg (line-cycle rms)	28.34 A	secondary conduction loss
I_{Cout} (line-cycle rms)	30.24 A	output bank ripple current

Table 12: Currents at the low equivalent corner (HB edge), full load. Peaks and worst-cycle rms are taken in the switching cycle at the line peak, $\theta = 90^\circ$; line-cycle rms values are over the whole line half cycle.

7.9 The transformer, as built

The tank asks for $L_r = 11 \mu\text{H}$, $L_m = 20 \mu\text{H}$ and $n_T = 7.50$. The transformer is one part: a 15-turn primary and two 2-turn secondaries, NS2 and NS3, joined at the centre tap. Several units in series (Section 6.21) are not needed here.

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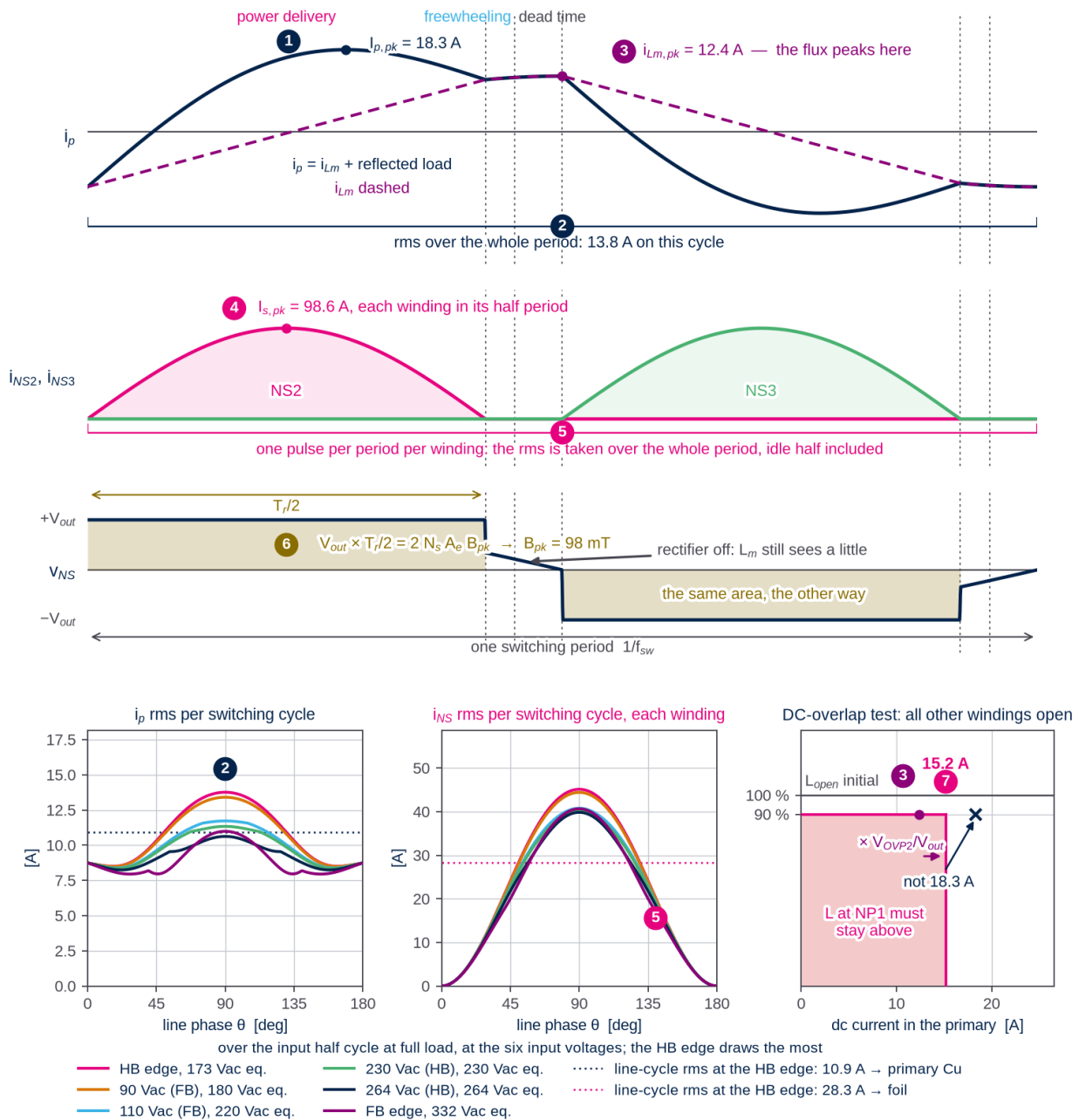


Figure 43: Where each number is read. Top: primary current, the two secondary winding currents and the secondary winding voltage over one switching period at the worst cycle: line peak at the HB edge (173 Vac equivalent), full load. Lower left and centre: the rms of the two winding currents over the input half cycle at the six input voltages; the HB edge draws the most in both windings, which is why the table is read there. Lower right: the DC-overlap bench test. The circled marks are the rows of Table 4.

Quantity	Value	Note
Turns	N_p 15 T; NS2 2 T, NS3 2 T; NAUX 3 T	$n_T = 15 / 2 = 7.5$; the auxiliary is ZCD sense only
Open-circuit inductance	31.0 μ H	$L_r + L_m$; across NP1, every other winding open
Short-circuit inductance	11.0 μ H	this is L_r , no separate resonant inductor; across NP1, NS2 and NS3 shorted, NAUX open

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Quantity	Value	Note
A_L	138 nH	open-circuit inductance / N_p^2 ; the core is gapped to it

Table 13: The transformer as wound and specified.

Pins. The B66368 coil former of the ETD 49/25/16DG core has 20 pins in two rows of 10. NP1 and the ZCD auxiliary take one row, NS2 and NS3 the other, two pins per secondary terminal. The centre tap is made on the board.

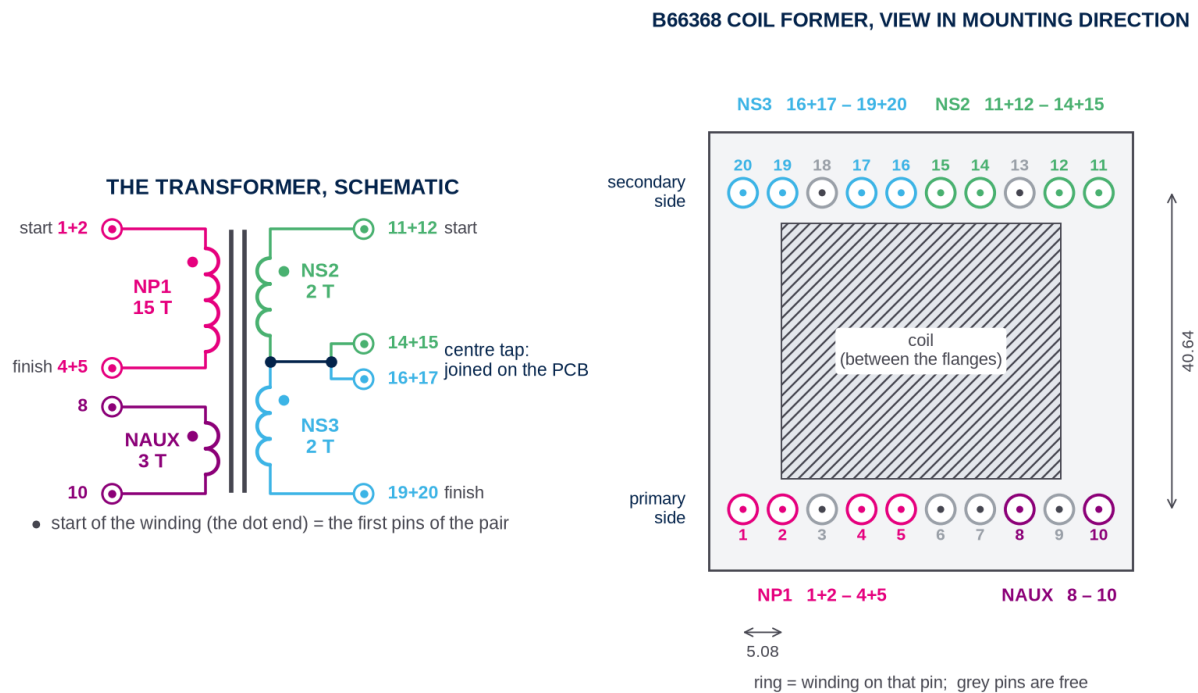


Figure 44: The schematic symbol with its pin numbers, and the same pins on the B66368 coil former (20 pins, pitch 5.08 mm, rows 40.64 mm apart). A ring in a winding's colour marks its pins; grey pins are free. The dot end is the first pin of each pair. The datasheet prints no pin numbers; the numbering shown counts along one row from pin 1 and back along the other, the usual convention. Confirm it against the bobbin drawing before the specification is released; the assignment does not depend on it, only the printed numbers do.

Winding	Pins (start – finish)	Turns	Conductor	Row
NP1 primary	1+2 – 4+5	15 T	Litz 310 × Ø0.10 mm	front row 1-10
NAUX auxiliary (ZCD)	8 – 10	3 T	any wire, sense only	front row 1-10
NS2 secondary	11+12 – 14+15	2 T	foil 0.20 × 10.5 mm, 3 in parallel	back row 11-20
NS3 secondary	16+17 – 19+20	2 T	the same foil, on top of NS2	back row 11-20
Centre tap	14, 15, 16 and 17, joined on the PCB	—	—	back row 11-20
Free	3, 6, 7, 9, 13, 18	—	not connected	both rows

Table 14: Winding-to-pin assignment.

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Polarity. With the dots as drawn, ZCD is positive while the low-side switch of leg 1 is on, as required. NS2 and NS3 are wound in the same sense; the tap joins the finish of NS2 to the start of NS3. Three more numbers follow from Table 13.

Peak flux density, at f_r :

Equation 24, repeated

$$B_{pk} = \frac{V_{out, eff}}{4 f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4 f_r N_s B_{max}}$$
$$B_{pk} = \frac{25.0 \text{ V}}{4 \times 151.75 \text{ kHz} \times 2 \times 211.0 \text{ mm}^2} = \mathbf{98 \text{ mT}}$$
$$A_e \geq \frac{25.0 \text{ V}}{4 \times 151.75 \text{ kHz} \times 2 \times 0.20 \text{ T}} = \mathbf{103 \text{ mm}^2}$$

Magnetising peak, over L_μ (it equals $i_{\mu, pk}$):

Equation 32, repeated

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_\mu}$$
$$I_{eq} = \frac{0.0976 \text{ T} \times 15 \times 211.0 \text{ mm}^2}{24.90 \text{ } \mu\text{H}} = \mathbf{12.41 \text{ A}} = i_{\mu, pk}$$

Saturation test current, the magnetising peak at the OVP2 output voltage:

Equation 33, repeated

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out, eff}}$$
$$I_{sat} = 12.41 \text{ A} \times \frac{30.61 \text{ V}}{25.0 \text{ V}} = 12.41 \times 1.2244 = \mathbf{15.2 \text{ A}}$$

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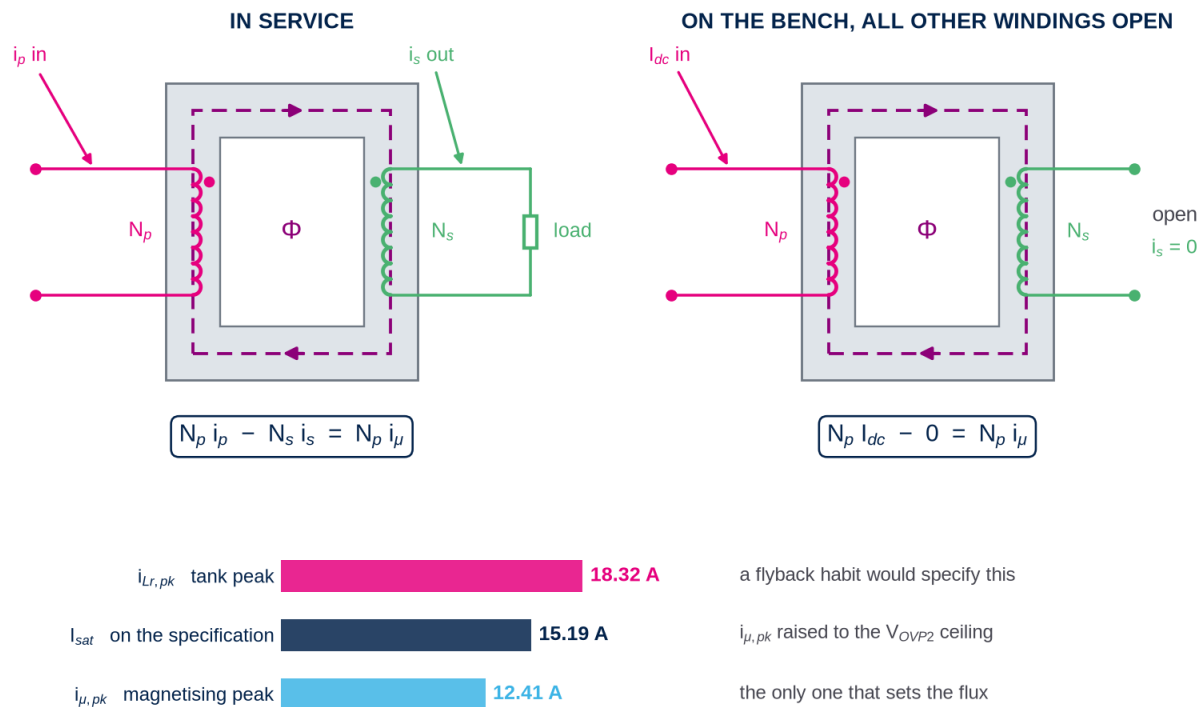


Figure 45: In service the secondary cancels most of the primary ampere-turns; with every other winding open, the whole test current magnetises the core. The design flux is therefore reached at 12.41 A on the bench, not at the 18.32 A tank peak.

Why OVP2, in this design's numbers. OVP1 is set at 28.16 V, 1.126 times the output, and OVP2 at 30.61 V, 1.224 times. At OVP1 the controller keeps switching, so the flux can reach 110 mT with the core still driven; at OVP2 it stops, and 120 mT is the largest flux this core is ever asked to carry. That is the number to hold against the hot B_s of N95 from the material data. The line on the specification reads: DC overlap, inductance at least 90 % of initial at 15 A, normal temperature (Table 17).

How to run it on this part. LCR meter across the primary (NP1, the pins of Table 14) with both secondaries, the centre tap and the auxiliary winding open. Small signal 100 kHz, 1 V, the same as the leakage test; bias current stepped from zero to past 15 A, inductance recorded at each step. Pass if L at 15 A is at least 0.9 of L at zero, where L at zero is $L_{open} \approx 31.0 \mu H$. Keep each step short: the dc heats the winding through its resistance, and a warm core saturates earlier. A part whose curve knees below 15 A has too little gap or the wrong core, even when L_{open} is on target; the margin against a hot core is in B_{pk} , not in the test.

If there were no OVP2. The current limit would be the only ceiling: OCP1 trips at 22.92 A, 1.51 times the OVP2-based 15.2 A. At the same L_μ that is 180 mT on this core, which has about twice the area the flux target needs; on a core sized to the 0.20 T target the same ratio would ask for 0.30 T, and the core area or N_s would have to grow by that ratio. The rule of thumb of 1.3 times $i_{\mu,pk}$ would give 16.1 A, near the OVP2 figure by coincidence and with no event behind it.

7.10 The core and the winding

The core is TDK **ETD 49/25/16DG**, N95 (core B66367, coil former B66368), centre leg ground to $A_L = 138$ nH, about 1.1 mm of gap. Figure 46 shows the section and the winding; Table 15 names each item and Table 16 gives the arithmetic.

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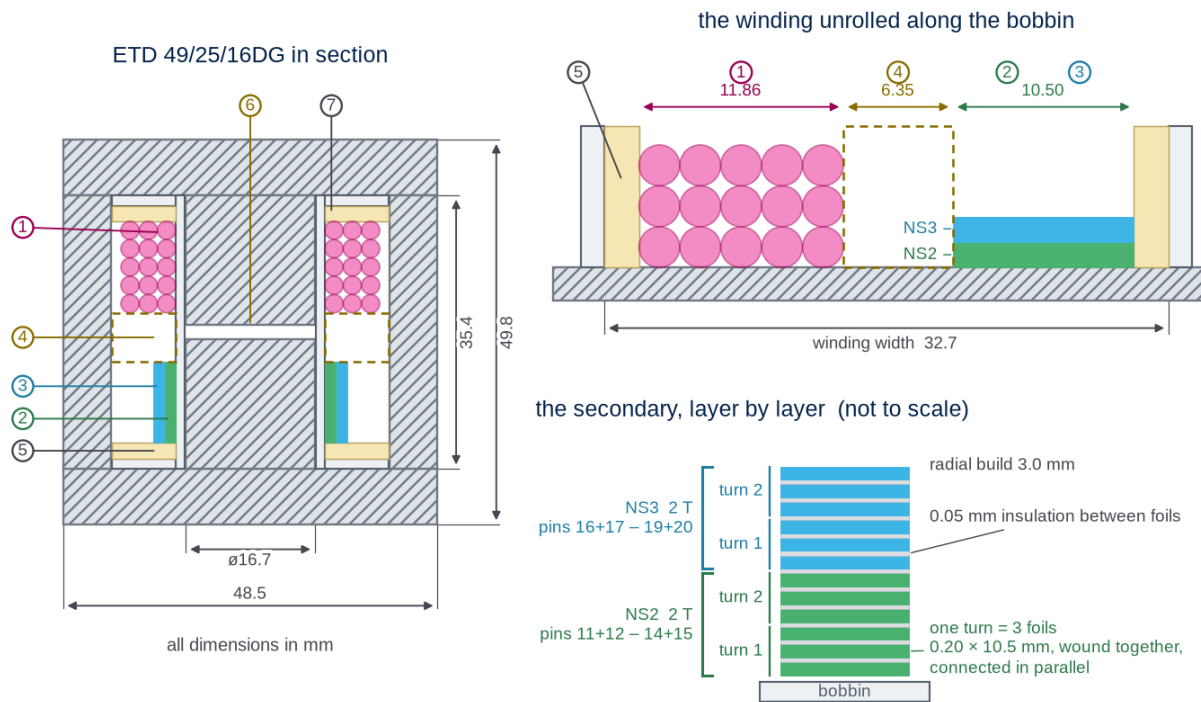


Figure 46: The core in section (left, to scale), the winding unrolled along the bobbin (top right, to scale) and the secondary layer by layer (bottom right, not to scale). Primary and secondary sit side by side with 6.35 mm between them. The circled numbers are the rows of Table 15.

Mark	Item	Turns	Conductor, placement
1	NP1 primary, pins 1+2 – 4+5	15 T	Litz 310 × Ø0.10 mm (bundle Ø2.37 mm), 3 layers of 5/5/5 turns
2	NS2 secondary, pins 11+12 – 14+15	2 T	foil 0.20 × 10.5 mm, 3 foils in parallel per turn; wound directly on the bobbin
3	NS3 secondary, pins 16+17 – 19+20	2 T	the same foil, on top of NS2
4	Separation	—	6.35 mm between primary and secondary; sets L_{short}
5	Margin tape	—	2.0 mm at each flange
6	Centre-leg gap	—	about 1.1 mm in total, ground to $A_L = 138 \text{ nH}$
7	Window	—	$A_N = 269 \text{ mm}^2$, 62 mm ² of copper in it
—	NAUX auxiliary, pins 8 – 10	3 T	any wire; ZCD sense only

Table 15: Items of the figure.

Three numbers in Table 16 are not simple divisions and are worked here first.

Skin depth at f_r , with copper at 100 °C ($\rho = 2.26 \times 10^{-8} \Omega\text{m}$):

Equation 28, repeated

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}}$$

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$$\delta = \sqrt{\frac{2.26 \times 10^{-8}}{\pi \cdot 151.7 \times 10^3 \cdot 4\pi \times 10^{-7}}} = 0.194 \text{ mm}$$

A conductor thicker than $2\delta = 0.39 \text{ mm}$ carries no more current, so the primary is Litz. The standard strand safely inside that is $d_s = \text{ø}0.10 \text{ mm}$, 3.9 times smaller than 2δ . Its copper area and the strand count are Equation 29:

Equation 29, repeated

$$a_s = \frac{\pi d_s^2}{4}, \quad n_s = \left\lceil \frac{A_{cu}}{a_s} \right\rceil$$

$$a_s = \frac{\pi \cdot (0.10)^2}{4} = 0.00785 \text{ mm}^2, \quad n_s = \left\lceil \frac{2.43}{0.00785} \right\rceil = 310$$

The served bundle's outside diameter, with $k_{litz} = 0.55$:

Equation 30, repeated

$$d_{litz} = \sqrt{\frac{4A_{cu}}{\pi k_{litz}}}$$

$$d_{litz} = \sqrt{\frac{4 \cdot 2.43}{\pi \cdot 0.55}} = 2.37 \text{ mm}$$

The secondary is foil. 6.30 mm^2 as round wire would be a bundle thicker than the whole build. At $t_f = 0.20 \text{ mm}$, Equation 31 gives

$$w_f = \frac{6.30}{0.20} = 31.5 \text{ mm}, \quad n_f = \left\lceil \frac{31.5}{12.0} \right\rceil = 3 \Rightarrow 3 \times 10.5 \text{ mm}$$

so one turn is **3 strips of 10.5 mm in parallel**, stacked radially.

Step	Primary NP1	Secondary NS2
Line-cycle rms current	10.93 A	28.34 A
Copper area at $J = 4.5 \text{ A/mm}^2$	$10.93 / 4.5 = \mathbf{2.43 \text{ mm}^2}$	$28.34 / 4.5 = \mathbf{6.30 \text{ mm}^2}$
Conductor ($\delta = 0.194 \text{ mm}$, Equation 28)	Litz of $\text{ø}0.10 \text{ mm}$ strands, each $a_s = 0.00785 \text{ mm}^2$ (Equation 29): $2.43 / 0.00785 = \mathbf{310 \text{ strands}}$, bundle $\text{ø}2.37 \text{ mm}$ (Equation 30)	foil 0.20 mm thick: $6.30 / 0.20 = 31.5 \text{ mm}$ wide, made as 3 strips of 10.5 mm in parallel
Turns	15, in 3 layers (5/5/5)	2
Width on the bobbin	$5 \times 2.37 = \mathbf{11.86 \text{ mm}}$	10.5 mm
Radial build	$3 \times 2.37 = 7.11 \text{ mm}$	NS2 + NS3: $12 \times (0.20 + 0.05) = 3.0 \text{ mm}$
Left between the windings	$32.7 - 2 \times 2.0 - 11.86 - 10.5 = \mathbf{6.35 \text{ mm}}$	

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Step	Primary NP1	Secondary NS2
Copper in the window	$15 \times 2.43 + 4 \times 6.30 = 61.6 \text{ mm}^2$, 76 % of A_N at $k_u = 0.30$	

Table 16: The winding, from current to copper. NS3 is the same as NS2.

Reduced to what a supplier can measure, this is the specification sheet. L_μ is deliberately absent (Section 6.10).

Item	Value	Condition
Core	ETD 49/25/16DG, N95, A_L 138 nH	core B66367, coil former B66368; distributed gap
Turns	N_p 15 T; NS2 2 T and NS3 2 T; NAUX 3 T	two secondary windings, centre tap outside the part
Open-circuit inductance	31.0 μH , -4.1 % at worst	measured across NP1; every other winding (NS2, NS3, NAUX) open; LCR meter 100 kHz, 1 V
Leakage inductance	11.0 $\mu\text{H} \pm 10 \%$	measured across NP1; NS2 and NS3 both shorted, NAUX open; LCR meter 100 kHz, 1 V
DC overlap	$\geq 90 \%$ of initial inductance	measured across NP1, every other winding open; dc 15 A overlapped on the 100 kHz, 1 V signal; normal temperature (mark 7)
Primary current	10.9 A rms / 18.3 A pk	line-cycle rms (mark 2) and composite peak (mark 1), both at the HB edge (173 Vac eq.), full load
Secondary current, each winding	28.3 A rms / 99 A pk	line-cycle rms (mark 5) and peak (mark 4), both at the HB edge, full load
Core area	$A_e \geq 103 \text{ mm}^2$	holds B_{pk} at or below 0.20 T at f_r (mark 6)
Switching frequency	127 to 249 kHz	at the line peak, full load, from the HB edge to the FB edge; near the zero crossings it falls towards $f_o = 90 \text{ kHz}$

Table 17: Transformer specification for the worked design.

7.11 The output bank, as sized

Both conditions of Section 6.23; the larger wins. **Ripple**, with Δv as a fraction:

Equation 35, repeated

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2}$$

$$C_{out} \geq \frac{657.5 \text{ W}}{2\pi \times 47 \text{ Hz} \times 0.05 \times 625 \text{ V}^2} = \mathbf{71.25 \text{ mF}}$$

Hold-up, starting half a ripple below V_{out} :

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Equation 36, repeated

$$C_{out} \geq \frac{2P_{out}T_{hold}}{\left(V_{out} - \frac{1}{2}\Delta v_{pp}\right)^2 - V_{o,min}^2}$$

$$C_{out} \geq \frac{2 \times 657.5 \text{ W} \times 0.012 \text{ s}}{24.41^2 - 19^2} = \mathbf{67.22 \text{ mF}} \quad (\text{started at } V_{out} : 59.77 \text{ mF})$$

Which wins, at $k = V_{o,min}/V_{out} = 0.760$:

Equation 37, repeated

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold}$$

$$0.373 > 0.354 \Rightarrow \text{ripple decides}$$

The two sides are only 1.060 apart; at 10 % ripple they would swap. The bank fitted is the next assembly above 71.2 mF: $470 \mu\text{F} \times 160 = 75.2 \text{ mF}$.

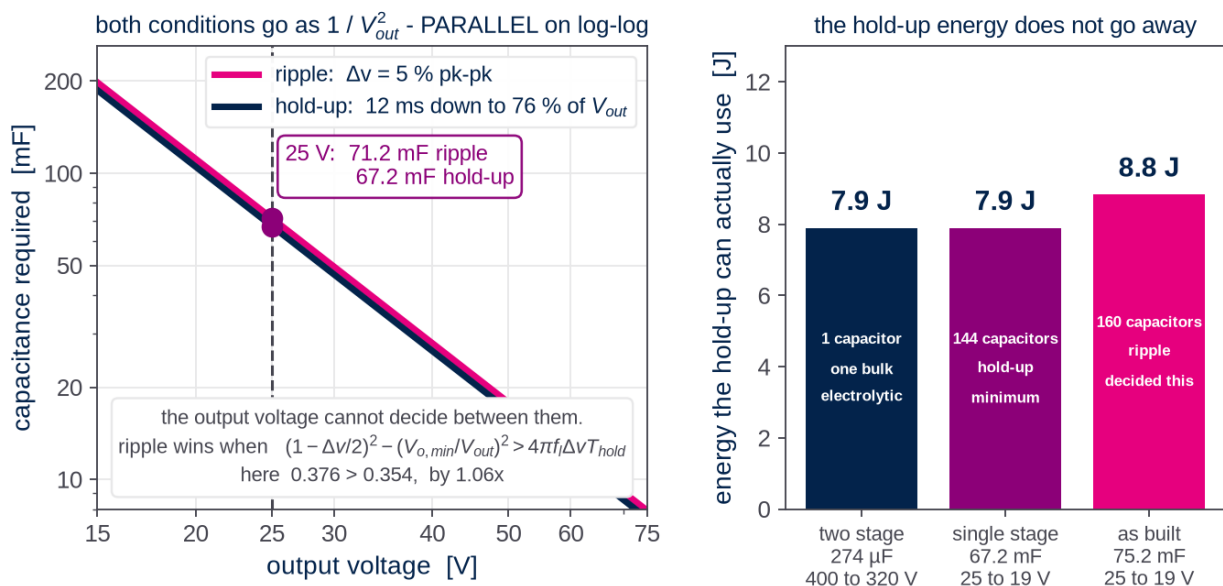


Figure 47: Left: both conditions fall as $1/V_{out}^2$; at 25 V ripple asks for 71.2 mF and hold-up for 67.2 mF. Right: the same 7.9 J of hold-up energy is a 274 μF part on a 400 V bus and 67.2 mF here, 245 times more. That ratio is the price of the architecture.

Quantity	Value	Against
Achieved ripple	1.18 V (4.74 %)	5 % allowed
Achieved hold-up	13.42 ms	12 ms required at 100 Vac, from the ripple trough down to 19 V — 15.10 ms if started at V_{out} instead of at the trough

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Quantity	Value	Against
Ripple current, switching component	23.8 A rms	—
Ripple current, $2f_i$ component	18.6 A rms	orthogonal to the above
Ripple current, total	30.24 A rms	0.19 A in each of 160 parts
Bank ESR	0.1875 mΩ	30 mΩ each, 160 in parallel
Ripple and noise	17.4 mV	with the 10 μF ceramic bypass

Table 18: What the selected bank then delivers, and what it has to survive.

7.12 Where the power goes

The specification allowed 27.3 W of loss, split three ways before any component existed. The device losses computed afterwards need not agree with that split, and here they do not.

Item	Value	Note
Budget — input bridge	5.52 W	synchronous bridge assumed
Budget — EMI filter	8.34 W	—
Budget — LLC stage	13.42 W	$\eta_{HB} = 98\%$ of $P_{in,LLC}$
Primary conduction, standing device	9.65 W	the half-bridge leg that never switches — the single largest item
Primary switching, three positions	4.83 W	not a problem
Secondary rectifiers	5.65 W	1.411 W per device, 2 in parallel per leg
Current-sense resistors	2.86 W	line-cycle average; 4.54 W at the worst switching cycle
Output capacitor ESR	0.162 W	—
Itemised total	23.15 W	and the magnetics are not in it

Table 19: The budget as assumed, and the device losses as computed. The second block is not a breakdown of the first.

The standing primary device, the low-side switch of the idle leg in half-bridge morphing, carries the whole line-cycle rms continuously, with $R_{DS(on)}$ **hot** at $k_T = 1.8$ times the 25 °C 45 mΩ (Section 6.24):

$$P_{cond} = I_{pri}^2 k_T R_{DS(on), 25} = (10.93)^2 \cdot 1.8 \cdot 45 \times 10^{-3} = 9.65 \text{ W}$$

Each secondary rectifier carries the leg current over the 2 devices in parallel, with $k_T = 1.9$ on 3.7 mΩ:

$$P_{SR} = \left(\frac{28.34}{2} \right)^2 \cdot 1.9 \cdot 3.7 \times 10^{-3} = 1.411 \text{ W per device}$$

The efficiency assumption is optimistic. The itemised device losses are 23.2 W against the 13.4 W that $\eta_{HB} = 98\%$ allows, before the transformer. Nothing electrical depends on it; the thermal design does, and that is settled by measurement.

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7.13 What the semiconductors have to be

Requirements, not part numbers (Section 6.24).

Item	Requirement
Primary drain-source voltage	$\geq 485 \text{ V}$, so a 600 V class part
Primary body diode	fast recovery: in full bridge the body diode conducts before turn-on
Primary $R_{DS(on)}$ per device	$\leq 25.2 \text{ m}\Omega$ hot to meet the loss budget
Primary $C_{o(tr)}$	small enough that the tank can swing the node inside 220 ns
Secondary drain-source voltage	$\geq 80 \text{ V}$ including the centre-tap doubling
Secondary rectifier current	28.34 A rms per leg, 98.6 A peak
Secondary package	check the lead and clip rating, not only the die

Table 20: Semiconductor requirements for the worked design.

7.14 The voltage loop, as built

The method of Sections 6.26 to 6.33, with this converter's numbers, in the order they are found.

Step 1 — the output divider, with $R_I = 100 \text{ k}\Omega$ chosen:

Equation 43, repeated

$$V_{out} = V_R \left(1 + \frac{R_I}{R_O} \right)$$

$$R_O = \frac{V_R R_I}{V_{out} - V_R} = \frac{2.495 \text{ V} \times 100 \text{ k}\Omega}{25.0 \text{ V} - 2.495 \text{ V}} = 11.09 \text{ k}\Omega \rightarrow \mathbf{11 \text{ k}\Omega}$$

$$V_{out} = 2.495 \text{ V} \times \left(1 + \frac{100}{11} \right) = \mathbf{25.18 \text{ V}}$$

Step 2 — the plant. $V_{FB} = K_{pwr} R_{CS} P_{in,LLC} = 0.167 \times 24 \text{ m}\Omega \times 670.9 \text{ W} = 2.689 \text{ V}$ (Equation 15). Then

Equation 39, repeated

$$G_{plant}(s) = \frac{v_{out}(s)}{v_{FB}(s)} = \frac{G_o}{s}, \quad G_o = \frac{P_{out}}{V_{out} V_{FB} C_{out}} \quad [\text{rad/s}]$$

$$G_o = \frac{657.5 \text{ W}}{25.0 \text{ V} \times 2.689 \text{ V} \times 75.2 \text{ mF}} = \mathbf{130.1 \text{ rad/s}}, \quad f_{cto} = \frac{130.1}{2\pi} = 20.7 \text{ Hz}$$

Step 3 — the gain allowed at $2f_p$, for 5 % third harmonic:

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Equation 40, repeated

$$\Delta V_{loop} = \frac{P_{out}}{V_{out}} \frac{1}{2\pi f_l C_{out}}$$
$$\Delta V_{loop} = \frac{657.5 \text{ W}}{25.0 \text{ V}} \cdot \frac{1}{2\pi \times 47 \text{ Hz} \times 75.2 \text{ mF}} = \mathbf{1.184 \text{ V}}$$

Equation 42, repeated

$$G_{EA}(2f_l) \leq \frac{4 V_{FB} D_3}{\Delta V_{loop}}$$
$$G_{EA}(2f_l) \leq \frac{4 \times 2.689 \text{ V} \times 0.05}{1.184 \text{ V}} = \mathbf{0.4541}$$

Step 4 — the K factor, for a target phase margin of 50° (tan 50° = 1.1918) and $\alpha_v = 1.2$:

Equation 48, repeated

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right]$$
$$K_v = \frac{1}{2 \times 1.2} \left[(1 + 1.2^2) \times 1.1918 + \sqrt{(1 + 1.2^2 \times 1.1918)^2 + 4 \times 1.2^2} \right] = \mathbf{2.7218}$$

Step 5 — the compensator gain constant:

Equation 49, repeated

$$EA_o = \frac{2\pi (2f_l) G_{EA}(2f_l)}{K_v^2}$$
$$EA_o = \frac{2\pi \times 94 \text{ Hz} \times 0.4541}{2.7218^2} = \mathbf{36.20 \text{ rad/s}}$$

Step 6 — zero and pole, with $\Gamma_v = 0.744 \times 264.0 / 173.24 = 1.1338$:

Equation 50, repeated

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v EA_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v}$$
$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{130.06 \times 2.7218 \times 36.20}{1.1338}} = \mathbf{16.92 \text{ Hz}}$$

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$$f_p = 2.7218 \times 16.92 = \mathbf{46.06 \text{ Hz}}, \quad f_z = \frac{16.92}{2.7218} = \mathbf{6.217 \text{ Hz}}$$

Step 7 — the bias parts, with the TL431 minimum current 0.8 mA, $V_Z = 12 \text{ V}$, $V_{Fo} = 1.15 \text{ V}$, $I_{FB,steady} = 150 \text{ }\mu\text{A}$, $I_{FB,max} = 300 \text{ }\mu\text{A}$, $CTR_s = 0.37$, $CTR_m = 0.53$:

Equation 44, repeated

$$R_P \leq \frac{V_{Fo}}{I_{min}}$$

$$R_P \leq \frac{1.15 \text{ V}}{0.8 \text{ mA}} = 1.437 \text{ k}\Omega \rightarrow \mathbf{1.3 \text{ k}\Omega} \text{ (rounded down)}$$

Equation 47, repeated

$$R_{B,max} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,steady}/CTR_s}$$

$$R_{B,min} = \frac{V_Z - (V_R + V_{Fo})}{V_{Fo}/R_P + I_{FB,max}/CTR_m}$$

$$R_{B,max} = \frac{12 - (2.495 + 1.15) \text{ V}}{1.15/1.3 \text{ mA} + 150 \text{ }\mu\text{A}/0.37} = \frac{8.355 \text{ V}}{0.885 \text{ mA} + 0.405 \text{ mA}} = \mathbf{6.48 \text{ k}\Omega}$$

$$R_{B,min} = \frac{8.355 \text{ V}}{0.885 \text{ mA} + 300 \text{ }\mu\text{A}/0.53} = \mathbf{5.76 \text{ k}\Omega} \rightarrow R_B = \mathbf{6.2 \text{ k}\Omega}$$

Step 8 — the compensation parts, with $R_{FB} = 35 \text{ k}\Omega$, $C_{opto} = 3 \text{ nF}$ and the high-frequency pole placed at 1000 Hz:

Equation 51, repeated

$$C_{Fo} = \frac{f_z}{f_p} \cdot \frac{CTR_s R_{FB}}{R_I R_B E A_o}, \quad C_F = C_{Fo} \left(\frac{f_p}{f_z} - 1 \right)$$

$$R_F = \frac{1}{2\pi f_z C_F}, \quad C_{fx} = \frac{1}{2\pi f_{pHF} R_{FB}} - C_{opto}$$

$$C_{Fo} = \frac{6.217}{46.06} \cdot \frac{0.37 \times 35 \text{ k}\Omega}{100 \text{ k}\Omega \times 6.2 \text{ k}\Omega \times 36.20} = 77.9 \text{ nF} \rightarrow \mathbf{82 \text{ nF}}$$

$$C_F = 82 \text{ nF} \times \left(\frac{46.06}{6.217} - 1 \right) = 526 \text{ nF} \rightarrow \mathbf{560 \text{ nF}}$$

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$$R_F = \frac{1}{2\pi \times 6.217 \text{ Hz} \times 560 \text{ nF}} = 45.7 \text{ k}\Omega \rightarrow \mathbf{47 \text{ k}\Omega}$$

$$C_{fx} = \frac{1}{2\pi \times 1000 \text{ Hz} \times 35 \text{ k}\Omega} - 3 \text{ nF} = 1.547 \text{ nF} \rightarrow \mathbf{1.55 \text{ nF}}$$

Step 9 — what the standard parts give:

Equation 46, repeated

$$EA_o = \frac{CTR R_{FB}}{(C_F + C_{Fo}) R_I R_B} \text{ [rad/s]}, \quad f_z = \frac{1}{2\pi R_F C_F}$$

$$f_p = \frac{1}{2\pi R_F C_{ser}}, \quad C_{ser} = \frac{C_F C_{Fo}}{C_F + C_{Fo}}, \quad f_{px} = \frac{1}{2\pi R_{FB} (C_{opto} + C_{fx})}$$

$$EA_o = \frac{0.37 \times 35 \text{ k}\Omega}{(560 + 82) \text{ nF} \times 100 \text{ k}\Omega \times 6.2 \text{ k}\Omega} = \mathbf{32.53 \text{ rad/s}}$$

$$f_z = \frac{1}{2\pi \times 47 \text{ k}\Omega \times 560 \text{ nF}} = \mathbf{6.047 \text{ Hz}}$$

$$C_{ser} = \frac{560 \times 82}{560 + 82} \text{ nF} = 71.5 \text{ nF}, \quad f_p = \frac{1}{2\pi \times 47 \text{ k}\Omega \times 71.5 \text{ nF}} = \mathbf{47.34 \text{ Hz}}$$

$$f_{px} = \frac{1}{2\pi \times 35 \text{ k}\Omega \times (3 + 1.55) \text{ nF}} = \mathbf{999 \text{ Hz}}$$

Step 10 — crossover and phase margin. $G_o EA_o = 4231$, so the iteration starts at $\omega_c = 65.0 \text{ rad/s}$ and settles at

Equation 53, repeated

$$\omega_c \leftarrow \sqrt{G_o EA_o A(\omega_c)}, \quad \text{starting from } \omega_c = \sqrt{G_o EA_o}$$

$$\omega_c = \mathbf{110.4 \text{ rad/s}}, \quad f_c = \frac{110.4}{2\pi} = \mathbf{17.57 \text{ Hz}}, \quad |T(\omega_c)| = 1.0003 \text{ (check : 1)}$$

Equation 54, repeated

$$\Phi_M = \arctan \frac{\omega_c}{\omega_z} - \arctan \frac{\omega_c}{\omega_p} - \arctan \frac{\omega_c}{\omega_{px}}$$

$$\Phi_M = \arctan \frac{110.4}{38.0} - \arctan \frac{110.4}{297.5} - \arctan \frac{110.4}{6279} = 71.0^\circ - 20.4^\circ - 1.0^\circ = \mathbf{49.6^\circ}$$

Step 11 — the gain margin:

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Equation 55, repeated

$$f_{180} = \sqrt{f_p f_{px} - f_z(f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})|$$

$$f_{180} = \sqrt{47.34 \times 999 - 6.047 \times (47.34 + 999)} \text{ Hz} = \mathbf{202 \text{ Hz}}$$

$$|T(f_{180})| = 0.0196, \quad GM = -20 \log_{10} 0.0196 = \mathbf{34.2 \text{ dB}}$$

Step 12 — the third harmonic of the built loop, at $2f_l = 94 \text{ Hz}$ ($\omega = 590.6 \text{ rad/s}$):

$$G_{EA}(2f_l) = \frac{32.53}{590.6} \cdot \frac{\sqrt{1 + (590.6/38.0)^2}}{\sqrt{1 + (590.6/297.5)^2} \sqrt{1 + (590.6/6279)^2}} = \mathbf{0.3843} \quad (\leq 0.4541)$$

Equation 41, repeated

$$D_3 = \frac{G_{EA}(2f_l) \Delta V_{loop}}{4 V_{FB}}$$

$$D_3 = \frac{0.3843 \times 1.184 \text{ V}}{4 \times 2.689 \text{ V}} = \mathbf{4.23 \%} \quad (\text{budget } 5 \%)$$

Step 13 — the feedback ripple at the burst point, with burst entry at 75 W:

Equation 56, repeated

$$\Delta V_{FB} = \frac{P_{in, BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l)$$

$$\Delta V_{FB} = \frac{75 \text{ W}}{25.0 \text{ V}} \cdot \frac{1}{2\pi \times 47 \text{ Hz} \times 75.2 \text{ mF}} \times 0.3843 = \mathbf{51.9 \text{ mV}}$$

Equation 57, repeated

$$\Delta R_{BM} = \frac{\Delta V_{FB}}{2 \times 0.01 \text{ V/k}\Omega}$$

$$\Delta R_{BM} = \frac{51.9 \text{ mV}}{2 \times 0.01 \text{ V/k}\Omega} = \mathbf{2.60 \text{ k}\Omega}, \quad R_{BM} = 30 - 2.60 = \mathbf{27.4 \text{ k}\Omega}$$

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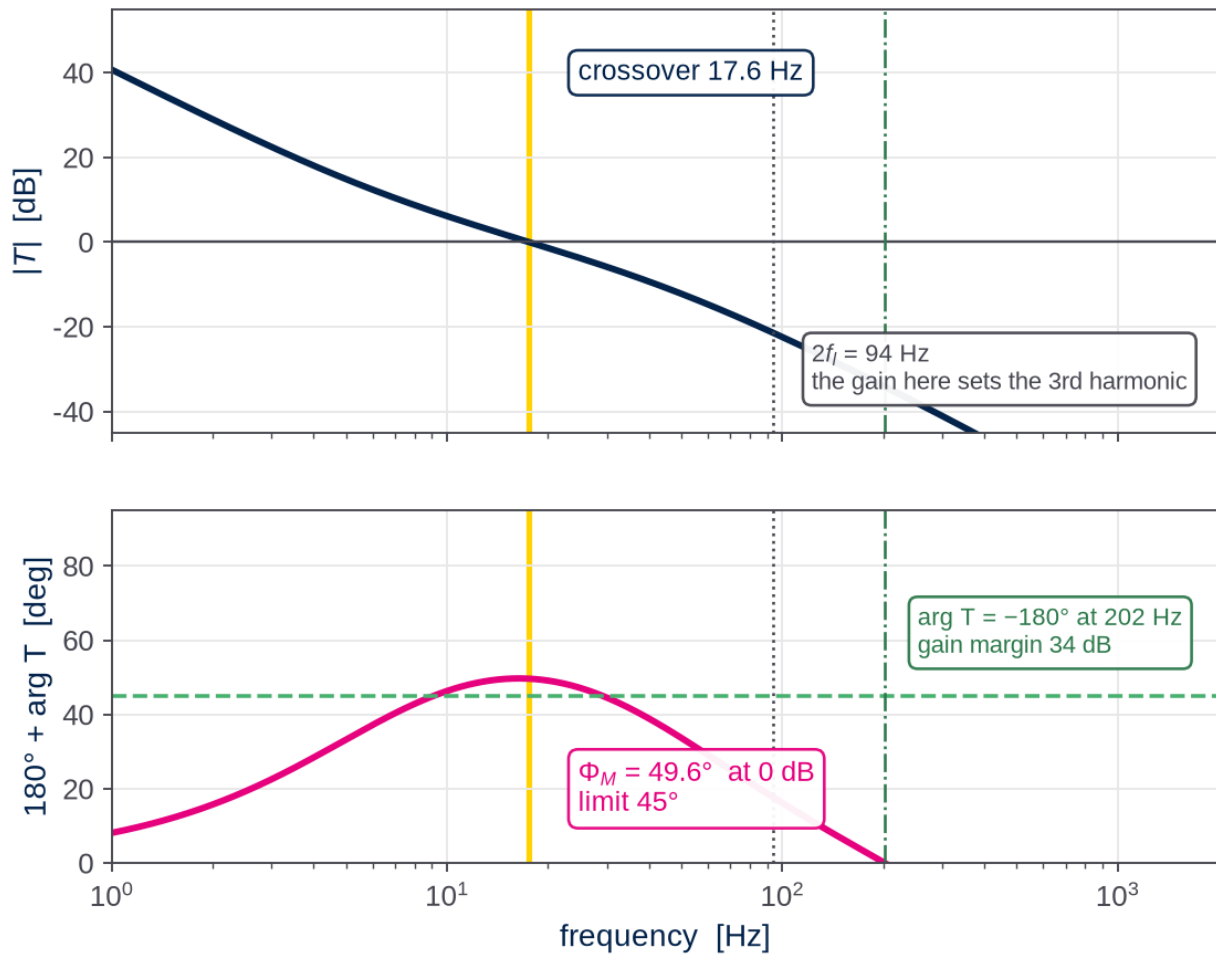


Figure 48: $|T|$ and $180^\circ + \arg T$ of the loop as built. The second trace is the phase margin only at the crossover; the gain margin is read at f_{180° .

Quantity	Value	Against
Divider R_I / R_O	100 / 11 k Ω	regulates to 25.18 V
Bias R_P, R_B	1.3 k Ω , 6.2 k Ω	$R_P \leq 1.44$ k Ω ; R_B in 5.76 to 6.48 k Ω
Compensation C_{F0}, C_F, R_F, C_{fx}	82 nF, 560 nF, 47 k Ω , 1.55 nF	zero 6.05 Hz, pole 47.3 Hz, pole 999 Hz
Crossover f_c	17.57 Hz	the 15 to 20 Hz this topology lands in
Phase margin	49.6°	target 50°, floor 45°
Gain margin	34.2 dB at 202 Hz	6 dB floor, 10 dB comfortable
Third harmonic on the input current	4.23 %	budget 5 %
Feedback ripple at the burst point	52 mV	clears the burst threshold if R_{BM} moves from the fitted 30 k Ω to 27.4 k Ω

Table 21: The voltage loop, as built.

7.15 The parts around the controller

The results; the following sections derive each value in the order it has to be done.

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Part	Value	Result
C_T / R_T	470 pF / 11 kΩ	f_{Min} 92.2 kHz, f_{Max} 319.6 kHz
R_{CS}	24.0 mΩ (120 mΩ × 5 in parallel)	OCP1 at 1.251 × the composite peak
R_{CFG}	30 kΩ	V_{BO} 120 V peak (84.8 Vac rms), morphing enabled
R_{BM}	30 kΩ	burst-mode entry point
ZCD divider	330 kΩ / 19 kΩ	OVP1 28.16 V, OVP2 30.61 V
C_{in}	2200 nF film	about 3.2 nF/W — there is no bulk capacitor
Compensation	82 nF / 560 nF / 47 kΩ / 1.55 nF	f_{cross} 17.57 Hz, Φ_M 49.64°

Table 22: Controller network for the worked design.

7.16 The oscillator: C_T first, then R_T

The VCO charges C_T from V_{ref}/R_T plus the error-amplifier current I_{EA} up to $V_{\text{ref}} = 1.5$ V, then adds a fixed idle time:

Equation 59

$$\frac{T_{\text{sw}}}{2} = \frac{C_T V_{\text{ref}}}{\frac{V_{\text{ref}}}{R_T} + I_{\text{EA}}} + T_{\text{idle}}$$

I_{EA} is the only control input. More current, higher frequency, less power. C_T is chosen first because it sets the span, R_T after it because it sets the floor:

Equation 60

$$C_{T, \text{max}} = \frac{I_{\text{EA}, \text{max}}}{2V_{\text{ref}}} \cdot \frac{(1 - 2T_{\text{idle}}f_{\text{sw}, \text{max}})(1 - 2T_{\text{idle}}f_{\text{sw}, \text{min}})}{f_{\text{sw}, \text{max}} - f_{\text{sw}, \text{min}}}, \quad C_{T, \text{min}} = \frac{1}{R_{T, \text{max}}} \left(\frac{1}{2f_{\text{sw}, \text{min}}} - T_{\text{idle}} \right)$$

Equation 61

$$R_{T, \text{ceil}} = \frac{1}{C_T} \left(\frac{1}{2f_{\text{sw}, \text{min}}} - T_{\text{idle}} \right)$$

with $f_{\text{sw}, \text{min}} = f_o = 90.4$ kHz, $f_{\text{sw}, \text{max}} = 225$ kHz, $I_{\text{EA}, \text{max}} = 400$ μA, $V_{\text{ref}} = 1.5$ V and $T_{\text{idle}} = 250$ ns:

$$C_{T, \text{max}} = \frac{400 \times 10^{-6}}{2 \cdot 1.5} \cdot \frac{(1 - 2 \cdot 250 \times 10^{-9} \cdot 225 \times 10^3)(1 - 2 \cdot 250 \times 10^{-9} \cdot 90.4 \times 10^3)}{(225 - 90.4) \times 10^3} = 703 \text{ pF}$$

and the floor condition, f_{Min} brought down to f_o with R_T at its 30 kΩ maximum, gives $C_{T, \text{min}} = 176$ pF. The datasheet allows 270 to 1000 pF, so the window is 270 to 703 pF, and the part is taken from its middle: $C_T = 470$ pF, C0G. Then

$$R_{T, \text{ceil}} = \frac{1}{470 \times 10^{-12}} \left(\frac{1}{2 \cdot 90.4 \times 10^3} - 250 \times 10^{-9} \right) = 11.24 \text{ kΩ} \rightarrow 11 \text{ kΩ}$$

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$R_{T,ceil}$ is a maximum. f_{Min} falls as R_T grows, so rounding up drops the clamp below f_o . Round it **down**.

The selected pair produces

Equation 62

$$f_{Min} = \frac{1}{2(C_T R_T + T_{idle})}, \quad f_{Max} = \frac{1}{2 \left(\frac{C_T}{\frac{I_{EA,max}}{V_{ref}} + \frac{1}{R_T}} + T_{idle} \right)}$$

$f_{Min} = 1/[2(470 \text{ pF} \times 11 \text{ k}\Omega + 0.25 \text{ }\mu\text{s})] = \mathbf{92.25 \text{ kHz}}$, which must clear f_o .

Quantity	Value	Limit	Margin
f_{Min}	92.2 kHz	above $f_o = 90.4 \text{ kHz}$	1.021
f_{Max}	319.6 kHz	above the highest operating $f_{sw} = 249.0 \text{ kHz}$ (FB edge, line peak)	1.283
f_{SU}	503.5 kHz	below the 675 kHz silicon ceiling	1.341
$R_T C_T$	5.17 μs	2.5 to 12 μs	2.068 / 2.321
R_T	11 k Ω	5 to 30 k Ω	2.200 / 2.727
C_T	470 pF	270 to 1000 pF	1.741 / 2.128

Table 23: Oscillator: what the two parts produce, and the limits each result has to clear.

T_{idle} is the weakest number in this chapter. The draft datasheet gives 700 ns in the text and a table that back-solves to 250 ns; this design uses 250 ns. At 700 ns f_{Min} lands on f_o and the floor margin is gone. Measure it first.

7.17 The current-sense resistor, which sets three things at once

Two conditions, the smaller wins:

Equation 63

$$R_{CS,max1} = \frac{16.8 \text{ }\Omega \cdot \text{W}}{P_{in}} \quad R_{CS,max2} = \frac{0.55 \text{ V}}{I_{Lr,pk}}$$

The first is the maximum-power law (Section 4.6), the second the OCP1 trip point:

$$R_{CS,max1} = \frac{16.8}{684.8} = 0.02453 \text{ }\Omega = 24.53 \text{ m}\Omega, \quad R_{CS,max2} = \frac{0.55}{18.32} = 0.03003 \text{ }\Omega = 30.03 \text{ m}\Omega$$

so the power law limits. Dissipation at the worst switching cycle fixes how many resistors share it:

$$P_{CS} = I_{pri,rms}^2 R_{CS} = (13.76)^2 \cdot 24.0 \times 10^{-3} = 4.54 \text{ W} \Rightarrow N = 5 \text{ in parallel}$$

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5 × 120 mΩ in parallel give $R_{CS} = 24.0 \text{ m}\Omega$.

R_{CS} sits in the bridge return, so $R_{CS,max2}$ uses the composite peak. $I_{trafo,pk} = 16.37 \text{ A}$ instead of $I_{Lr,pk} = 18.32 \text{ A}$ would loosen the over-current protection by about 12 %.

Result	Value	Against
Maximum input power	700.0 W	$P_{in} = 684.8 \text{ W}$
OCP1 trip	22.92 A	composite peak 18.32 A, margin 1.251
OCP2 trip	31.25 A	switching stops at once; restart after 50 μs at f_{SU}
Sense dissipation, worst switching cycle	4.54 W total, 0.909 W each	1 W parts, margin 1.100

Table 24: What the selected R_{CS} then fixes.

7.18 Burst mode: one resistor, read once at power-up

Equation 64

$$R_{BM} = 16.7 \frac{\text{k}\Omega}{\text{V}^2} R_{CS} P_{in, BM}, \quad V_{BM, eq} = \frac{R_{BM}}{100 \text{ k}\Omega} + 0.5 \text{ V}$$

Burst starts at $P_{in, BM} = 11 \% \times 684.8 \text{ W} = 75.3 \text{ W}$:

$$R_{BM} = 16.7 \cdot 24.0 \times 10^{-3} \cdot 75.3 = 30.1 \text{ k}\Omega \rightarrow 30 \text{ k}\Omega, \quad V_{BM, eq} = \frac{30}{100} + 0.5 = 0.800 \text{ V}$$

Valid range 15 to 140 kΩ; tied to ground, burst mode is off. Check this threshold against the feedback ripple (Section 6.33) or the converter chatters in and out of burst.

7.19 Brown-out and bridge configuration: the CFG pin

One resistor, two jobs, both read at power-up:

Equation 65

$$V_{BO, pk} = R_{CFG} \times 4 \frac{\text{V}}{\text{k}\Omega}, \quad V_{BO, rms} = \frac{V_{BO, pk}}{\sqrt{2}}$$

Brown-out is compared against the *peak* of the mains, so the $\sqrt{2}$ is easy to drop. Brown-out just below the lowest input voltage:

$$R_{CFG, max} = \frac{\sqrt{2} \cdot 90}{4} = 31.82 \text{ k}\Omega \rightarrow 30 \text{ k}\Omega$$

That is a maximum, so it rounds down; the selected part sets

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$$V_{BO, rms} = \frac{30 \cdot 4}{\sqrt{2}} = 84.85 \text{ V}_{rms}$$

against a 90 Vac minimum, margin 1.061. Rounding up would stop the converter starting at the lowest input voltage.

R _{CFG}	LOUT2	Configuration
15 kΩ	open	morphing, fixed brown-out: off below 60 V peak, on above 70 V peak
15 to 47 kΩ	open	morphing, adjustable brown-out — this design
47 to 100 kΩ	open	fixed full bridge
15 kΩ	to GND	fixed half bridge, split C _r , fixed brown-out
15 to 100 kΩ	to GND	fixed half bridge, split C _r , adjustable brown-out

Table 25: What R_{CFG} and LOUT2 select together. The 235 and 245 V thresholds are fixed inside the IC and cannot be moved.

For universal input the window is 15 kΩ to 31.8 kΩ; the 47 kΩ morphing limit never applies. Hang nothing on LOUT2: a pull-down under about 8 kΩ reads as fixed half bridge.

7.20 Output sensing and over-voltage: the ZCD divider

The divider ratio alone sets both over-voltage thresholds; the absolute values only set the bias current:

Equation 66

$$R_{ZCD, L} = \frac{2.3 \text{ V}}{I_{bias}}, \quad R_{ZCD, H} = R_{ZCD, L} \left(\frac{n_{aux}}{n_{sec}} \frac{V_{OVP1, out}}{2.3 \text{ V}} - 1 \right)$$

Equation 67

$$V_{OVP1} = \frac{2.3 \text{ V}}{n_{aux}/n_{sec}} \left(\frac{R_{ZCD, H}}{R_{ZCD, L}} + 1 \right), \quad V_{OVP2} = \frac{2.5}{2.3} V_{OVP1}$$

OVP1 10 % above the output is 27.50 V. The lower resistor comes from the bias current, 120 μA, the upper from the ratio, with $n_{aux}/n_{sec} = 1.5$:

$$R_{ZCD, L} = \frac{2.3}{120 \times 10^{-6}} = 19.17 \text{ k}\Omega \rightarrow 19 \text{ k}\Omega$$

$$R_{ZCD, H} = 19 \left(\frac{1.5 \cdot 27.5}{2.3} - 1 \right) = 321.8 \text{ k}\Omega \rightarrow 330 \text{ k}\Omega$$

The upper resistor is the *next E24 value above* the calculation, so OVP1 lands clear of the ripple crest. The fitted pair gives

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$$V_{OVP1} = \frac{2.3}{1.5} \left(\frac{330}{19} + 1 \right) = 28.16 \text{ V}, \quad V_{OVP2} = \frac{2.5}{2.3} \cdot 28.16 = 30.61 \text{ V}$$

Start-up hands over to the loop at 16.53 V of output.

Swapping the two resistors makes OVP1 trip below a volt of output, so the converter never starts. Reversed winding polarity makes the bridge hard switch from the first pulse.

The datasheet turns-ratio ceiling applies only when the auxiliary winding supplies V_{CC} . Here it senses only (37.5 V nominal, 45.9 V at OVP2), so the only constraint is whole turns.

7.21 HVSU and V_{CC}

- **HVSU connects ahead of the bridge**, on the ac side, through one 1000 V diode per line; behind the bridge X-capacitor discharge and brown-out detection both stop working.
- The V_{CC} capacitor must hold the IC from $V_{CCon} = 17 \text{ V}$ to $V_{CCoff} = 8 \text{ V}$ until the supply takes over; the start-up unit stops 120 ms after V_{CCon} regardless.
- A diode between a linear regulator and the pin keeps the start-up current out of the regulator. 100 nF at the pin.

7.22 Controller pin rules

Pin	Rule
ISEN	No filter and no series resistor. The maximum-power law and the over-current threshold share this pin.
CFG, BM	No capacitor. Both are strapped at power-up. An out-of-range CFG resistor latches the controller off.
DRV_EN, RT	No capacitor. Both are sampled briefly every few milliseconds and a capacitor is read as a fault.
LOUT2	No pull-down if morphing is used. A few kilohms to ground reads as a fixed half bridge.
HVSU	Connect ahead of the input bridge, on the ac side. Behind the bridge, both X-capacitor discharge and brown-out detection stop working.
ZCD	Auxiliary winding polarity must make ZCD positive when the low side of leg 1 is on. Reversed, the bridge hard switches.
HOUTx, LOUTx	Logic-level outputs, not gate drivers. External half-bridge drivers are required.

Table 26: Pin rules that must not be broken.

7.23 Protections, and where each threshold is set

Each threshold is fixed by a resistor already chosen, so a change made for one reason moves a threshold set for another.

Protection	Set by	This design
Brown-out, on the rectified mains	R_{CFG} , read at power-up	V_{BO} 120 V peak = 84.8 Vac rms, clear of the 90 Vac minimum by 1.061
Cycle-by-cycle over-current, OCP1	R_{CS} — the same resistor as the maximum-power law	22.92 A against a composite peak of 18.32 A, a margin of 1.251

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Protection	Set by	This design
Second-level over-current, OCP2	R_{CS} , fixed ratio to OCP1	31.25 A
Output over-voltage, OVP1	the ZCD divider on the auxiliary winding	28.16 V, above the ripple crest and below OVP2
Second-level over-voltage, OVP2	the same divider	30.61 V
Capacitive-mode protection	internal; it watches the ZCD edge against the gate	the last line of defence only — the frequency floor f_{Min} is the first
Burst mode at light load	R_{BM} , read at power-up	30 k Ω ; the $2f_1$ feedback ripple is 52 mV, so see Section 6.33
Loss of ZVS at the worst corner	not a protection — a design margin	swept, not estimated; Section 6.5

Table 27: Protections and the component that sets each one.

7.24 System design rules

The same material as a checklist.

- **Design the tank at the equivalent range**, 173.2 to 332.3 Vac, not at the mains range.
- **Expect λ near 0.5**. A two-stage inductance ratio will not start at the lowest input voltage.
- **Keep f_{Min} above f_o** ; the anti-capacitive protection is the last line, not the first.
- **Size the output bank from ripple and hold-up**, take the larger, then check the rms ripple current separately.
- **Crossover in the tens of hertz**; handle the surviving feedback ripple with R_{BM} , not a faster loop.
- **Measure both morphing edges** at full and light load.
- **Check the turns ratio as $nV_{o,eff}$** , never as a bare ratio.
- **Judge over-current on the composite tank peak**.
- **Include the $2f_1$ component** in the capacitor ripple current; judge a centre tap at the output node.
- **Carry $\sqrt{d}$ into the secondary rms**; dropping it overstates the loss by about 26 % here.
- **Put L_{open} and L_{short} on the transformer drawing** beside the turns.
- **ESR at the switching frequency**, not 120 Hz.
- **Never round L_m up**.
- **Verify ZVS by sweeping the selected tank**; the closed form has no fixed error sign.
- **Record calculated and selected values separately**, and make every downstream check read the selected one.

7.25 What to measure first on hardware

In order; the first four decide whether the design is sound.

- **$f_{sw}(\theta)$ over a line half cycle**. It settles T_{idle} and both oscillator clamps, the thinnest margin in the design.
- **Cold start into the full 75.2 mF bank**. The start-up window is finite and this risk is new to the architecture.
- **ZVS at the half-bridge edge** (245 V_{pk}, full load): calculated 460 ns against 220 ns. Then the worst point of the sweep, 25 % load at 332.3 Vac equivalent, 401 ns, which is the figure this design is held to.

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- **Switching frequency at the full-bridge edge** ($235 V_{pk}$, full load): calculated 249.0 kHz; it must not hit the VCO ceiling.
- **Input current and THD at 90 Vac, full load**: the zero-crossing dead zone.
- **Output ripple against what the load tolerates**.
- **Ripple current and temperature rise in the bank**: 30.24 A rms calculated, 0.19 A per part.
- **Device temperatures, in half-bridge morphing** above $245 V_{pk}$, where the standing device dissipates most. This decides the accepted loss-budget shortfall.
- **Feedback ripple against the burst threshold** at light load.
- **An ac dropout at the ripple trough**: output still above 19 V after 12 ms.
- **A step across the morphing band**, both directions.

8 List of symbols

The symbols used in this note. The controller's own electrical parameters are in its datasheet.

Symbol	Meaning
The tank and its gain	
C_r, L_r, L_m	resonant capacitor, series inductance and magnetising inductance of the tank model
L_p	$L_r + L_m$, the inductance the tank shows at no load
$Z_0, Z_{0,design}$	characteristic impedance $\sqrt{L_r/C_r}$; the value the tank is designed to, $R_{ac} Q_{ZVS}$
Z_{in}	input impedance the bridge drives; $\arg Z_{in} = 0$ is the capacitive boundary
f_r, f_o	series resonance, and the lower resonance with L_m included
f_{sw}, f_n	switching frequency, and the same normalised to f_r
$f_{sw,max}, f_{sw,min}$	the specified maximum and minimum switching frequency; $f_{sw,max}$ is an input to λ
T_{sw}, T_r	switching period $1/f_{sw}$, and the resonant period $1/f_r$
λ, m	L_r/L_m , and $(L_r+L_m)/L_r = 1 + 1/\lambda$
$\lambda_{act}, \lambda_{req}$	the λ of the selected parts, and the λ the design asked for
$\lambda_1, \lambda_2, \lambda_{TD}, \lambda_3$	the four candidate λ : minimum gain, the same with the frequency ceiling, the dead-time condition, and the practical floor; the largest binds
Q, Q_{pk}, Q_{ZVS}	quality factor Z_0/R_{ac} ; its value at the line peak; the cap the ZVS condition puts on it
R_{ac}	the rectifier and load as one resistance at the fundamental
M	tank gain, $n V_{o,eff}$ over the drive
M_{req}, M_{pk}	the gain the operating point demands, and its value at the line peak
M_{min}, M_{max}	the gain range the tank must cover, from the two equivalent-input corners
M_{low}, M_{high}	the required gain at the low and the high equivalent corner
$M_{no load}, M_{\infty}$	the gain curve at $Q = 0$, and the floor $1/(1+\lambda)$ it flattens onto
M_z	the locus of the gain peaks, which is the capacitive/inductive boundary
x, u, q	the substitutions that turn the gain equation into a cubic: $1/f_n^2$, $\sin^2\theta$, and $Q_{pk}^2 u^2$
d	conduction ratio f_{sw}/f_r of the secondary, 1 above resonance

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Symbol	Meaning
n, n_T	equivalent-model turns ratio, and the physical (wound) turns ratio
N_{rect}	devices in the secondary conduction path: 1 centre tap, 2 full bridge
V_{drive}	the amplitude the bridge applies to the tank: the rectified mains in full bridge, half of it in half bridge
V_{refl}	reflected output voltage $n V_{o,\text{eff}}$
$V_{o,\text{eff}}$	$V_{\text{out}} + N_{\text{rect}} V_f$
V_f	forward drop of one secondary rectifier
The mains, and power factor	
θ	line phase angle
$f_l, f_{l,\text{min}}, \omega_l$	line frequency, its lowest specified value, and $2\pi f_l$
$v_{\text{ac}}, V_{\text{ac}}, I_{\text{ac}}$	the instantaneous mains voltage, and the rms voltage and current drawn
$V_{\text{rms}}, I_{\text{rms}}, I_{1,\text{rms}}$	rms input voltage and current, and the rms of the current's fundamental alone
$p_{\text{in}}(t), P$	instantaneous and average input power
PF, THD	power factor, and total harmonic distortion of the input current
ϕ_1, D	displacement angle of the fundamental, and the distortion factor $I_{1,\text{rms}}/I_{\text{rms}}$
D_3	third harmonic on the input current, as a fraction of the fundamental
$V_{\text{in}}, V_{\text{pk}}$	the rail the bridge works from, and the peak of the mains
$V_{\text{ac,eq}}, V_{\text{eq}}, V_{\text{ac,eq,low}}, V_{\text{ac,eq,high}}$	equivalent input voltage the tank sees after morphing, and its value at the two morphing thresholds
$V_{\text{eq,min}}, V_{\text{eq,max}}$	the lowest equivalent input (the HB edge) and the mains maximum, as the ST tool's margin factor Γ_v uses them
Power and loss	
$P_{\text{in}}, P_{\text{out}}$	input power, output power
$P_{\text{LLC}}, P_{\text{EMI}}, P_{\text{BR}}$	the three budgeted losses: the LLC stage, the EMI filter, the input bridge
$P_{\text{in,LLC}}$	power into the tank, $P_{\text{out}}/\eta_{\text{HB}}$
η_{HB}	assumed efficiency of the LLC stage
$P_{\text{cond}}, P_{\text{mos,dc}}$	conduction loss of the primary device that stands on in half-bridge morphing
$P_{\text{SR}}, P_{\text{SR,leg}}$	secondary rectifier loss, per device and per rectifier leg
P_{CS}	dissipation in the current-sense resistors
P_{budget}	the loss a device position is allowed, before parts are chosen
P_v	core loss per unit volume, from the material curve
E, C, V, V_{min}	the energy a capacitance C gives up between two voltages, in the hold-up expression
Currents and times of one switching cycle	
$i_{\text{Lr}}, I_{\text{Lr,pk}}$	tank current, and its peak: reflected load plus magnetising
$i_{\text{trafo}}, I_{\text{trafo,pk}}$	the reflected load current alone, and its peak

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Symbol	Meaning
$i_{Lm}, I_{Lm,pk}$	magnetising current of the tank model, and its peak
$i_{\mu}, i_{\mu,pk}$	magnetising current of the physical transformer, and its peak
i_p, i_s	the instantaneous primary and secondary winding currents
$I_{sec,pk}$	peak secondary current per rectifier leg
$I_{pri,rms}$	primary rms over the worst switching cycle
$I_{lc}, I_{pri}, I_{rect}$	a line-cycle rms: the rms of each switching period, averaged in I^2 over the line half cycle; for the primary and for one rectifier leg
I_{Cout}, I_{out}	ripple current in the output bank, and the load current
$t_D, T_{ZC}, T_{ZC,min}$	bridge dead time; the time the tank current takes to reach zero after the transition, and its smallest value over the operating space
v_d, V_{ds}	bridge mid-point voltage, and a device's drain-source voltage
$S_1, S_2, S_3, S_4, D_1, D_2$	the four bridge switches, and the two secondary rectifiers
The transformer and its core	
$B(t), B_{pk}, B_{max}$	flux density, its peak, and the ceiling the design keeps under
N	turns of the winding a voltage is applied to, in Faraday's law
A_e, A_{min}	core effective area, and its narrowest section
A_N	winding window area of the coil former
A_L, g, μ_0	inductance factor of the gapped core, the total centre-leg gap, and the permeability of free space
N_p, N_s, N_x	primary turns, secondary turns per winding, number of units in the assembly
$N_{aux}, n_{aux}/n_{sec}$	auxiliary winding turns, and its turns ratio to one secondary
L_{open}, L_{short}	inductance across the primary with every other winding open; and with every secondary shorted and the auxiliary open
I_{dc}	the dc current the DC-overlap test overlaps on the LCR signal
$I_{p,pk}, L_p$	in the flyback comparison only: the primary peak current and the primary inductance of a flyback
B_s	saturation flux density of the core material, from the material data at temperature
L_{μ}, L_{L1}, L_{L2}	physical magnetising inductance and the two leakage inductances
L_1, L_2	the open-circuit inductance measured at the primary and at one secondary
$L, \Delta L$	an inductance and the change of it a tolerance allows
I_{eq}, I_{sat}	open-circuit current that reproduces the operating flux; the DC-overlap test current
l_N	mean length of one turn on the coil former
The copper	
A_{cu}, J	copper cross-section one winding needs, and the current density it is sized to
$N_w, A_{cu,w}$	turns and copper area of winding w, in the sum over all the windings
k_u	window utilisation: the share of A_N that ends up copper

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Symbol	Meaning
$\delta, \rho, \rho_{20}, f$	skin depth, the resistivity of copper at the working temperature and at 20 °C, and the frequency the depth is evaluated at
d_s, a_s, n_s	Litz strand diameter, the copper cross-section of one strand, and the number of strands a winding needs
d_{litz}, k_{litz}	outside diameter of the served bundle, and the share of that area which is copper
$t_f, w_f, w_{f,max}, n_f$	foil thickness, the width one turn needs, the widest strip the bobbin takes, and the number of strips in parallel
Semiconductors	
$R_{DS(on)}, R_{DS(on),25}, k_T$	on-resistance, its 25 °C headline value, and the multiplier from 25 °C to $T_{j,max}$
$T_j, T_{j,max}, T_a$	junction temperature, its rating, and ambient temperature
$C_{oss}, C_{o(er)}, C_{o(tr)}, Q_{oss}$	the small-signal, energy-related and time-related output capacitance of a MOSFET, and the charge that time-related one stands for
The controller and its network	
V_{FB}, v_{FB}	feedback voltage above its 0.5 V offset at rated power, and its small-signal part
$V_{os}, K_{HV}, K_M, K_{FF}$	the offset and the three internal gains of the feedback law, quoted from the datasheet block diagram
K_{pwr}	slope of the power command, P_{out} per volt of V_{FB}
$R_{CS}, R_{CS,max1}, R_{CS,max2}$	current-sense resistor, and the two conditions that bound it: the maximum-power law and the OCP1 trip
I_{OCP1}, I_{OCP2}	over-current thresholds set by R_{CS}
C_T, R_T	oscillator timing capacitor and resistor
$C_{T,min}, C_{T,max}, R_{T,ceil}, R_{T,max}$	the window those two have to be chosen inside
$V_{ref}, I_{EA}, I_{EA,max}$	the oscillator reference, the error-amplifier current that is the only control input, and its maximum
f_{Min}, f_{Max}, f_{SU}	oscillator floor, ceiling and start-up frequency set by R_T, C_T and T_{idle}
T_{idle}	oscillator idle time
$R_{BM}, r_{BM}, V_{BM,eq}, P_{in,BM}$	burst-mode resistor, the fraction of rated power burst starts at, the equivalent FB threshold, and the input power that corresponds to
$R_{CFG}, R_{CFG,max}$	configuration resistor, and the largest value the lowest input voltage allows
$V_{BO}, V_{BO,pk}, V_{BO,rms}$	brown-out threshold, as a mains peak and as an rms voltage
$R_{ZCD,H}, R_{ZCD,L}, I_{bias}$	the ZCD divider, and the current it is designed to draw
$V_{OVP1}, V_{OVP2}, V_{OVP1,out}$	the two over-voltage thresholds, and the output voltage OVP1 is aimed at
$V_{CC}, V_{CCon}, V_{CCoff}$	the IC supply, and the two thresholds of its under-voltage lockout
The voltage loop	
$G_{plant}(s), G_{EA}(s), T(s)$	plant, compensator and loop gain; $T = G_{plant} G_{EA}$
G_o, EA_o	gain constant of the plant integrator and of the compensator, in rad/s
$\omega_z, \omega_p, \omega_{px}, \omega_c$	compensator zero, pole, high-frequency pole, and the crossover, in rad/s
$f_z, f_p, f_{px}, f_{180}$	the same zero and poles in hertz, and the frequency where $\arg T = -180^\circ$

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Symbol	Meaning
$f_c, f_{\text{cross}}, \Phi_M, GM$	loop crossover frequency, phase margin, gain margin
f_{cto}	where the bare plant integrator would cross 0 dB, $G_o/2\pi$
$f_{\text{PHF}}, f_{\text{MB}}$	the geometric mean of the two poles, and the frequency the K-factor method balances the loop about
K_v, Γ_v, α_v	K factor of the Type II compensator, and the input-voltage margin factor and weighting constant it is built from
Z_f	feedback impedance of the compensator, C_{F0} in parallel with R_F+C_F
R_I, R_O	output divider of the compensator
C_{F0}, C_F, R_F, C_{fx}	compensator parts: the two capacitors and the resistor of the TL431 network, and the FB-pin capacitor
$C_{\text{opto}}, C_{\text{ser}}$	the optocoupler's own output capacitance, and C_F+C_{F0} in series
$R_B, R_{B,\text{max}}, R_{B,\text{min}}, R_P, R_{FB}$	optocoupler LED resistor and the window it has to sit in, the TL431 bias resistor, and the pull-up inside the FB pin
$CTR, CTR_s, CTR_m, i_{\text{LED}}$	optocoupler current transfer ratio, its value at the steady-state and at the maximum LED current, and the LED current itself
$I_{\text{FB,steady}}, I_{\text{FB,max}}, I_{\text{min}}$	FB-pin current at steady state and at maximum, and the least current the TL431 needs to stay in regulation
V_R, V_Z, V_{F0}	TL431 reference, the regulated rail that feeds the LED, and the LED forward drop
v_{out}, v_C	small-signal output voltage, and the voltage on a capacitor in a waveform
$\Delta V_{\text{loop}}, \Delta V_{\text{FB}}$	$2f_i$ output ripple seen by the loop, and the ripple it leaves on the FB pin
The output, and hold-up	
$C_{\text{out}}, C_{\text{in}}$	output capacitor bank, input film capacitor
$V_{\text{out}}, V_o, V_{\text{out,eff}}, I_{\text{out}}$	output voltage and load current; $V_{\text{out,eff}}$ is $V_{o,\text{eff}}$ spelled out
$\Delta v, \Delta v_{\text{pp}}$	allowed and achieved $2f_i$ output ripple
$T_{\text{hold}}, t_{\text{hold}}, V_{o,\text{min}}$	required and achieved hold-up time, and the lowest output allowed at its end
ESR	equivalent series resistance of the bank, taken at the switching frequency
Verification margins	
$k, X, X_{\text{act}}, X_{\text{req}}$	a verification margin $X_{\text{act}}/X_{\text{req}}$, where X is whatever quantity the row is about; every k in this note is ≥ 1 when the design passes
$k_{\text{floor}}, k_{\text{ceil}}$	how far f_{Min} clears f_o , and how far f_{Max} clears the highest operating frequency
k_{Ploss}	the loss budget of a device position over the loss computed for it
w_k, θ_k	the weights and the nodes of the five-point Simpson rule the line-cycle rms values are integrated with

Table 28: Symbols.

9 References

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10 Errata and open items

Everything here is worked around rather than solved, and most of it only the released datasheet or a prototype can close. Re-read it before a production release.

10.1 Datasheet: the draft is not self-consistent

Item	What the draft says	What is used here, and why
Oscillator idle time T_{idle}	Section 5.3.2 says 700 ns; its Table 5 back-solves to about 250 ns from the frequency expressions	250 ns. The two clamps bracket the operating range at 250 ns and stop doing so at 700 ns, so this is a measurement, not a rounding
Brown-out expression	$V_{BO} = \min(60 \text{ V}, R_{CFG} \cdot 4 \text{ V/k}\Omega)$ — read literally this is 60 V for every resistor, which contradicts the configuration table	Read as $R_{CFG} \cdot 4 \text{ V/k}\Omega$ with 15 k Ω as the floor. That is the only reading consistent with the table
Brown-out units	the threshold is a <i>peak</i> voltage; board notes and design tools quote rms	Converted explicitly wherever it is compared ($\sqrt{2} = 1.414$)
Thermal resistance	TBD	No junction temperature can be predicted from the datasheet alone; the loss budget is checked against measured rise instead
ZCD absolute maximum	lower limit given as TBD	The divider is sized from the OVP thresholds only
Driver naming in Section 5.3.1	LOUT1 and LOUT2 are transposed in one sentence against the block diagram and the configuration table	The block diagram and the table are taken as correct: LOUT2 is the pin that is strapped and held high in half bridge

Table 29: Points where the draft datasheet contradicts itself or leaves a value open. Every one of them must be re-checked against the released document.

10.2 Traps in the obvious way of building a design sheet

Places where the obvious way to set up a design sheet gives a plausible wrong number that nothing flags.

Where	The trap	Consequence if missed
Equivalent input range	taking the maximum from the ac maximum instead of the full-bridge morphing edge	The frequency corner is never evaluated. Here it is the difference between 264 and 332.3 Vac equivalent, and it also sets λ

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Where	The trap	Consequence if missed
λ from the minimum-gain condition	the same corner again, one step earlier	The required λ comes out far too small — a factor of over a hundred in one case — and L_m with it
Line-cycle rms currents	evaluating one phase, usually $\theta = \pi/4$, and calling it the line-cycle value	The output bank ripple current and the rectifier loss are both understated. A Simpson rule over the half cycle costs nothing and is exact enough
Secondary rms above resonance	squaring a full half sine when the current is truncated	Up to 8 % at the high corner. Below resonance the factor is exactly 1, which is why it can sit unnoticed for a long time
Primary device rating	rating on the reflected load current $I_{\text{trafo,pk}}$	About 12 % under-rated — the switch carries the <i>composite</i> tank current
Loss and thermal resistance	computing loss from the 25 °C $R_{\text{DS(on)}}$ and then asking for a heatsink that holds 125 °C	The required thermal resistance comes out only about half as demanding as it really is
ZVS check	the closed-form shortcut, at the design Q and with the design λ	Error of either sign; on this tank 36 % conservative, on another about 23 % optimistic. Optimistic is the dangerous direction
Selected versus calculated	a check that reads the calculated value while the board carries the selected one	Every downstream margin is reported for a design that was not built

Table 30: Traps that produce a plausible wrong number.

10.3 Constants used here without a derivation

- The $\pi^2/8$ in the dead-time form of the λ condition. It behaves like a correction for the fundamental content of a square wave, but no source states it. It is indicative; the binding check is the ZVS sweep.
- The **exponent 5** in the closed-form ZVS shortcut. This is a fitted number, and it is the main reason the shortcut errs unpredictably near the boundary.
- The **0.744** in the input-voltage margin factor of the compensator design. It scales the crossover target and nothing downstream is sensitive to it at the percent level.
- The **16.8 $\Omega \cdot W$** maximum-power constant is consistent with the feedback span and the multiplier gains quoted in the datasheet — $2.8 \text{ V} / 0.167 = 16.8$ — so it is derived rather than assumed, and is listed here only because the gains themselves are draft values.

10.4 Open items in this design

Item	State	What closes it
T_{idle}	two candidate values, 250 and 700 ns	Measure $f_{\text{sw}}(\theta)$ on the first board
Primary conduction loss	0.311 against the budget — not met, and accepted	No single 600 V device meets a 3 W budget in the standing position. Whether it needs a heatsink is a thermal measurement, not a calculation
Secondary loss budget	1.063 — essentially exhausted	One more device in parallel per leg recovers it; the decision waits on the measured temperature
Transformer inductance tolerance	the tank tolerates a 4.1 % fall in open-circuit inductance and the usual specification asks for ± 10 %	Either tighten the low side with the supplier, or lower R_T to lift f_{Min} first. Before the transformer is ordered

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Item	State	What closes it
Output bank volume and height	75.2 mF, 160 parts	A mechanical question, not an electrical one, and it can send the whole architecture back to the output voltage
Standby power	burst entry assumed at 75 W	Measure, then trade R_{BM} against the feedback ripple
Efficiency assumption	$\eta_{HB} = 98\%$ assumed	Optimistic. Taking 95 % instead moves R_{CS} and R_{ac} by about 3 %, so nothing downstream is sensitive — but it should be replaced by a measurement

Table 31: What is not settled, and what would settle it.

Every cross-check here is a consistency check, not a correctness check. Three implementations agreeing means they implement the same equations, not that the equations describe the hardware. The first prototype is the correctness check.